Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Muñoz-EDSA

In Partial Fulfillment
of the Requirement for the Degree
Bachelor of Science in Information Technology

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EXECUTIVE SUMMARY

The study titled **Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.** focused on the development and implementation of an automated solution to address the company's challenges in timekeeping and payroll management. Assero Global Inc., a private security service provider operating across multiple regions in the Philippines, previously relied on manual Daily Time Records (DTRs) and Excel-based payroll computation, which often resulted in errors, payroll delays, and inefficiencies in managing employee data.

To resolve these issues, the developers developed a web-based and mobile-integrated system that streamlined attendance tracking and payroll processing. The mobile application allowed security personnel to log their attendance using facial recognition for identity verification and GPS tracking to confirm their location at designated posts. This ensured accurate and secure attendance monitoring while eliminating fraudulent practices such as buddy punching. The recorded data were automatically synchronized with the web-based system, which enabled administrators to compute payroll, manage employee information, and generate reports in real time.

The system was developed using ReactJS and React Native for cross-platform accessibility, Google APIs for facial recognition and GPS integration, and Supabase for database management and secure cloud storage. The Prototyping Model was utilized throughout the development process to allow continuous testing and feedback, ensuring that the final system met the operational needs of Assero Global Inc.

The implementation of the system significantly improved the company's efficiency in processing payroll, reduced administrative workload, and enhanced data accuracy. Moreover, it promoted transparency and employee satisfaction by providing a reliable and accessible platform for timekeeping and payroll information.

ENDORSEMENT FORM FOR FINAL DEFENSE

TITLE OF RESEARCH: **Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.**

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APPROVAL SHEET

This capstone project proposal titled **Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.** prepared and submitted by **Daniel I. Tarnate, Jason D. Selerio, Jimwell Andrey T. Concepcion, Lhorea Anne R. Oliveros and Maria Christina S. Resurreccion** in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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ABSTRACT

Title of Research:	Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.
Developers:	Daniel I. Tarnate Jason D. Selerio Jimwell Andrey T. Concepcion Lhorea Anne R. Oliveros Maria Christina S. Resurreccion
Degree:	Bachelor of Science in Information Technology

Keywords:	Web-based system; payroll automation; timekeeping; facial recognition; GPS tracking; mobile application; HRIS; security agency; attendance monitoring; Supabase; React Native.

This study developed a web-based timekeeping and payroll system with a mobile application using facial recognition and GPS tracking for the security guards of Assero Global Inc. The project addressed problems caused by manual DTR submission and Excel-based payroll processing, which often led to errors, delays, and inconsistent records. The system enabled secure attendance logging through biometric verification and location validation, with data automatically synchronized to a centralized payroll platform. Results showed improved accuracy in attendance records, faster payroll processing, reduced administrative workload, and enhanced employee access to payslips. Overall, the system significantly increased efficiency, transparency, and reliability in Assero Global Inc.'s operations.

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INTRODUCTION

As technology continues to advance, many companies, including Assero Global Inc., still use traditional methods. However, to keep up with today's fast-changing environment, businesses need to modernize. One area that has improved through technology is payroll and timekeeping. According to Doc Byte (2024), digital transformation in industries is a smart move that brings many benefits, such as better efficiency, higher income, and lower expenses. This shift is not just a trend; it gives companies a strong advantage in the digital world. Assero Global Inc. is a leading provider of private security services in the Philippines, serving various sectors like engineering, construction, education, and business process outsourcing (BPO). The company operates in Metro Manila, Central Luzon, Region 4A, the Visayas, and Mindanao. Despite its wide reach, the company still faces problems with its current timekeeping and payroll systems. Based on interviews, the company still uses manual daily time records (DTRs) and Excel spreadsheets, which are prone to errors because of repeated manual data entry. These problems lead to mistakes in employee records, incorrect deductions and benefits, delays in payroll, and violations of labor laws, all of which hurt employee morale and client satisfaction.

Because of the company's large operations, it needs a more efficient and automated system to handle timekeeping and payroll. The lack of a centralized database adds to the problem, as different users manually manage data in different ways. Also, differences in client needs, regional wages, and government-mandated contributions make it even more important to have a flexible and scalable solution. If these problems are not addressed, the company risks losing productivity, violating labor regulations, and damaging its reputation.

According to People Managing People (2024), automated systems can reduce the time spent handling data and allow HR and finance staff, like Krishna Tacadena, to focus on more important tasks. These systems also reduce errors and help the company follow rules, saving money eventually. Wardhana et al. (2024) stated that facial recognition makes attendance logging faster and more efficient. Nallusamy et al. (2023) also pointed out that using GPS and AI-powered location tracking ensures that security personnel are in the right place when they check in, which improves both accuracy and efficiency.

Project Context

Assero Global, formally known as AGISSS Security Services Inc., a provider of private security services in the Philippines, currently faces challenges associated with traditional timekeeping methods, including manual daily time records (DTR) and spreadsheet-based payroll management. Interview conducted by the developers with the finance and administrative departments reveal frequent discrepancies due to manual input errors, delayed pay slip distribution, and inefficiencies in handling employee deductions and benefits.

Given these pressing challenges, it becomes hard for Assero Global to transition towards an automated and technologically inclined solution. The company's need for an advanced timekeeping and payroll system is further emphasized by its operational scale, with employees dispersed across multiple regions and varying work schedules, requiring precise and timely monitoring. A failure to address these issues may result in continued inefficiencies, compliance risks and declining employee morale, compromising service quality, and client satisfaction.

Hence, this study proposes the development of a "Digitalized Timekeeping and Payroll System," tailored to Assero Global's unique operational requirements. This web and mobile-based solution will leverage advanced technologies, such as facial recognition for secure employee identification and GPS tracking for precise location verification, to enhance the reliability and transparency of attendance monitoring. By automating payroll processes, the system aims to minimize manual errors, ensure compliance with regulatory standards, and improve the timeliness of payroll delivery.

By modernizing timekeeping and payroll processes, this project addresses critical inefficiencies while promoting employee welfare and organizational resilience. The proposed system not only offers a practical response to immediate operational challenges but also establishes a scalable framework that can adapt to future growth, ensuring long-term benefits for both employees and management. This study, therefore, represents a strategic convergence of technological innovation and organizational need, poised to deliver measurable improvements in efficiency, compliance, and overall workplace satisfaction.

Purpose and Description

The project revolves around developing a comprehensive Payroll system integrated with digitalized timekeeping specifically designed for Assero Global Inc., a security services provider with personnel dispersed across various regions of the Philippines. The main goal of the project is to resolve existing issues related to collecting accurate Daily Time Records (DTRs), which currently result in payroll inaccuracies and delayed salary distributions.

The proposed solution includes a mobile application that integrates facial recognition and GPS tracking to accurately verify personnel identity and confirm their presence at designated work locations at the correct times. This technology eliminates manual attendance recording, significantly reducing errors and administrative workload.

Collected attendance data will be automatically transmitted to a centralized, web-based payroll system accessible at the company's main office. This payroll system efficiently processes the received data, facilitating accurate and timely payroll calculations. By automating the entire process, the system enhances the accuracy and reliability of payroll, thereby improving employee satisfaction and trust in the company.

What makes this project particularly valuable and unique is its integration of advanced biometric technology (Facial Recognition) with real-time location tracking (GPS), tailored specifically for businesses with geographically dispersed personnel. This innovative combination not only ensures accuracy and accountability but also significantly simplifies operational management. The ease of use provided by the mobile application, combined with the efficiency of web-based payroll processing, ensures that both field personnel and office administrators benefit from the system.

TThe project provides an effective, innovative, and relevant solution to Assero Global Inc.’s operational challenges, significantly improving payroll accuracy, employee accountability, and overall organizational efficiency.

General Problem

How to develop a web-based timekeeping and payroll system with mobile solutions utilizing facial recognition and GPS tracking for the security guards of Assero Global Inc.?

Assero Global Inc. continues to encounter significant challenges due to a stagnant and error-prone manual payroll and timekeeping process. The current system relies on the submission of Daily Time Records (DTRs) through Messenger, email, or physical delivery, depending on the personnel assigned to each site. This fragmented approach leads to frequent delays in processing and errors in data records.

The Owner has expressed frustration with the burdensome process of managing payroll and attendance through non-standardized channels. The Finance and Human Resource sections, responsible for processing payroll and attendance records, face ongoing difficulties in ensuring accuracy, timeliness, and consistency due to the lack of centralized reporting.

Specific Problem

- *How to develop an employee information management module that will serve as the data source for payroll?*

Assero Global Inc. currently experiences physical storage constraints due to the continuous accumulation of hardcopy of of employee records. With over 500 security personnel, the office shelves are filled with folders, each containing physical documents required for HR and payroll processes. This reliance on paper-based filing consumes significant office space and creates inefficiencies in record retrieval and maintenance. The absence of a digitized employee management system increases the risk of misplaced files, delays in information access, and vulnerability to document loss or damage.

According to interviews conducted with administrative and finance personnel, all employee records are kept as physical copies stored in folders on office shelves, with one folder for each security guard. This amounts to more than 500 folders occupying limited office space. Additionally, the current method of record keeping is entirely manual and decentralized, contributing to difficulties in tracking, securing, and updating employee profiles.

• *How to develop a module that will utilize GPS Tracking for the attendance of the security personnel of Assero Global Inc.?*

Assero Global Inc. currently operates without an automated system for monitoring the attendance of security personnel. Attendance tracking is conducted manually through Daily Time Records (DTRs), which are submitted via printed forms, Facebook Messenger, email, or physical delivery. This inconsistent and decentralized approach results in frequent delays, verification difficulties, and record discrepancies, particularly during payroll processing.

Based on interview findings, security personnel are required to manually fill out DTRs, which are then submitted to the Finance section through informal and fragmented channels. These methods overload communication platforms and result in time-consuming manual encoding into Excel payroll templates. The Finance Officer has reported an overwhelming volume of messages during salary cutoffs, which complicates the already tight window for finalizing payslip.

Furthermore, interviews revealed persistent issues with body punching and falsified entries on DTRs, which are often identified too late for correction. These incidents are frequently reported verbally and lack formal documentation, further complicating payroll accuracy and accountability. The Owner acknowledged the absence of any centralized or digital attendance verification method. Given the geographic dispersion of assigned posts and the dynamic scheduling of personnel, the current attendance system does not provide real-time monitoring or location verification. The lack of geolocation and biometric features has led to challenges in validating actual time-in and time-out records, thereby affecting the integrity of payroll computations.

- *How to develop a module that will automate the payroll of Aserro Global Inc.'s security personnel?*

Assero Global Inc.'s current payroll process relies exclusively on Microsoft Excel templates. This manual data entry method significantly increases the risk of miscalculations, delayed payroll release, and employee grievances. WWith ovverr 500 security personnel deployed nationwide, officers in the Human Resource and Finance sections are already experiencing data overload.

The Finance Officer reported issues with the receipt of Daily Time Records (DTRs) via Facebook Messenger, resulting in an overwhelming volume of messages from various clients. These DTRs must then be manually encoded into the Excel-based payroll template. Given that salary cutoffs fall on the 10th and 25th of each month, the Finance section is allotted only five days to finalize the payslips for over 500 security personnel. This process has also led to the Finance section receiving numerous complaints from employees due to delays and inaccuracies.

An interview with the Owner confirmed that the manual encoding of payroll data for hundreds of guards leads to calculation errors, especially when handling deductions such as SSS, PhilHealth, and Pag-IBIG, as well as regional and client-specific variations. Additional complexities arise from varying wage rates, contractual requirements, and allowance structures, all of which must be entered manually placing further strain on the Finance section and increasing the likelihood of inconsistencies.

The interview further revealed that no centralized system currently exists for managing employee records and payroll. Data is stored only on USB flash drives, exposing the process to risks of data loss, delayed recovery, and administrative inefficiencies.

- *How to develop a module that will provide the employee's payslip?*

Assero Global Inc. currently manages payslip generation through manual encoding from various satellite offices, which are submitted to the main office for processing and physical printing. This process is time-consuming, increases operational costs, and diverts HR and Finance personnel from more strategic functions. Additionally, printing payslips involves costs for paper, ink, and equipment maintenance, contributing to environmental waste and a higher carbon footprint. Delays in processing frustrate employees who rely on timely and accurate payment information, and errors in manual encoding may lead to incorrect salary releases that require additional resources to rectify.

According to interviews with finance personnel, payslips are often released a week after payroll is processed, leading to frustration among employees who expect timely access to their compensation details. Complaints directed at the Finance section were specifically attributed to delays in payslip distribution.

- *How to develop a module that will generate reports for Assero Global Inc.?*

At Assero Global Inc., payroll reports and payslip summaries are currently prepared manually using printed documents and Excel spreadsheets. This manual process is slow and prone to errors, as each report must be reviewed individually by the finance and administrative staff. Because payroll data must be encoded, checked, and reconciled by hand, the workflow becomes time-consuming, delays payroll processing, and increases the likelihood of miscalculations.

The absence of an automated system that can generate accurate payslips or provide payroll summarization restricts the company's ability to make timely financial decisions. It also makes it difficult to detect salary discrepancies, delayed submissions, or issues with deductions and contributions. Without a centralized reporting tool, management relies heavily on manual consolidation, which limits oversight and slows down compliance preparation.

Interviews with the owner revealed that payroll reports and payslips are stored and processed manually through Excel files saved in USB flash drives. These backup files are retrieved every few days and forwarded to accounting for tasks such as BIR-related documentation. The finance team also reported that payroll reporting errors occur around three (3) times every two (2) months, often caused by late or incomplete DTR submissions and inconsistent salary computations.

General Objective

To develop a web-based timekeeping and payroll system with mobile solutions utilizing facial recognition and GPS tracking for the security guards of Assero Global Inc.

The proposed Web-Based Timekeeping and Payroll System with Mobile Solutions for Assero Global Inc. aims to streamline attendance tracking, automate payroll processing, and securely manage employee records. The system will feature a mobile application that allows security personnel to clock in and out only when physically present at their assigned locations, utilizing GPS-based geofencing to prevent fraudulent entries. This real-time location verification, combined with automated payroll calculations based on attendance data and employee salary details, will help ensure correct salary computation and reliable attendance records. The time-out process will include confirmation prompts to minimize errors, while employees will have secure access to their payslips, allowing flexible viewing and downloading cutoff periods, monthly, or annually.

The system will be designed to support the operational scale of more than 500 security personnel deployed across multiple locations nationwide, addressing current issues such as processing delays, mismatched records, and the lack of centralized access to accurate and complete attendance and payroll data.

Additionally, the system will generate comprehensive reports on attendance and payroll metrics, customizable for management to facilitate better workforce oversight and strategic decision-making. With secure authentication protocols and data encryption in place, the system will protect sensitive employee information and ensure compliance with data privacy standards. Overall, this technology-driven solution will improve operational efficiency, reduce manual workloads, and promote transparency, empowering both employees and management at Assero Global Inc.

Specific Objective

- *To develop an employee information management module that will serve as the data source for payroll.*

The proposed system aims to develop an employee information management module that will serve as the primary data source for payroll. This module will store key details such as employee identification, assignment location, schedule, and salary rate. In addition to supporting payroll computations, the system will also act as a secure digital repository for employee-related data. By digitalizing and centralizing information, the system will improve data accuracy, allow for faster retrieval, and simplify maintenance, while minimizing the risks commonly associated with paper-based records such as damage, loss, or misplacement.

This system will improve how employee data is stored and protected. It will reduce the chances of losing important information due to problems with physical storage, such as fire, theft, or damage, by offering a safe and digital way to keep records. Payroll processing will become faster and more accurate through automatic data collection, which helps avoid errors and delays caused by manual entry. In addition, the system will help the company follow record-keeping rules and make audits easier by keeping files well-organized and easy to access. Overall, this solution will boost productivity in the organization and increase employee trust in how payroll is handled.

An integrated employee information management system provides a structured digital repository that improves data reliability and retrieval speed. Centralizing these records not only supports accurate payroll computation but also strengthens compliance with internal controls and labor regulations (Akisha K. & Dr. Thiyagarajan, 2023).

• *To develop a module that will utilize GPS Tracking for the attendance of the security personnel of Assero Global Inc.*

To solve this problem, the developers will develop a mobile attendance module where facial recognition and GPS verification integrated, in this system, employees will be required to arrive at their assigned location before they can clock in using the mobile app. This geolocation feature ensures that time-in can only be recorded when the employee is physically present at the designated site, thereby enhancing accountability and accuracy in attendance tracking. The time-out function will allow employees to log into their departure at any time, with payroll computation based on their total hours worked. To prevent accidental time-out entries, the system will include a confirmation prompt, ensuring that employees can verify their actions before they are finalized.

The attendance automation module is expected to greatly improve the correctness of attendance records for security personnel. By using location-based time-in, the system will help prevent buddy punching and time fraud, leading to more accurate payroll calculations. The confirmation step for time-out will help avoid mistakes, making sure employees are paid fairly based on the actual hours they worked. The mobile app will also make it easier for employees to log their attendance directly from their phones.

Overall, this system will encourage honest attendance reporting, make operations more efficient, and support a more clear and trustworthy payroll process at Assero Global Inc. Implementing a mobile attendance system that integrates facial recognition and GPS tracking can significantly reduce fraudulent practices like buddy punching. Such systems ensure that employees are physically present at designated locations when clocking in or out, enhancing accountability and accuracy in attendance records (TimeTrak, 2025)

- *To develop a module that will automate the payroll of Aserro Global Inc.'s security personnel.*

The proposed module aims to automate the payroll process of Assero Global Inc.'s security personnel. Manual payroll calculations will be removed, reducing the risk of errors and allowing for timely salary disbursements. The system will integrate employee attendance, leave, and overtime data collected from mobile applications to accurately compute wages, deductions, and bonuses. Detailed payroll reports will be generated, and secure records will be maintained for auditing and compliance purposes. A user-friendly interface will enable HR personnel to manage payroll operations efficiently, while employees will have transparent access to payroll information through the mobile application.

The implementation of this payroll automation system is expected to significantly improve operational efficiency at Aserro Global Inc. by minimizing administrative workloads and accelerating payroll processing times. It will enhance employee satisfaction by providing accurate and on-time payments and increasing transparency in payroll management. Additionally, the system will reduce the likelihood of payroll disputes and facilitate compliance with labor laws and company policies. Furthermore, this system will contribute to a more organized and accountable payroll process, fostering a positive work environment and supporting the company's growth objectives.

Automated payroll systems reduce manual errors, accelerate processing time, and ensure accurate computation of employee wages, deductions, and contributions. This improvement in payroll efficiency also contributes to higher employee satisfaction and organizational trust (Dipo, Ifeanyi, Amos et al., 2020).

- *To develop a module that will provide the employee's payslip.*

The proposed system aims to develop a module that will display the employee's payslip. This module will provide employees with secure access to their detailed payroll information through an intuitive interface. Employees will have the freedom to view or download their payslips according to their preferred criteria, including every cutoff period, specific months, or consolidated annual summaries. The system will ensure that the payslip data is accurate, up-to-date, and presented in a clear, easy-to-understand format. Additionally, it will incorporate secure authentication to protect sensitive payroll information and maintain employee privacy.

The implementation of this payslip module will empower employees by providing them with immediate and convenient access to their compensation details, fostering transparency and trust within the organization. By allowing flexible viewing and downloading options, including every cutoff period, the system will enhance employee satisfaction and reduce administrative workload related to payslip distribution. The secure and organized management of payroll information will also contribute to better compliance with labor regulations and auditing requirements. Overall, the module will strengthen payroll communication and support a more engaged and informed workforce.

Digital payslip platforms empower employees by providing immediate and secure access to their salary details, reducing administrative burdens, and improving transparency. This system enhances data privacy while supporting employee engagement (Finesse Resources, 2022).

- *To develop a module that will generate reports for Assero Global Inc.*

The proposed system aims to create a reporting module that automatically generates accurate and organized payroll summary reports. This module will compile essential payroll information—including basic pay, deductions, adjustments, bonuses, allowances, and net pay—into clear, easy-to-read summaries. It will also generate digital payslip reports that provide an overview of salary disbursements for each cutoff or pay period.

The reporting module will offer filtering and customization options, enabling management to generate summaries based on specific date ranges, employee groups, or payroll cycles. These reports will be available in structured formats and can be exported for accounting use, internal review, and compliance documentation.

By automating payroll report generation, the system will provide Assero Global Inc. with timely and reliable insights into payroll expenses, recurring trends, and potential discrepancies. This improvement will enhance financial monitoring, reduce manual consolidation efforts, and support auditing and regulatory processes. Overall, this module will strengthen transparency, operational efficiency, and strategic financial planning within the organization. It will improve transparency, operational efficiency, and strategic planning within the organization.

A robust payroll reporting module enables organizations to generate comprehensive reports on employee compensation, tax summaries, and payroll expenses. These insights are crucial for informed decision-making and optimizing payroll processes (WebHR, n.d.).

Scope

The project centers on the development and implementation of a Web-Based Timekeeping and Payroll System with Mobile Solutions for Assero Global Inc. This system is designed to streamline attendance monitoring and payroll processing through automated, secure, and scalable technology. It consists of both front-end and back-end modules that serve the needs of security personnel, administrators, and payroll officers.

System Modules:

- *Mobile Attendance Module* – Enables security personnel to clock in and out using facial recognition and GPS location tracking via mobile devices.
- *Web-Based Monitoring Module* – Allows HR staff and managers to monitor attendance logs and real-time location data.
- *Employee Payroll Module* – Automates payroll based on validated attendance and location data.
- *Payslip Preview Module* **–** Let the employees preview their payslip beforehand.
- *Employee Records and Attendance Management Module* - maintains a secure, unified database for employee –information, attendance, leave balances, and absenteeism, while enabling HR to update records, monitor trends, and enforce role- based access control for data security.
- *Compliance Module* – Calculates government-mandated deductions (SSS, Pag- IBIG, PhilHealth) and generates compliance reports.
- *Scheduling Module* – Manage work schedules for client-assigned security guards and relievers. Set time ranges, locations, and assigned days. configure client-wide defaults or individual employee overrides, all within client assignments
- *Cloud Storage and Backup Module* – Securely stores employee timekeeping and payroll records for up to 10 years.

Beneficiaries:

- *Security Personnel* – Can efficiently log attendance on-site using the mobile app.
- *HR/Payroll Admins* – Gain centralized access to real-time attendance and salary data.
- *Finance Department* – Ensures accurate, automated payroll processing without the need for manual DTR (Daily Time Record) management.
- *Management* – Can oversee deployment data and payroll analytics remotely.

The system will operate across both mobile and desktop platforms, accessible through verified credentials. Real-time syncing will ensure both security and flexibility. Users will benefit from a simplified and user-friendly interface that accommodates all levels of technical experience.

Limitations

The developers acknowledge the following limitations that are outside the system's control or scope:

- *User Device Availability* – The system requires security personnel to have smartphones compatible with facial recognition and GPS functionalities. The unavailability of such devices may hinder usage.
- *GPS Accuracy* - depends on the employee's mobile device hardware and environmental factors (signal interference in urban areas or indoor locations)
- *Regulatory Compliance* - The system will adhere to configurable payroll and tax rules, but compliance with local labor laws and tax regulations is the responsibility of Assero Global Inc.'s HR team. Updates to legal requirements may require manual configuration changes
- *Employee Device Compatibility* - The mobile app's performance depends on the employee's device specifications (e.g., OS version, GPS capability). Older devices may experience compatibility issues or reduced functionality.
- *Data Security* - while the system implements standard security measures (e.g., ecryption, authentication), it cannot control external security breaches caused by user behavior (e.g., sharing login credentials) or third-party infrastructure vulnerabilities. Physical security of employee devices is outside the system's control.

These limitations are not defecienciese but rather reflect the natural boundaries of software systems that depend on user, hardware, or infrastructure conditions. The system is desgined to mitigate many of these risks through automation, logging, and verification protocols. However, it is important to recognize that not all real-world variables can be fully managed through technology alone.

Review of Related Literature

In today's digital economy, organizations are increasingly dependent on technology to address complex challenges in workforce management. Traditional manual systems used for timekeeping and payroll processing have long been plagued by inefficiencies, human error, and compliance issues. These problems are magnified in industries with distributed or mobile workforces, such as the security services sector. In response, companies are adopting digitalized systems powered by facial recognition, GPS tracking, and automated payroll technologies. This chapter provides a comprehensive review of foreign and local studies that explore these technologies, establishing a conceptual framework for the proposed project at Assero Global Inc. The review is organized thematically to highlight innovations in biometric systems, GPS integration, payroll automation, data security, and regulatory compliance.

Local Literature

Evaluating the Effectiveness of Human Resource Information Systems in Enhancing Administrative Efficiency: A Case Study of the Zamboanga City Local Government

Delos Santos-Suñga, C., & Moreno, F. (2024). This study assessed the implementation of a Human Resource Information System (HRIS) in the Zamboanga City local government. The findings revealed significant improvements in administrative operations, such as faster data retrieval, enhanced reporting accuracy, and reduced reliance on manual HR tasks. The study highlighted that streamlined data handling processes reduced clerical errors and improved compliance with record-keeping regulations.

This study is highly relevant to the proposed system for Assero Global Inc., as it supports the use of HRIS platforms to improve administrative efficiency and data accuracy. The successful implementation of HRIS in a government context demonstrates its applicability in private security firms facing similar operational inefficiencies, particularly in managing voluminous employee records and automating workflows.

Adopting Human Resource Information System (HRIS)-Enabled Government Transformation: Perspective of MBHTE Employees

Salah, H. D., Abdulgani, M. A., Aliuden, M. F. M., Mantikayan, J. M., Guiamalon, T. S., Dilna, S. G., Mohamad, H. A., & Ferolino, M. F. L. (2022). This research analyzed how HRIS adoption by the Ministry of Basic, Higher, and Technical Education (MBHTE) improved administrative efficiency in the Bangsamoro Autonomous Region. Key findings included improved HR management, automated reporting, and employee satisfaction with the system's functionality. The system offered dashboards for easy tracking of personnel records and integrated leave, attendance, and payroll processing.

This literature supports the implementation of automated HR and payroll modules in the proposed system for Assero Global Inc. It shows the transformative impact of HRIS even in complex and underserved regions, highlighting its adaptability to various institutional structures and its ability to promote user-centric design that fosters operational excellence.

Effectiveness on the Implementation of Human Resource Information System to the User's Satisfaction at Yngen Datacom Corporation

Caparas, J. C. W., Fernando, F. D. P., Lucido, K. M. L., Mapili, N. M. M., Marasigan, K. M., & Salo, J. R. G. (2023). This undergraduate thesis from the Polytechnic University of the Philippines examined the effectiveness of an HRIS implementation at Yngen Datacom Corporation. Results indicated high levels of user satisfaction and efficiency in streamlining HR tasks such as leave processing, employee data management, and payroll tracking. The system replaced several fragmented processes with an all-in-one platform.

The thesis aligns with the current study's goal of digitizing HR operations at Assero Global Inc., especially in ensuring end-user satisfaction. The integration of HRIS in a private tech company reflects how digital systems can streamline employee-related services across industries. This is important for a security agency that handles numerous employees in various locations.

Designing and Implementing e-Justice Systems: An Information Systems Approach to Regional Trial Court Case Docket Management in Northern Mindanao, Philippines

Grepon, B. G. (2021). This research tackled the development of an electronic justice docket management system in Northern Mindanao courts. It successfully demonstrated how automation reduced delays in document handling, increased access to real-time information, and minimized record-keeping errors. It also facilitated improved monitoring of case timelines and simplified the management of legal documents.

This study is relevant to the timekeeping and payroll project because it shows the potential of automated automated systems to address long-standing issues with document handling and processing, which are parallel to the DTR and payroll documentation challenges at Assero Global Inc. Automation improves not only accuracy but also traceability and organizational capacity.

Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines

Grepon, B. G., Baran, N. T., Gumonan, K. M. V. C., Martinez, A. L. M., & Lacsa, M. L. E. (2021). This study focused on the implementation of an e-school system that automated student records and administrative processes. The system centralized operations and improved access to data for both students and administrators. Real-time data synchronization allowed different departments to collaborate more effectively.

Its relevance to the current study lies in its focus on digitizing institutional management workflows. The centralized approach of the e-school system mirrors the goals of the proposed HR and payroll platform to consolidate and streamline personnel operations. It demonstrates the effectiveness of centralized platforms in reducing inter-departmental miscommunication and redundancy

An HR and Payroll SaaS Platform for a Logistics and Supply Chain Company

Yusen Logistics Philippines (YLPH). (n.d.). This case study examined the implementation of a software-as-a-service (SaaS) HR and payroll platform by YLPH. The adoption resulted in reduced payroll processing time, real-time tracking of employee records, and better compliance management. The cloud-based setup allowed remote access and helped reduce system downtime.

This study supports the system development goals of Assero Global Inc., especially regarding cloud-based deployment for mobile staff. It validates the scalability and adaptability of SaaS platforms in labor-intensive industries like logistics and security, both of which require real-time access to personnel data.

Enhancing Employee Experience with HRIS: A Philippine Perspective

Payday PH (2023). This article featured a BPO company's use of HRIS to improve employee satisfaction through faster payroll processing and streamlined leave and attendance management. The company reported improved retention rates and reduced disputes. Automation also provided a clearer view of individual performance metrics.

It shiws how HRIS systems directly affect employee morale and productivity, affirming the importance of transparency and accessibility in payroll services. These features are core to Assero Global Inc.'s proposed mobile-access payslip and attendance platform, which aims to give employees control over their own data.

How HRIS Reduces Payroll Processing Time and Costs

Payday PH (2023). This article analyzed how a mid-sized Cebu-based retail company saw a 50% decrease in payroll processing time after implementing an HRIS. It emphasized reduced costs and better adherence to labor regulations. Employee access to payslips and forms was cited as a major advantage.

This literature is useful for justifying the cost-efficiency and compliance advantages of the proposed payroll system. Assero Global Inc. can expect similar gains by automating its payroll processes. The ability to instantly access payroll data reduces delays and enhances transparency.

HR and Payroll Processing in the Philippines: Ultimate Guide

Darwinbox. (2023). This guide outlined the legal, administrative, and technical standards for payroll processing in the Philippines. It explained common payroll errors, tax rules, and the advantages of cloud-based automation. It included best practices for system integration and multi-department access.

The study serves as a reference for ensuring regulatory compliance and best practices in designing the system. It is particularly helpful for incorporating government-required forms and tax computations into Assero Global Inc.'s payroll system, aligning operations with national labor policies.

Integrating HRIS in Local Government Units: A Case of Pasig City's Digital Payroll System

Rosales, M. A. (2023). This study explored the transition of Pasig City's payroll operations from manual to digital platforms through the implementation of a Human Resource Information System. The system improved data accuracy, enhanced report generation, and allowed real-time monitoring of employee attendance.

The findings offer direct relevance to Assero Global Inc.'s initiative, particularly in the context of shifting from Excel-based processes to an integrated, real-time payroll system. It affirms that digital transformation significantly boosts operational transparency and efficiency in payroll administration. It also underscores the importance of institutional readiness in system rollout.

SSS Contribution Table

The Social Security System (SSS) plays an essential role in employee compensation management in the Philippines, particularly in computing mandatory deductions in payroll systems. The SSS contribution schedule is the basis for determining the employer's and employee's monthly contribution rates according to the employee's salary bracket. It ensures that members receive benefits related to sickness, maternity, retirement, disability, and death. As of the latest schedule published by the Social Security System (2025), the contribution rate is set at 14% of the employee's monthly salary credit (MSC), shared between the employer and the employee. This rate is divided into 9.5% employer share and 4.5% employee share, with the minimum MSC at ₱4,000 and the maximum MSC at ₱30,000. These rates are vital for payroll systems to compute accurate deductions and ensure compliance with government-mandated contributions (Social Security System, 2025). This information supports the development of automated payroll systems, as integrating SSS computation modules based on official government data enhances accuracy and reduces manual errors during payroll processing.

Foreign Literature

Development of a Secure Web-Based Client Management System

Ramadas, K., & Abdullah, S. (2023). This research assessed the effect of this article presented a robust design and evaluation of a web-based payroll system capable of reducing payroll errors and streamlining compensation cycles. The developers concluded that the system significantly minimized manual workload and ensured timely salary distribution across departments. Key features include auto-generated payslips, employee records management, and report generation.

This study supports the development of Assero Global Inc.'s system by showing the operational gains possible through similar web-based payroll platforms, thus providing a framework that can be localized for Philippine security agencies.

Web-based payroll management system: Design, implementation, and evaluation

Ahmed, A. M., Mohammed, C. N., & Ahmad, A. M. (2023). This article presented a robust design and evaluation of a web-based payroll system capable of reducing payroll errors and streamlining compensation cycles. The developers concluded that the system significantly minimized manual workload and ensured timely salary distribution across departments. Key features include auto-generated payslips, employee records management, and report generation.

This study supports the development of Assero Global Inc.'s system by showing the operational gains possible through similar web-based payroll platforms, thus providing a framework that can be localized for Philippine security agencies. The functionality discussed is in line with the project's objectives to eliminate manual errors and improve payroll accountability.

Impact of computerized payroll systems on employee productivity

Tepede, D., Chukwulobebe, I., Fayemi, T. A., & Ojuawo, O. (2024). This research assessed the effect of computerized payroll systems on employee satisfaction and work efficiency. Results indicated that streamlined payroll processes boosted morale, reduced complaints, and shortened payroll cycles. The developers used survey tools to analyze feedback from employees and found a strong correlation between automated payroll access and reduced administrative disputes.

Its significance to this study lies in reinforcing the argument that digital payroll implementation contributes not only to operational efficiency but also to improved employee trust, essential in the security services industry. It also supports the use of employee self-service tools such as digital payslips, as proposed in the mobile application module.

Employee information and payroll management system.

This study outlines the development of an employee information and payroll management system that integrates employee records with payroll functionality. The project emphasized automation of key HR processes including attendance tracking, employee database management, and payslip generation. The evaluation showed improved operational accuracy and significant time savings in HR workflows.

This is relevant to the current system design as it demonstrates how integration of employee data into a centralized digital platform facilitates seamless payroll processing. Assero Global Inc. can benefit from a similar structure where employee profiles and timekeeping data directly inform payroll calculations.

Employee time tracking app with GPS.

TimeTrak's GPS-enables time tracking app allows businesses to confirm employee clock-ins based on their physical location. The app is especially useful for mobile and remote teams, allowing managers to monitor punctuality and presence through location-stamped entries. Many companies adopting this system have reported better accuracy in timesheet validation and fewer instances of time theft.

This aligns closely with the mobile attendance module of the proposed system, which aims to use GPS tracking to validate the location of security personnel at clock-in. Implementing similar functionality will help Assero Global Inc. reduce buddy punching and ensure legitimate attendance records

Digital payslips made easy.

This online article introduces a cloud-based tool for digital payslip distribution that offers secure access, historical viewing, and mobile compatibility. The system has been adopted by many small and medium-sized businesses to cut costs on paper printing, minimize payroll disputes, and improve employee satisfaction through 24/7 access.

This literature supports the development of a mobile payslip viewing module for Assero Global Inc., providing evidence that digital delivery reduces both costs and employee concerns. Integrating this feature into the system will ensure accessible and paperless payslip retrieval for deployed personnel.

The Role of Cloud Technology in Enhancing Remote Workforce Management and Collaboration.

Achoe, D. (2025). This study investigates how cloud-based solutions improve workforce oversight and communication across geographically dispersed operations. The study reports enhanced employee engagement and streamlines administrative workflows due to centralized data sharing and remote collaboration tools. Use cases involving businesses deploying security and field personnel in multiple cities.

The study substantiates the importance of using cloud infrastructure in the context of Assero Global Inc., which deploys personnel across various locations. Cloud-based systems are crucial for ensuring synchronized operations, real-time updates, and secure record-keeping. The relevance lies in its emphasis on centralized access to HR and payroll data, which is foundational to the proposed web-based platform.

Automated HR report generation using business intelligence tools: Case study in Latin America.

Martins, C. & Dias, M. (2022). This study examined the implementation of business intelligence tools to automate HR reporting in multinational companies. The system significantly reduced reporting time from day to minute and improved decision-making accuracy. Visual dashboards and auto-updated summaries were cited as key innovations.

The insights from this study are relevant for the reporting module in the Assero Global Inc. system. Automating the generation of payroll and attendance reports will enable managers to act on workforce trends promptly and more efficiently.

The role of cloud technology in enhancing remote workforce management and collaboration is important.

Shehu, S., Ibrahim, M., & Musa, Y. (2024). This study explored the impact of cloud-based systems on workforce coordination and productivity, focusing on remote work settings. Findings showed significant improvements in data accessibility, communication speed, and task visibility among decentralized teams. The authors highlighted that cloud platforms provided a unified environment where HR, finance, and operational data could be managed in real-time.

This is relevant to Assero Global Inc., where the workforce is geographically dispersed. Integrating a cloud-based timekeeping and payroll system will enable real-time updates, reduce administrative lag, and support the seamless management of data across regions. The study strengthens the rationale for using cloud technologies to achieve operational coherence in distributed organizations.

Synthesis

The reviewed studies collectively emphasize the need for automated systems that enhance accuracy, security, and efficiency in timekeeping and payroll management. Research on biometric technologies such as facial recognition highlights their effectiveness in minimizing identity fraud and ensuring that only authorized personnel can record attendance. Similarly, studies on GPS-based tracking systems demonstrate their usefulness in verifying employee locations, particularly for organizations with field-based workers like security guards. These technologies, when integrated into web and mobile applications, allow for real-time monitoring, reduced human error, and improved transparency in attendance reporting and payroll computation.

The literature also reveals that combining web-based platforms with mobile solutions significantly improves accessibility and operational efficiency. Web applications support administrative tasks such as payroll computation, employee record management, and report generation, while mobile applications enable employees to record attendance and access personal data conveniently. Overall, the related studies support the development of a Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc., as this integration of technologies addresses the company's challenges in monitoring, verifying, and managing employee data efficiently and securely.

These limitations are not deficiencies but rather reflect the natural boundaries of software systems that depend on user, hardware, or infrastructure conditions. The system is designed to mitigate many of these risks through automation, logging, and verification protocols. However, it is important to recognize that not all real-world variables can be fully managed through technology alone.

TECHNICAL BACKGROUND

Overview of Current Technologies to be used in the System

The proposed system for Assero Global Inc. will be developed using a combination of mobile and web-based technologies to ensure accessibility, scalability, and security. The system will be composed of a mobile application for attendance tracking and a web platform for monitoring, payroll processing, and data management. To build this system, the developers will use modern, industry-standard development environments and APIs that support cross-platform functionality and integration.

For the mobile application, Android Studio will be utilized as the primary development environment. It will incorporate face-api.js for facial recognition-based attendance logging, ensuring biometric verification and minimizing the possibility of fraudulent clock-ins. additionally, the mobile app will leverage various APIs including Cally, Leaflet, Moments, and React Big Calender to enhance timekeeping accuracy and location validation. These APIs will also support features such as real-time syncing, deployment verification, and local environmental awareness, which are crucial for field-based personnel.

The web-based components of the system will be developed using Visual Studio, combining front-end technologies such as HTML, CSS, JavaScript, and Bootstrap for responsive design and intuitive user experience. These technologies will power the system's administrative dashboard, payroll module, reporting features, and compliance automation tools. The web platform will serve as the main interface for HR, payroll officers, and management, allowing them to review real-time attendance, compute salaries, generate reports, and manage employee data securely.

By integrating these technologies, the proposed system will offer a comprehensive digital solution to address the operational inefficiencies currently faced by Assero Global Inc., ensuring reliable attendance tracking and accurate payroll processing through secure and user-friendly digital tools.

Prototyping Model

The proposed system will adopt the Prototyping Model, which is a software development model in which a prototype is built, tested, and reworked until an acceptable prototype is achieved. It also creates a base to produce the final system or software. It works best in scenarios where the project's requirements are not known in detail. It is an iterative, trial, and error method which takes place between developer and client.

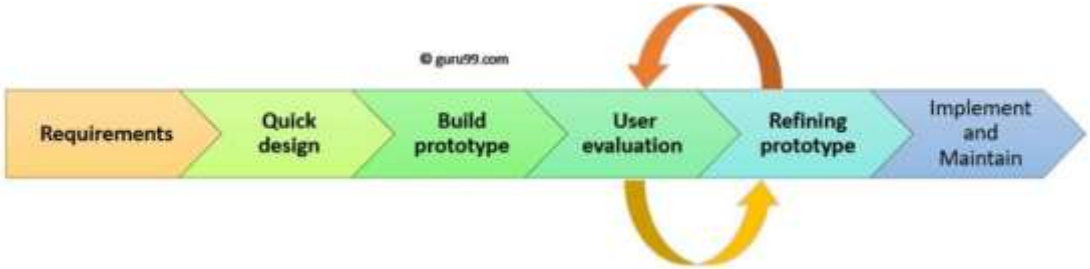


Figure 1. Prototype Model

Taken from: https://www.guru99.com/software-engineering-prototyping-model.html

Requirements gathering and analysis

The first phase focuses on understanding the detailed functional and non-functional requirements of Assero Global Inc.'s payroll and timekeeping process. This includes interviews with the company's Finance Officer, HR personnel, and selected field security personnel to identify pain points such as delayed payroll processing, manual DTR submissions via Messenger, inconsistencies in attendance logging, and the absence of centralized records. The goal is to clearly define expectations and identify the features needed in the final system, such as facial recognition, GPS tracking, digital payslip access, and report generation modules.

Quick design

A preliminary or “quick” design of the system is created, showcasing the basic interface and structural layout of the application. For Assero Global Inc., this includes rough wireframes of the mobile attendance module, employee dashboard, payroll computation screens, and admin access panel. While not fully functional, this phase enables stakeholders to visualize the general framework of the solution and provide early-stage input.

Build prototype

Using the outputs of the quick design, the development team constructs a working prototype of the core modules. For this project, the prototype may include a functional facial recognition login screen, GPS-based time-in and time-out process, and a sample digital payslip viewer. This prototype mimics real interactions with the application and allows users to test features directly on their mobile devices or browsers.

Initial user evaluation

The initial prototype is presented to the Thesis Adviser for evaluation, serving as the primary user representative during this phase. The prototype includes core system functionalities such as facial recognition login, GPS-based attendance tracking, payroll computation, and employee payslip display. The adviser provides critical feedback on system usability, alignment with the project’s objectives, and areas requiring improvement. This stage helps the developers identify gaps between the prototype and the intended solution for Assero Global Inc., ensuring that the design remains aligned with academic rigor and real-world application. The adviser’s suggestions also guide refinements before final implementation.

Refining prototype

Based on the evaluation, the prototype undergoes revision. For example, if users report difficulties in GPS signal acquisition or errors in deduction computation (SSS, PhilHealth, Pag-IBIG), the development team adjusts the system accordingly. This loop of feedback and improvement continues until all user-defined requirements are satisfied. It ensures that the final design not only works but meets the real-world demands of Assero Global Inc.'s field deployment and compliance needs.

Implement and Maintain

Once the prototype is refined and approved, the fully integrated system is developed. This includes backend database configuration, secure login and role-based access, automated payroll computation, and real-time report generation. After final testing, the system is deployed within the company's actual operations. Ongoing maintenance ensures data integrity, system uptime, and adaptation to regulatory updates or business process changes, safeguarding the long-term reliability of Assero Global Inc.'s digital payroll and attendance solution.

Calendar of Activities

January – The developers brainstormed and searched for a possible host company.

February—The developers interviewed a company that initially created two (2) titles for one (1) company. The developers also interviewed an organization, thereby creating another title. The developers gathered information and interviewed employees and organization members to complete chapters one (1) and two (2) and prepare for mock defense.

March – The developers conducted a survey, completed the checklist given, and obtained more evidence by interviewing and requesting it from the host company.

April – The developers started revising the documents and are still getting new evidence from the company, as the panelist requested. The developers have also created formal letters requesting company policies and organizational structure. The developers conducted another interview with the host company's finance department. The developers have also started designing the UI system, specifically for presentation purposes.

May – The developers are revising more and starting to design the system's UI for the Final Mock Defense.

June – The developers continued the designing of the Mobile Application.

July – The developers presented the Mobile Application to their adviser to seek further comments and revisions.

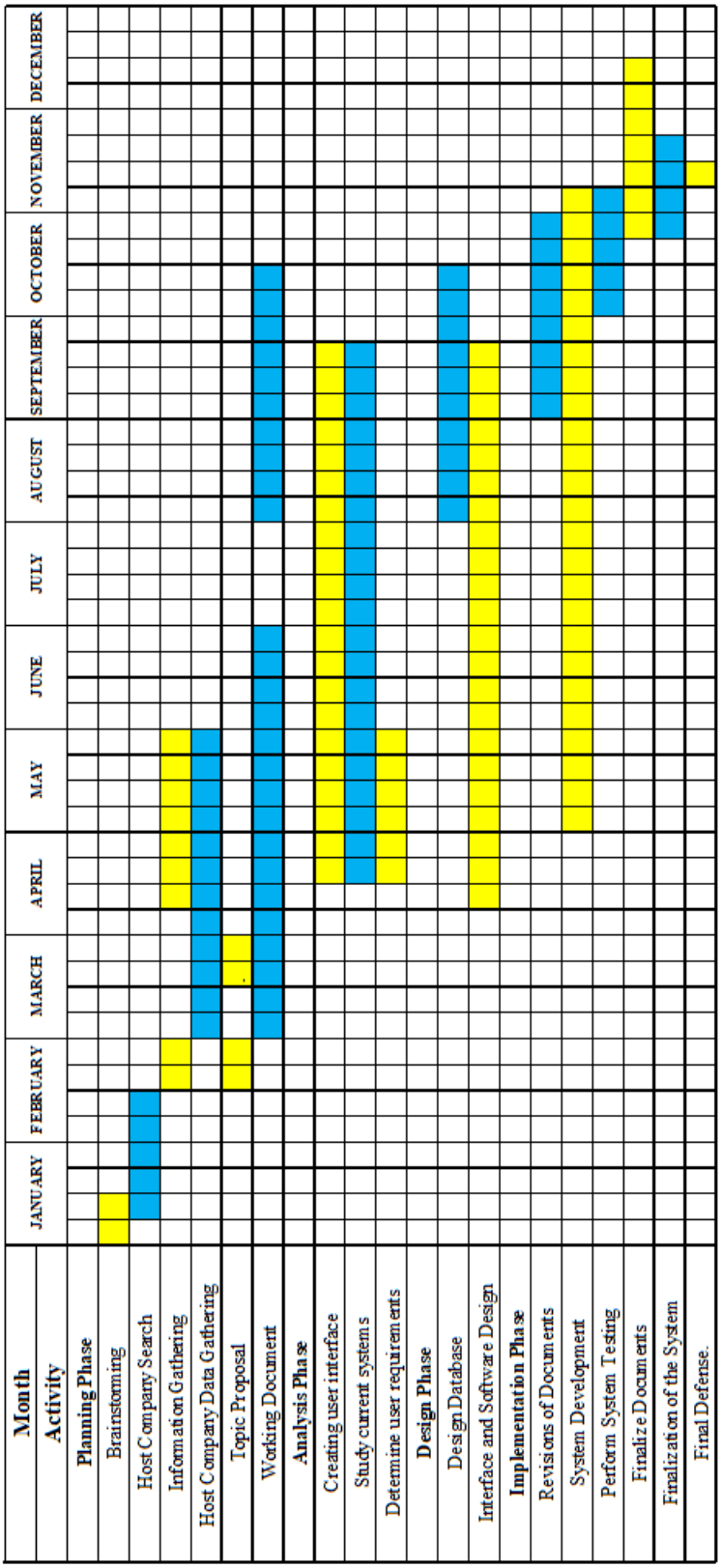
August – The developers resumed the documentation, did some revisions for the Web Applications and find a proper dynamic hosting for their web application.

September – The developers started to work on the system's database and web application modules.

October – The developers started to gather data using the questionnaire among Assero Global Inc.'s employees.

November – The developers started the finalization of the documents and the entire system.

Gantt's Chart of Activity



RESOURCES

Hardware

Desktop

The developers require laptops or desktop computers with a minimum Intel Core i5 or equivalent processor, 8GB of RAM, and at least 256GB of SSD storage. These specifications are essential to run integrated development environments (IDEs) such as Visual Studio Code and Android Studio smoothly, which are used for front-end, back-end, and mobile application development. A stable internet connection and support for virtualization (for Android emulators) are also recommended.

Android Smartphone

The mobile application will be deployed and tested on Android smartphones with at least a Quad-Core 1.4 GHz CPU, 4GB RAM, and 16GB internal storage. These devices must run Android Nougat (version 7.0) or later to support critical APIs such as Google Face Mesh, GPS location services, and calendar synchronization. These specifications ensure smooth operation of face detection algorithms and reliable location tracking in the field.

Software

Visual Studio Code

According to Code Visual Studio (n.d.), Visual Studio Code (VS Code) is a free, lightweight code editor that runs on macOS, Linux, and Windows operating systems. It offers quick installation and minimal system requirements, allowing developers to start coding in just a few minutes. VS Code also supports a large ecosystem of extensions, making it adaptable for various programming languages and development environments. Its integrated debugging, version control, and IntelliSense features contribute to efficient and productive software development.

React Native

According to ReactnativeDev (n.d.), React Native is a framework that enables developers to build native mobile applications using React and JavaScript. It allows React developers to reuse existing knowledge to create mobile interfaces that compile to native components. At the same time, native developers can use React Native to unify platform-specific features across iOS and Android. This strategy reduces development time while maintaining high performance and native-like user experience.

Expo

Expo is a production-grade framework built on top of React Native, designed to simplify mobile app development. It provides a set of development tools, file-based routing, and a standard library of prebuilt native modules that eliminate complex configuration steps. Expo also includes over-the-air updates, a device preview app, and optimized build services, making development faster and more accessible. Its unified workflow helps developers build, test, and deploy React Native applications efficiently.

React JS

According to Meta Open Source (n.d.), ReactJS is a free and open-source JavaScript library used for building user interfaces, especially for single-page applications requiring fast and dynamic interactions. It promotes a component-based architecture that allows developers to build reusable UI elements. React also uses a virtual DOM to optimize rendering performance and update components efficiently as data changes. Its flexibility, scalability, and large ecosystem have made React one of the most widely adopted libraries in modern web development.

HTML, CSS, JavaScript, and Bootstrap

According to Mozilla Developer Network (MDN), HTML provides the structural foundation of a webpage, while CSS defines its styling, layout, and visual presentation. JavaScript enables interactivity and dynamic behavior, allowing web pages to respond to user input in real time. Bootstrap enhances the development process by offering ready-made components and responsive grid systems, allowing developers to build visually appealing interfaces more efficiently. Together, these technologies form the basis of most modern web applications.

Heroicons

Heroicons is a React package that provides a collection of free, MIT-licensed SVG icons created by the team behind Tailwind CSS. The icons come in various styles such as outline, solid, and mini, offering flexibility for different design requirements. Developers can easily import icons as React components and customize them as needed. Heroicons integrate seamlessly into modern interfaces, especially those using Tailwind CSS.

Lordicon

Lordicon is a React wrapper for animated icons provided by the Lordicon platform. These icons are powered by Lottie animations, enabling smooth, interactive, and visually engaging effects. Developers often use Lordicon to elevate user experience through animated feedback, loading indicators, and interactive UI elements. Its extensive library and customization options make it a valuable addition for modern user interfaces.

Lucide React

Lucide React is an open-source icon library offering clean and customizable SVG icons specifically for React applications. Each icon can be modified through React props such as size, color, and stroke width, allowing flexible design adjustments. As a community-driven alternative to Feather Icons, it expands the available set of modern minimalist icons. Its lightweight nature makes it ideal for maintaining performance while enhancing UI design.

React Big Calendar

React Big Calendar is a full-featured calendar component library designed for React applications. It supports features such as event scheduling, drag-and-drop interactions, and multiple viewing modes (month, week, day). Its integration with React state and props enables dynamic calendar updates based on user actions or backend data. This tool is commonly used in dashboards, booking systems, and employee scheduling applications.

React Router

React Router is a widely used routing library for managing navigation in React applications. It supports dynamic routing, nested routes, and browser history manipulation, enabling smooth transitions between views. React Router helps developers structure their application as separate pages while still maintaining the single-page application (SPA) architecture. Its flexibility and ease of use make it an essential tool for multi-page React projects.

Cally

Cally is a JavaScript library designed to handle event scheduling and calendar-related functionalities. It helps developers organize and manage events, timelines, and date-based tasks efficiently. The library simplifies common calendar operations, making it suitable for applications that require task scheduling, to-do lists, or organizational tools. Its lightweight nature contributes to smooth performance even in complex scheduling scenarios.

Chart.js

Chart.js is a versatile JavaScript library used for creating interactive and visually appealing charts. It supports various chart types, including line, bar, pie, and radar, making it suitable for many types of data visualization. Its configuration system is beginner-friendly yet powerful enough for advanced use cases. Developers commonly use Chart.js to display analytics and dashboard metrics in modern web applications.

Leaflet

Leaflet is a lightweight, open-source JavaScript library for building interactive maps. It provides features such as markers, popups, map layering, and tile rendering with excellent performance on both mobile and desktop devices. Developers use Leaflet extensively in mapping and geospatial applications because of its simplicity and extensibility. Its well-documented API makes it accessible even to developers who are new to mapping systems.

React-Leaflet

React-Leaflet is a React wrapper for the Leaflet mapping library, allowing developers to use Leaflet maps as React components. It adapts Leaflet's map elements—such as markers, layers, and popups—into reusable components managed through React's state and props. This makes map rendering more predictable and easier to maintain in React applications. It is essential when integrating Leaflet into any React-based frontend.

Face-api.js

Face-api.js is a JavaScript library that provides face detection, facial recognition, and facial landmark tracking directly in the browser. Built on top of TensorFlow.js, it enables real-time face analysis without relying on a backend server. Developers commonly use it for authentication systems, user identification, and interactive camera-based experiences. Its client-side processing capabilities enhance performance and privacy.

Moment.js

Moment.js is a widely used library for handling dates and times in JavaScript. It simplifies tasks such as parsing, formatting, validating, and manipulating date objects. Although now in maintenance mode, it remains prevalent in many existing applications due to its reliability and extensive functionality. Developers often use it to manage timestamps, time zones, and user-friendly date displays.

Node.js

Node.js is an open-source, cross-platform JavaScript runtime environment designed for creating scalable server-side applications. It uses Google's V8 engine to execute JavaScript outside the browser, enabling backend development with a familiar language. Node.js is known for its event-driven, non-blocking architecture, which allows it to handle many simultaneous connections efficiently. Its large ecosystem of packages makes it highly suitable for API development, real-time systems, and backend services.

Express.js

Express.js is a fast, minimalist web framework built for Node.js. It provides routing, middleware support, and HTTP utility methods, allowing developers to build APIs and web servers with ease. Express simplifies common backend tasks such as request handling, data parsing, and authentication. Because of its flexibility and extensive plugin ecosystem, it is one of the most widely used frameworks in Node.js development.

Dotenv

Dotenv is a Node.js library used to load environment variables from a .env file into a project's runtime environment. This allows developers to securely store sensitive configuration values such as API keys, database credentials, and secret tokens. It promotes clean code practices by separating configuration from application logic. Using Dotenv enhances maintainability and security across development environments.

Nodemailer

Nodemailer is a Node.js module used to send emails from server-side applications. It supports various email transport methods, including SMTP and OAuth2 authentication, making it compatible with providers like Gmail and Outlook. Developers commonly use Nodemailer for password resets, verification of emails, and automated notifications. Its simplicity and reliability have made it a standard tool for backend email functionality.

TensorFlow.js

TensorFlow.js enables machine learning models to run directly in the browser or in Node.js using JavaScript. Developers can use it to train new models or load pre-trained ones for tasks such as image classification, predictions, and pattern recognition. This allows advanced machine learning features without requiring Python or external servers. TensorFlow.js makes machine learning more accessible to web developers while enabling real-time in-browser processing.

Supabase

Supabase is an open-source Backend-as-a-Service (BaaS) platform that provides authentication, database management, storage, and real-time capabilities. Unlike Firebase, which uses a NoSQL database, Supabase is powered by PostgreSQL—a robust relational database known for reliability and scalability. It offers APIs, client libraries, and tools that simplify backend development. Supabase enables developers to build full-stack applications more quickly by handling many server-side concerns automatically.

SQLite

SQLite is a lightweight, serverless, self-contained SQL database engine widely used in mobile applications, embedded systems, and small to medium-sized software projects. It stores data in a single file, making it easy to implement without requiring a separate server to process. Despite its small footprint, SQLite supports most SQL standards and is highly reliable for local data storage.

Hostinger

Hostinger is a global web hosting company that offers affordable hosting services such as shared hosting, VPS hosting, cloud hosting, and domain registration. It is known for its beginner-friendly control panel, fast server performance, and customer-focused features. Hostinger also provides tools and resources that help individuals and businesses easily build and manage their websites.

Requirements Analysis

This section details the requirements needed to address the operational challenges faced by Assero Global Inc. through the development of a Web-based Timekeeping and Payroll System with Mobile Solutions. The analysis identifies key functions, business needs, and technical considerations to guide system development and ensure alignment with stakeholder expectations.

A. Requirement

Personnel must be able to log attendance via mobile application using facial recognition and GPS. The system must transmit attendance data to a web-based backend that will automatically compute payroll, manage employee records, and generate reports. The system must provide administrative access to authorized staff and allow real-time monitoring of all activities.

B. Business / User Requirement

The system must assist Assero Global Inc. in managing over 500 deployed personnel, eliminating reliance on Excel files and manual DTRs. The proposed software should handle real-time employee logins and logouts, track location, and support HR in payroll processing, compliance, and report generation.

System Integration for Timekeeping and Payroll

1. The system must replace manual DTR submission processes.
2. The system should accurately track attendance and integrate it with payroll computations.

Mobile Attendance App (Android)

- Security personnel will log in attendance through facial recognition.
- The system will verify their attendance via GPS.

Web-based HR Dashboard

- Admins will monitor attendance logs, employee data, and payroll via a secure web portal.
- Admins can generate payroll summaries and compliance reports.

C. System Requirements

Major System Capabilities

- The system will function online (requires internet connection).
- The system will support biometric facial recognition.
- The system will support GPS location tracking.
- The system will store attendance and payroll records securely.
- The system will allow role-based user access.
- The system will generate payroll summaries and reports.
- The system will provide secure data encryption and backups.
- The system shall include an employee information management component to maintain accurate records that serve as the bases for payroll and attendance processing.

Major System Conditions

- Each personnel must use a verified device and registered credentials.
- Each account is unique to each employee and cannot be shared.

D. System User Characteristics

User:

- **Security Personnel** – Users of the mobile application for time-in and time-out via facial recognition and GPS.
- **Finance/Admin/HR Staff** – Web-based users who manage attendance logs, employee data, salary computations, and reports.

E. Functional Requirements

1. Manage Employee Record

1.1 Admins shall manage employee profiles.

1.2 System shall log historical attendance and payroll entries.

2. Manage Attendance

2.1 Employees shall log in attendance through face recognition and GPS.

2.2 The system shall validate attendance based on location and time.

2.3 Attendance data shall be submitted in real-time to the admin dashboard.

3. Manage Payroll

3.1 Admins shall access attendance logs for payroll computation.

3.2 The system shall calculate salary based on configured rules.

4. Generate Preview of Payslips

4.1 The system shall generate a preview of the security guard's payslip.

4.2 The payslip should only just be a preview the salary based on their attendance the deductions not yet included.

5. Generate Reports

5.1 The system shall generate biweekly reports on attendance and payroll.

5.2 Reports shall be exportable in PDF format.

F. Non-Functional Requirements

1. Operational Requirements

   1.1 The system will be accessible via web and Android platforms.

2. Performance Requirements

   2.1 Real-time syncing of data between mobile and web systems.

   2.2 High availability and minimal downtime.

3. Security Requirements

   3.1 All user data will be encrypted.

   3.2 Role-based access controls will be enforced.

   3.3 The system will log all access and updates for audit trails

Requirements Documentation

The proposed system aims to replace and optimize the manual timekeeping and payroll processes currently used by Assero Global Inc. The system will be accessible through both mobile and web platforms, providing a user-friendly interface designed for field personnel, HR staff, and payroll administrators. The intuitive UI will support ease of navigation across user roles and allow secure data input and monitoring in real-time.

WEB APPLICATION

Login Page - serves as the system's authentication interface where users securely enter their credentials to access their designated web or mobile application accounts. ￼

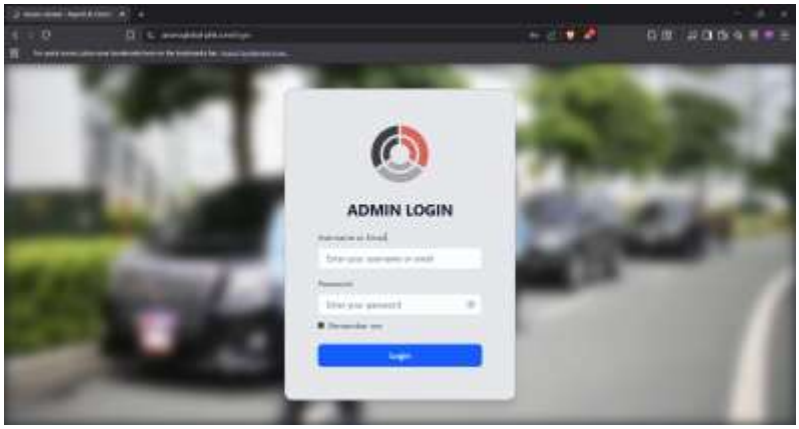


Figure 2. Login Page

Dashboard - Overview of key information such as employee statistics, attendance summaries, and payroll insights for efficient monitoring and decision-making.

Figure 3. Dashboard

Finance Module

Payroll - is responsible for automatically computing employee salaries, deductions, and allowances based on attendance data, ensuring accurate and efficient payroll processing.

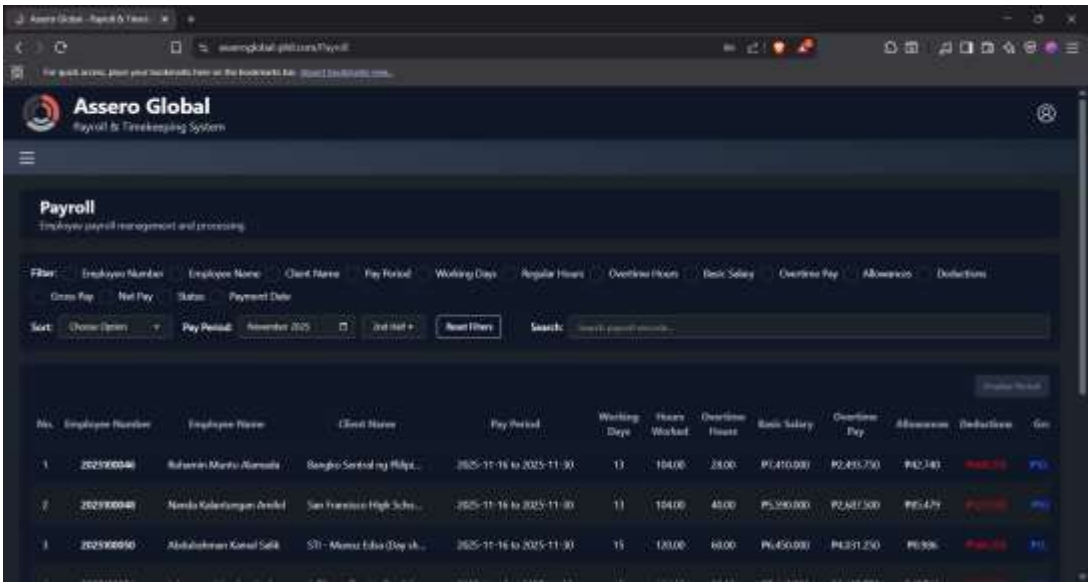


Figure 4. Payroll Sub-Module

Leave Management - allows employees to file leave requests and enable administrators to review, approve, and monitor leave records efficiently within the system.

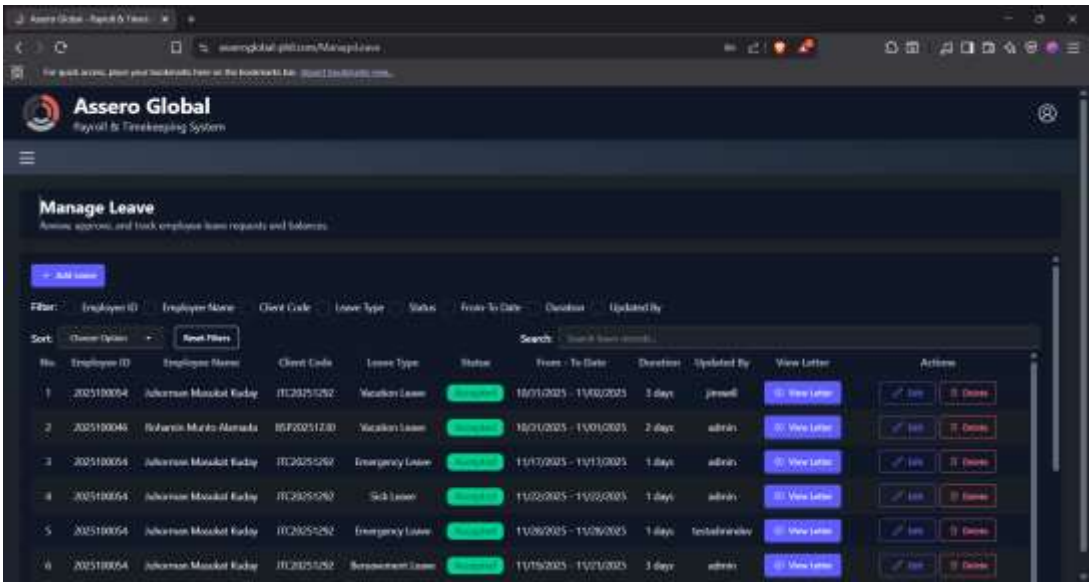


Figure 4.1. Leave Management Sub-Module

Calendar - serves as a visual tool that displays important company dates, payroll cutoffs, and events to help users stay informed and organized within the system.

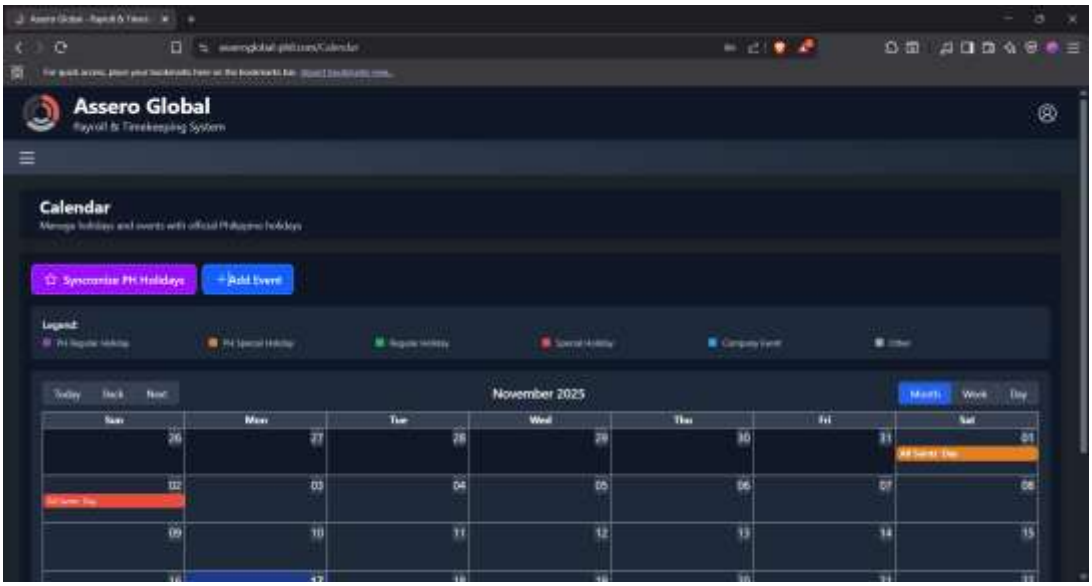


Figure 4.2. Calendar Sub-Module

Reports - used to view and generate payroll records and employee payslips.

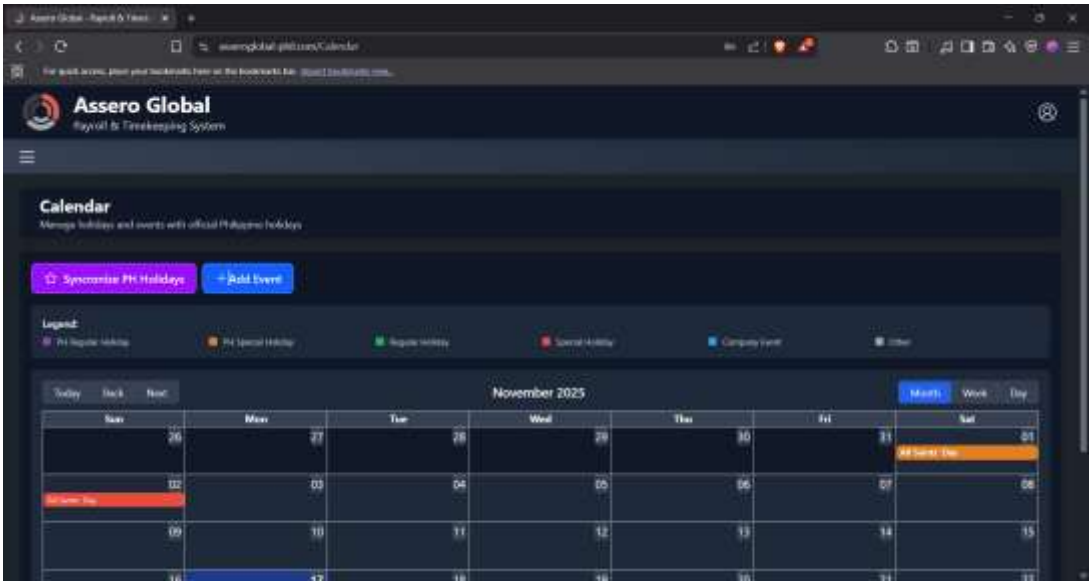


Figure 4.3 Reports Module

Attendance - records and monitors employee time-in and time-out using facial recognition and GPS tracking to ensure accurate and verified attendance data.

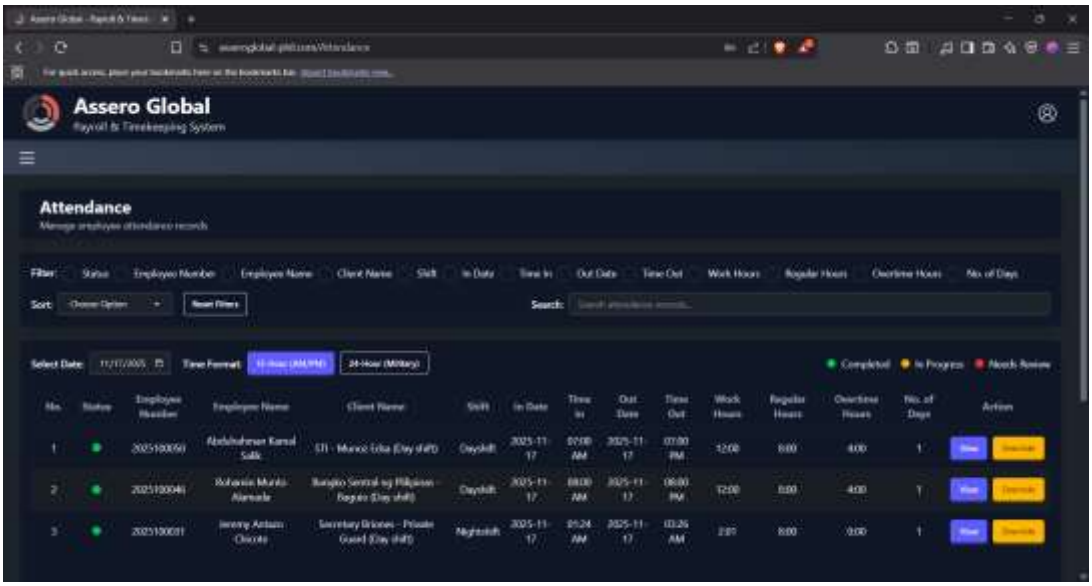


Figure 5. Attendance Module

Schedule - allows administrators to create, assign, and manage employee work schedules to ensure proper deployment and shift organization.

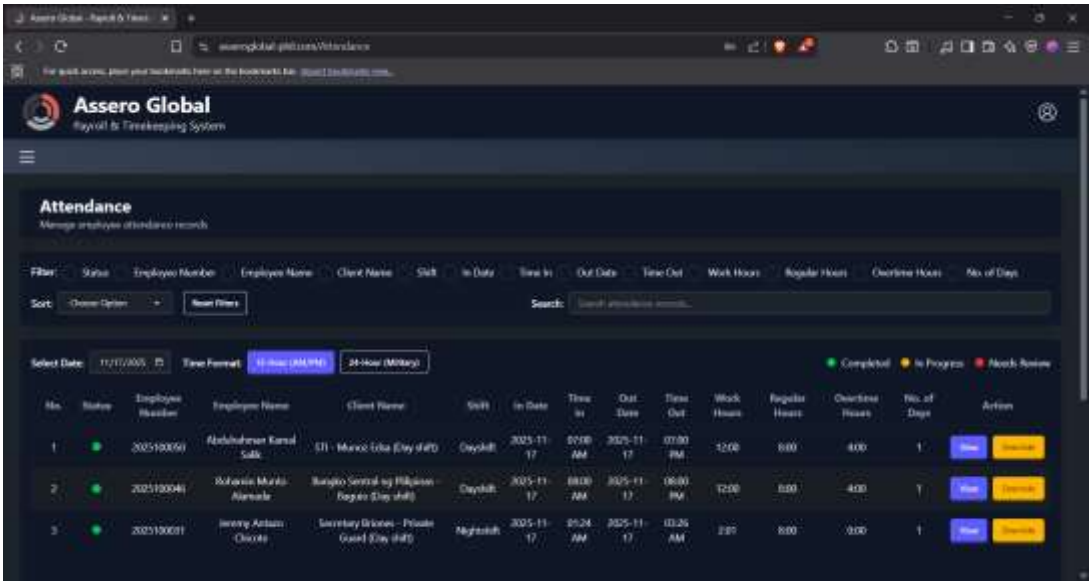


Figure 6. Schedule Module

User Management Module

Add Employee Module - Allows users to create and register a new employee profile in the system.

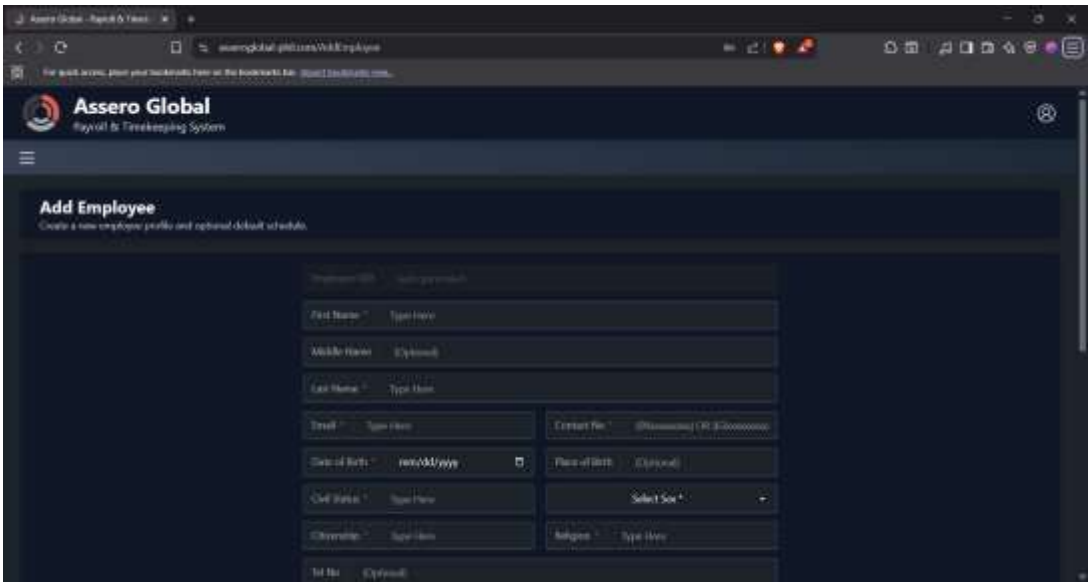


Figure 7. Add Employee Module

Manage Employee - Allows managers to add, update, monitor, and organize employee information within the system.

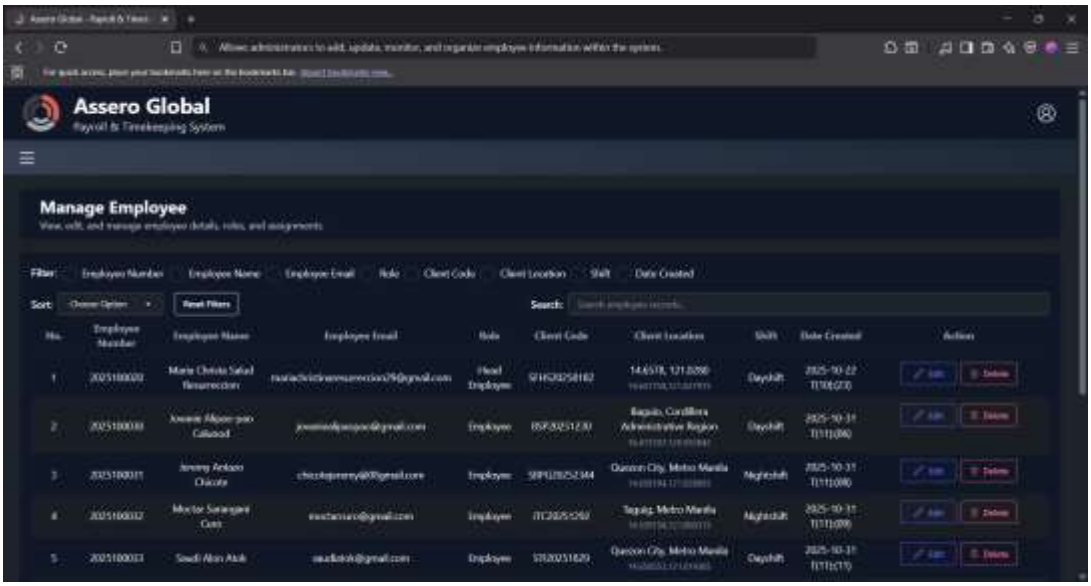


Figure 7.1. Manage Employee Module

Add Client - Enables users to input and save new client information for future transactions or services.

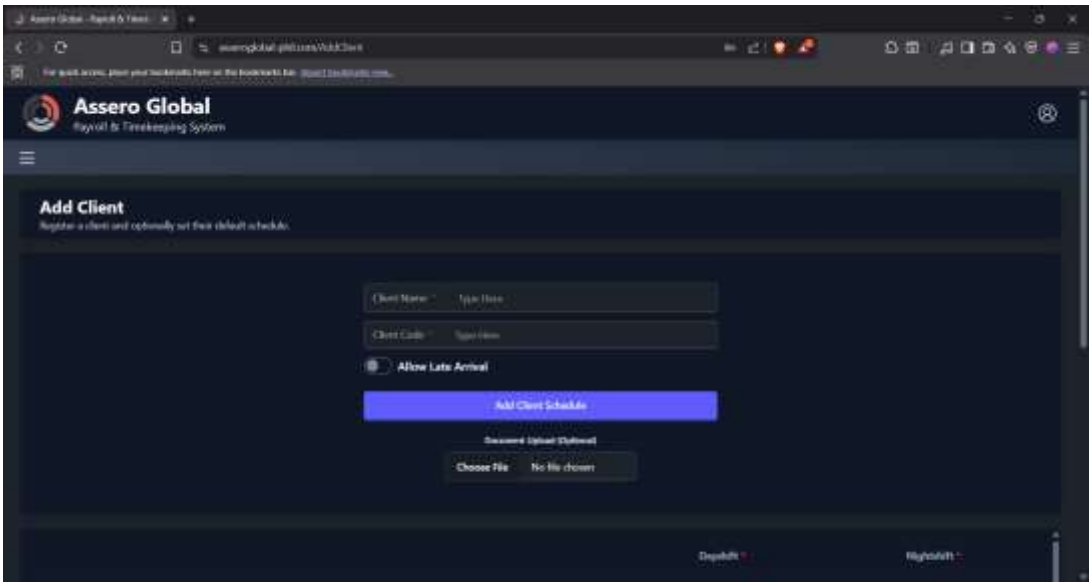


Figure 7.2. Add Client Module

Manage Client - Enables users to record, track, and maintain client details and service-related information for efficient management.

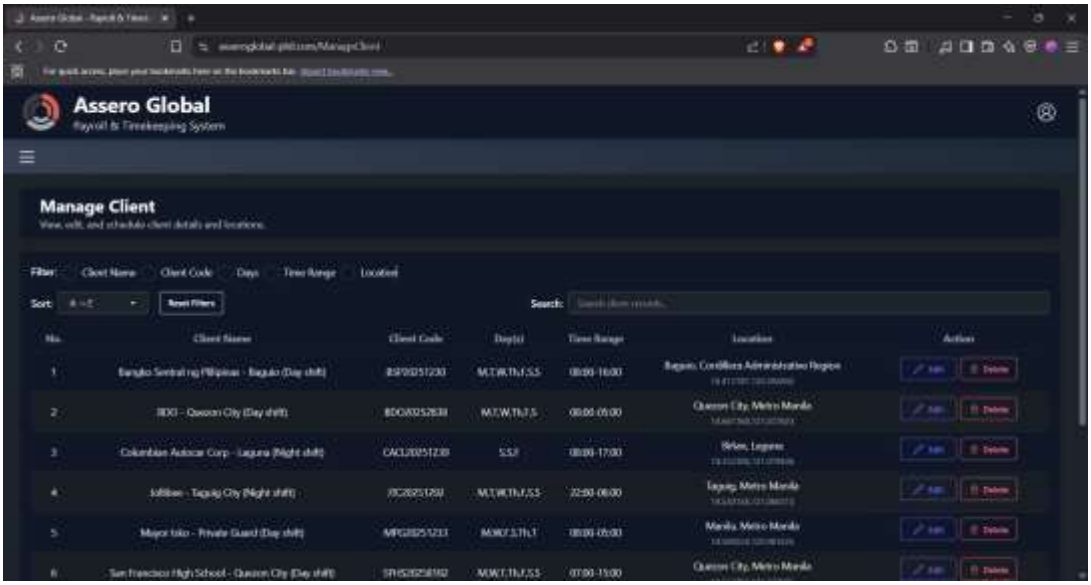


Figure 7.3 Manage Client Module

Add Manager - Lets users add and set up a new manager account along with their required details and permissions.

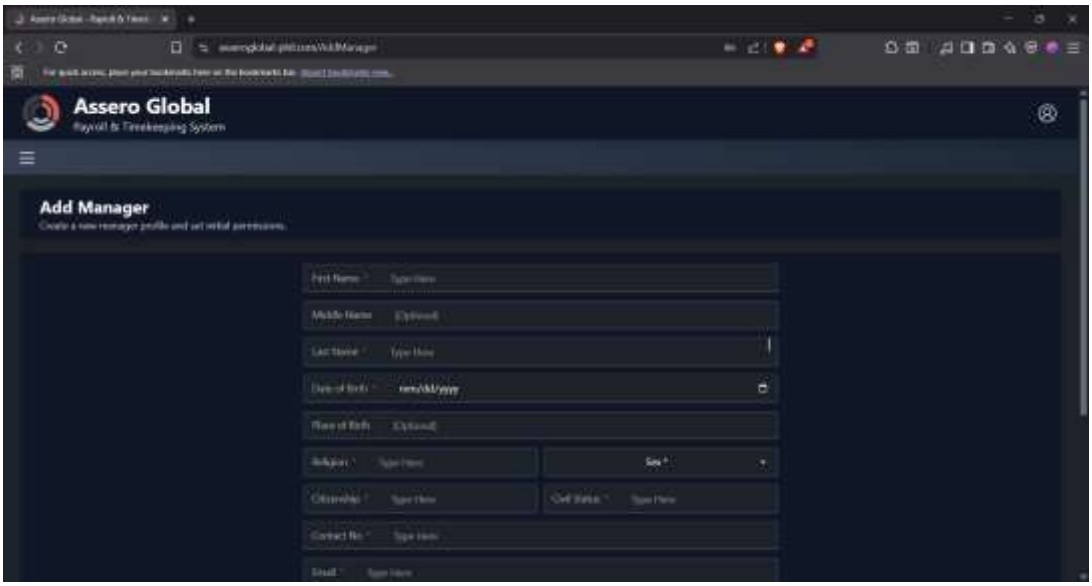


Figure 7.4 Add Manager Module

Manage Manager - Provides functionality to create, edit, and oversee manager profiles and their associated roles or permissions.

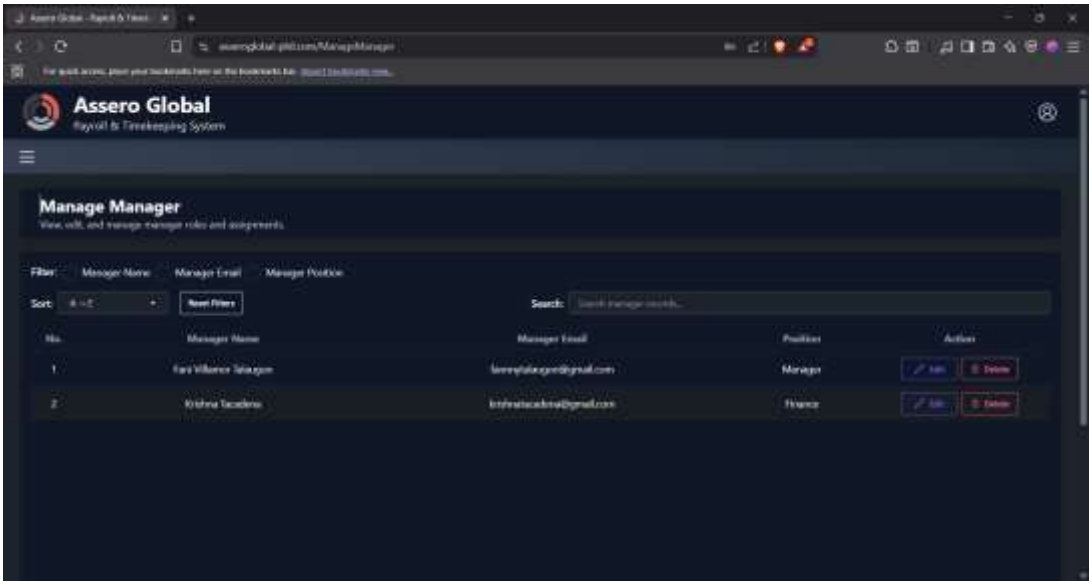


Figure 7.5. Manage Manager Module

Admin Invite - Allows the system to send an invitation email so a new admin can securely create their account, set their credentials, and be added to the platform.

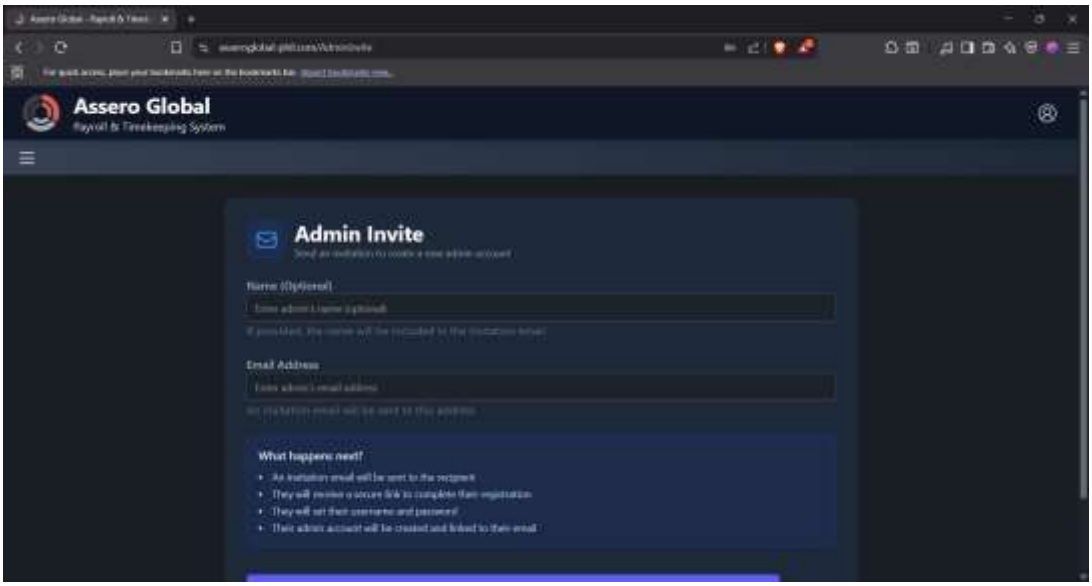


Figure 7.6. Admin Invite Module

Settings

User Permission - This module allows administrators to configure and manage role-based permissions for each finance-related feature in the system.

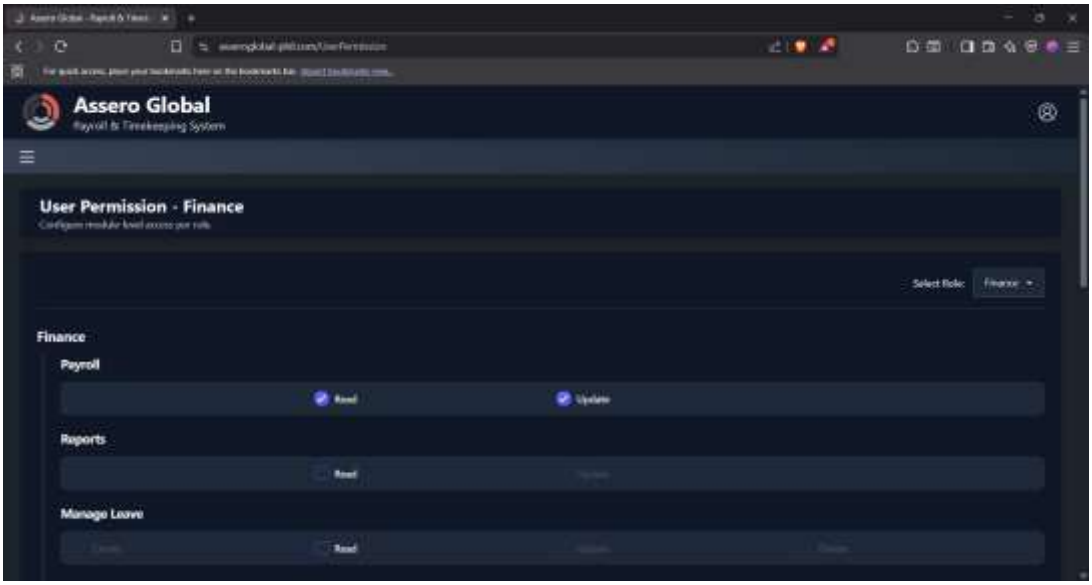


Figure 8. User Permission Module

Adjust Deductions - This module allows administrators to configure and update the system's SSS, PhilHealth, and Pag-IBIG deduction parameters used in automated payroll calculations.

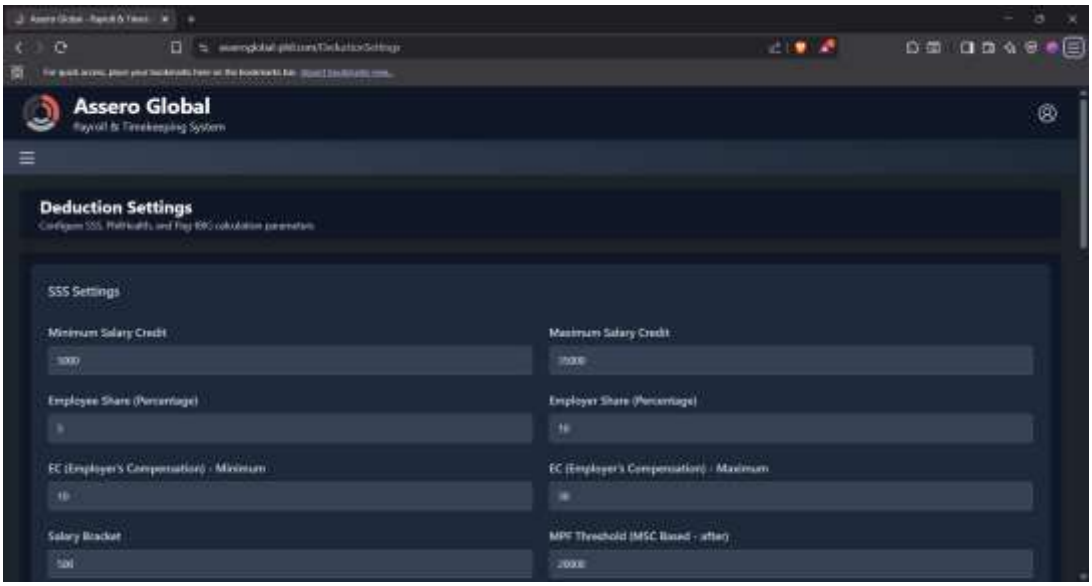


Figure 8.1. Deduction Adjustment Module.

Activity Log - records and tracks all user actions and system activities to ensure transparency, accountability, and security within the application.

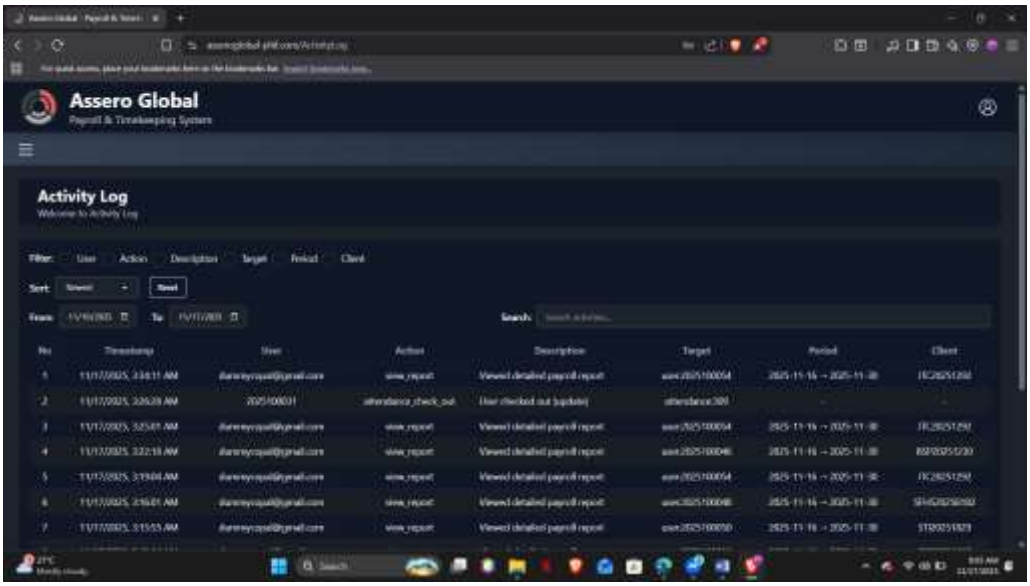


Figure 8.2. Activity Log

Backups - securely stores and preserves system data copies to ensure easy recovery and protection against data loss or system failure.

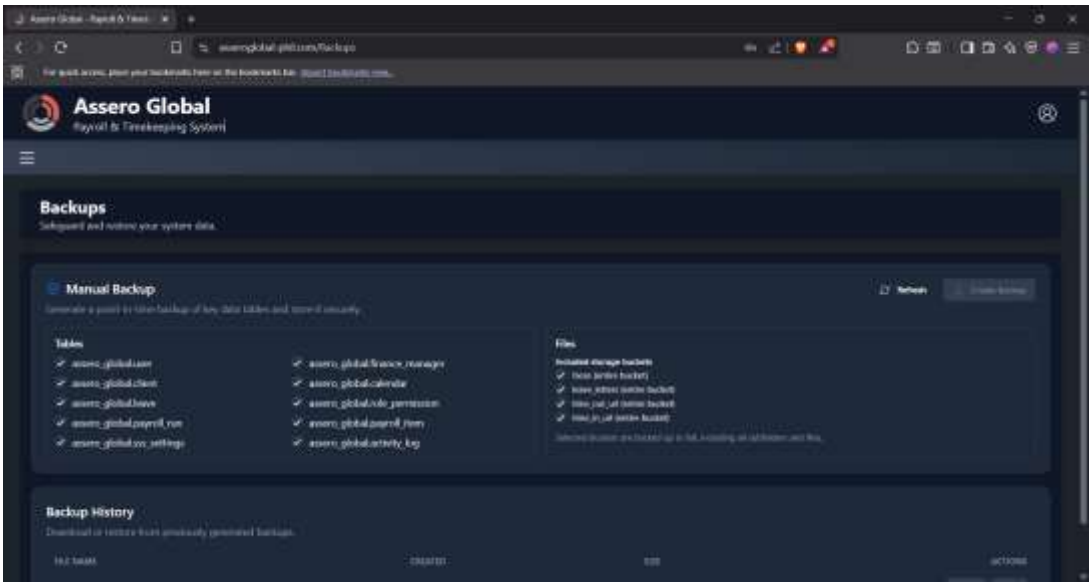


Figure 8.3. Backups

Contacts - This module allows administrators to edit and update the company's official contact details displayed on the system's public landing page.

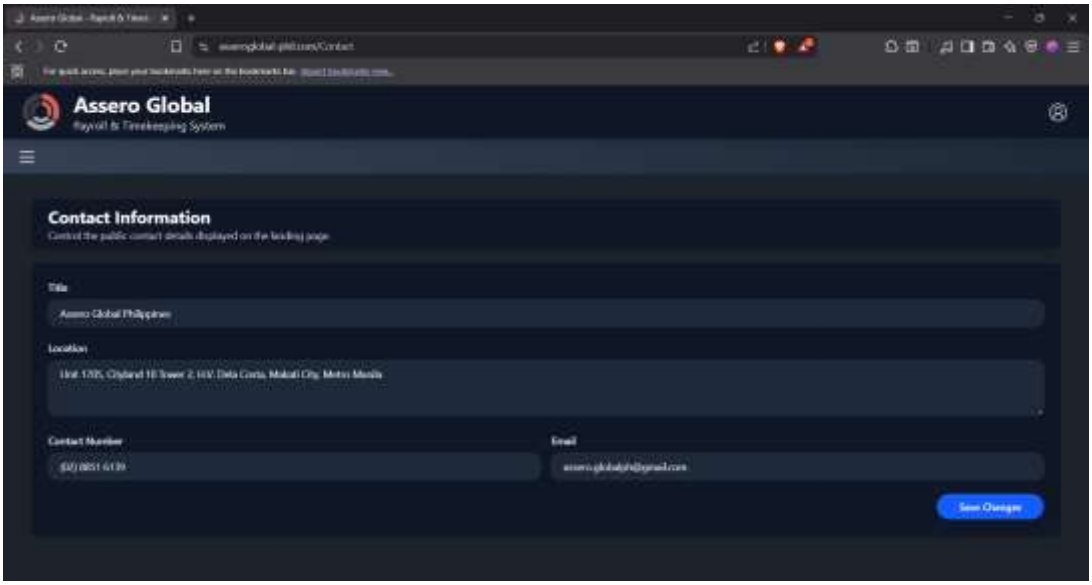


Figure 8.4. Contacts

MOBILE APPLICATION

Login Page - allows employees to securely access their accounts by entering their registered credentials before using the system's features.

Figure 9. Mobile App: Login Page

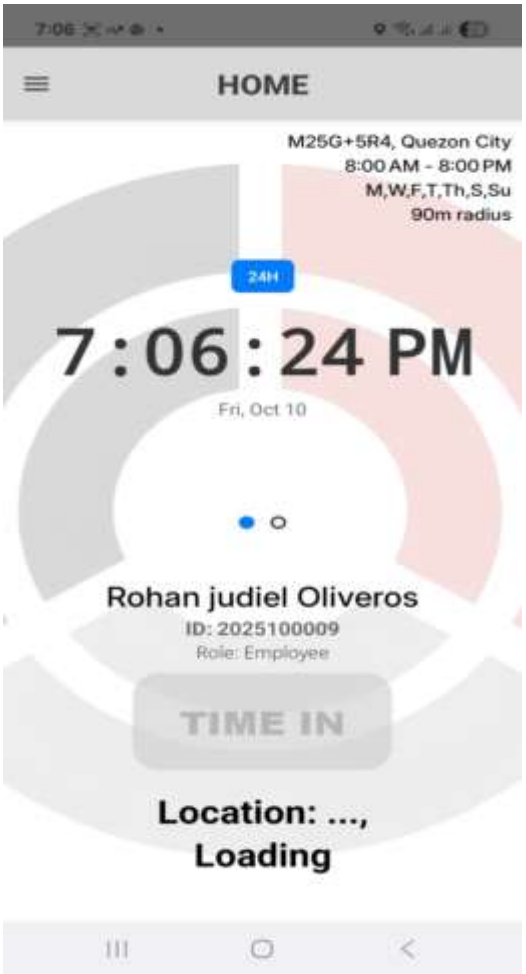


Figure 10. Mobile App: Home Page

Home Page - serves as the main interface where users can easily navigate to different features such as attendance, payslip, and profile

Reports - displays the employee's time tracking data, including total entries, time-ins, time-outs, and total work hours, allowing users to monitor their attendance records over specific periods.

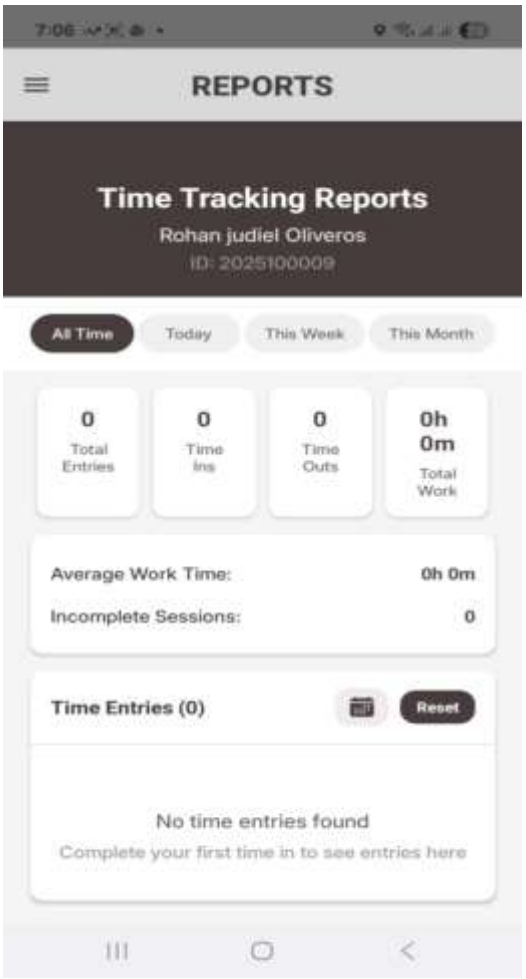
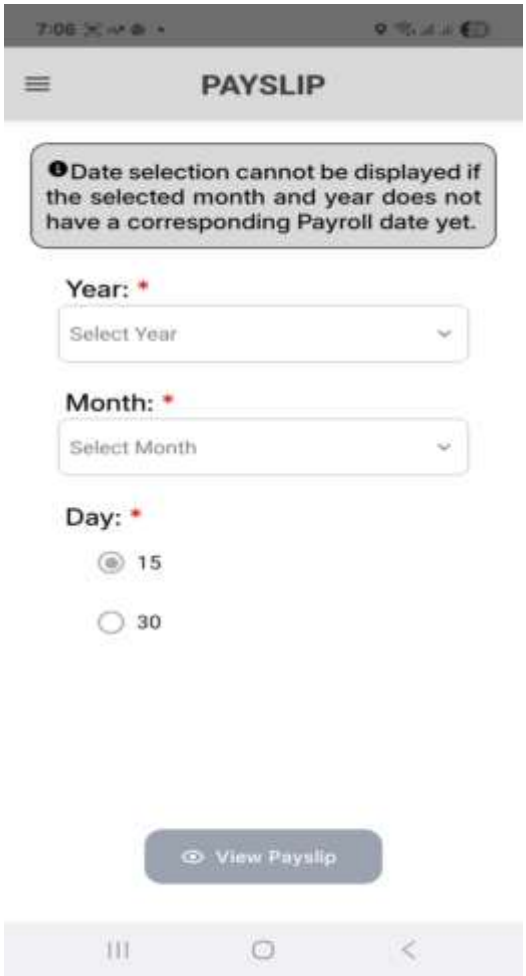


Figure 11. Mobile App : Reports



Payslip - allows employees to view their payroll details by selecting the corresponding year, month, and cutoff date to access their generated payslip.

Figure 12. Mobile App: Payslip

Settings - allows users to manage their account preferences, update personal information, and access application options for a personalized experience.

Figure 13. Mobile App: Settings

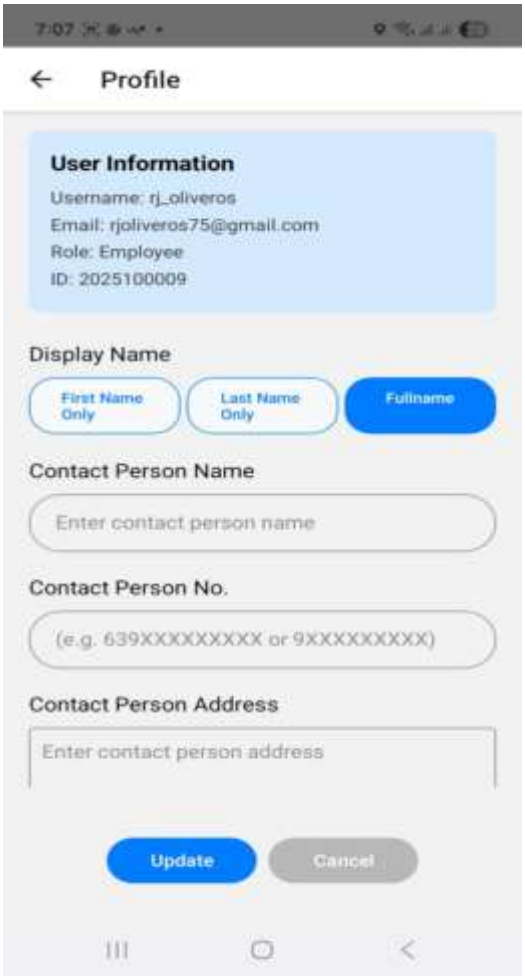


Figure 14. Mobile App: Profile

Profile - displays and manages the employee's personal information, account details, and work-related data within the mobile application.

Facial Recognition - allows employees to record their attendance by scanning their faces, ensuring accurate and secure identity verification before logging time-in and time-out.

Figure 9.6. Facial Recognition

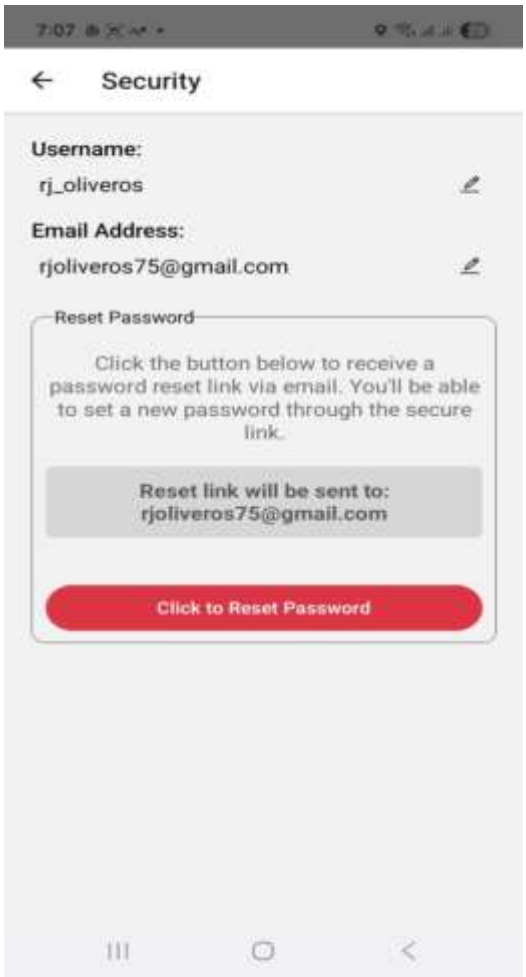


Figure 9.7. Security

Security - allows users to update their usernames and reset their passwords, ensuring account safety and secure system access.

Members - allows head employees to track the attendance of employees in their jurisdictions.

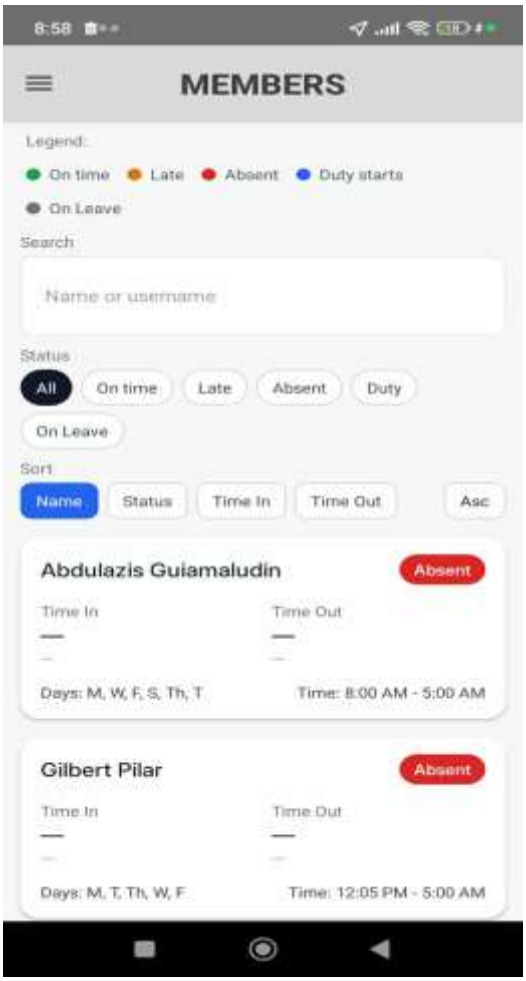


Figure 9.8. Members

Calendar - allows employees to view the incoming or on-going Events and Holidays.

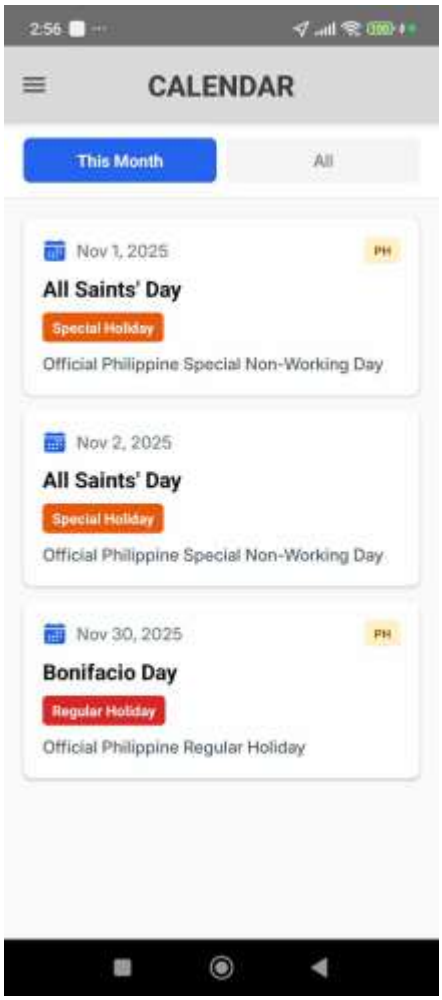


Figure 9.9. Calendar

Design of Software, System, Product, and/or Processes

Figure 10. Context Flow Diagram of a Web - Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

Figure 10 illustrates the flow of data between the web-based system, the mobile application, and its users, namely the Administrator, Manager, Finance, and Employees. The Administrator manages employee, manager, and client accounts, updates schedules, monitors attendance and leave, and oversees automated payroll generation and reports. The Manager authorizes and manages leave requests, employee and client information, and calendar activities, while the Finance department handles payroll auditing, employee and client lists, and rate management. Through the mobile application, employees can log in, record attendance using facial recognition and GPS tracking, and view their payslips and reports. All data transmitted from the mobile app is synchronized with the web system, ensuring accurate timekeeping, payroll processing, and centralized monitoring for Assero Global Inc.

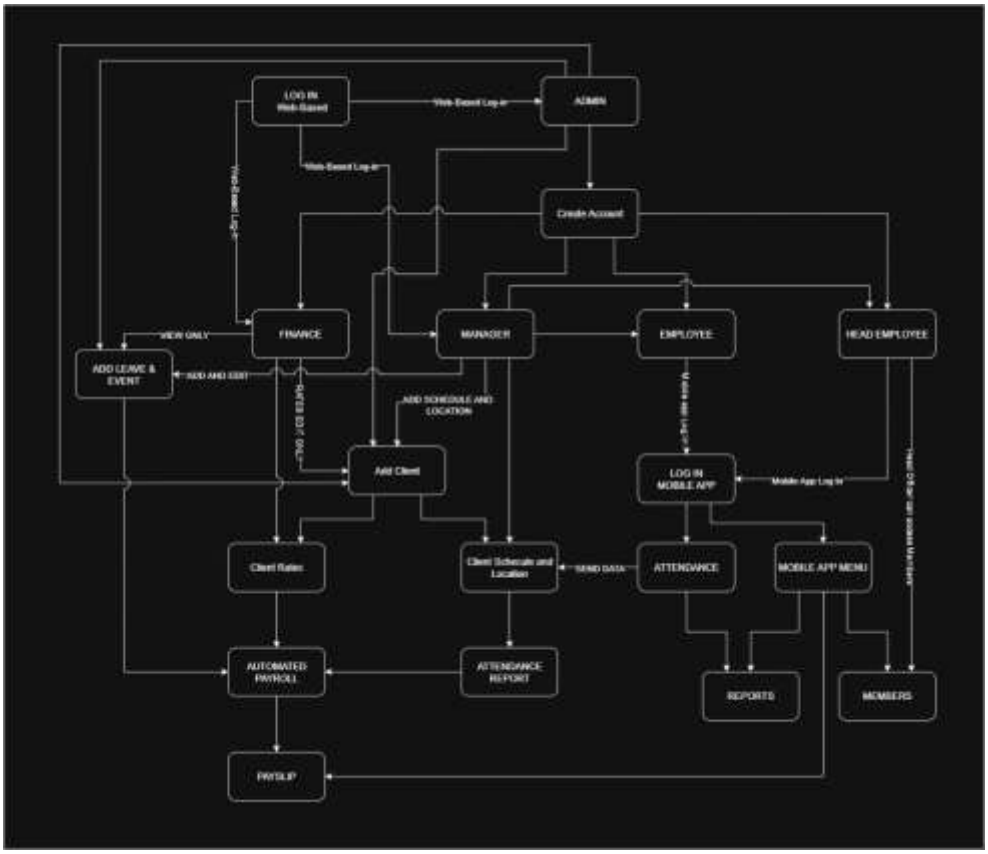


Figure 11. Data Flow Diagram of a Web - Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

Figure 11 illustrates the flow of data between users and processes within the system, showing how information moves from input to output across both the web and mobile applications. The Administrator initiates account creation and oversees data management, including employee, client, and schedule records. The Manager adds schedules, assigns client locations, and monitors employee activities, while the Finance Department manages client rates, audits payroll, and monitors leave and event records. The Head Employee and Employee access the system primarily through the Mobile Application, where they log in, record attendance, and send data such as time, date, and location. These inputs are processed to generate attendance reports, automated payroll, and payslips. The diagram demonstrates the interconnected flow of data among all modules—ensuring synchronized operations between web-based management and mobile attendance tracking for efficient monitoring and payroll processing.

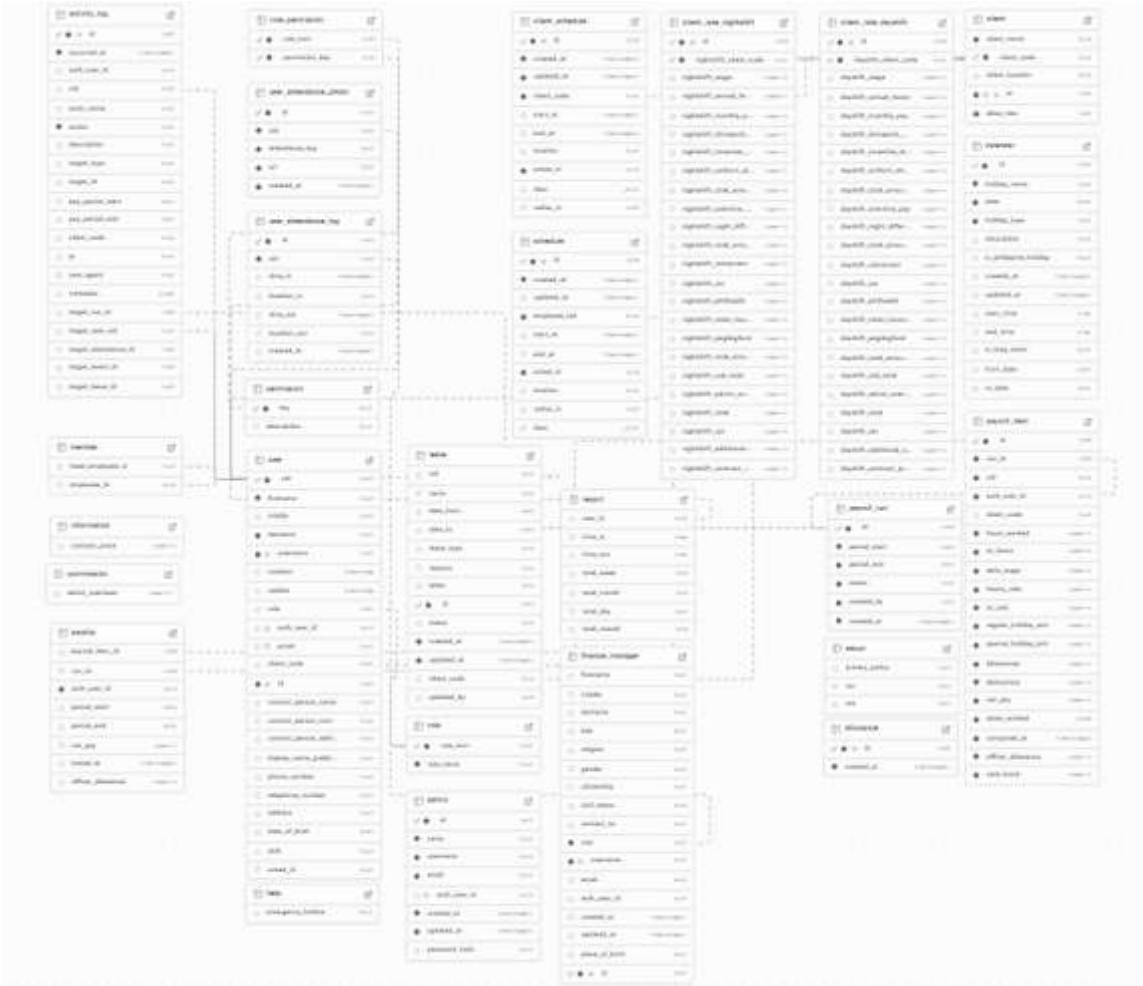


Figure 12 Entity Relationship of a Web - Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

Figure 12 presents the Entity Relationship Diagram (ERD) of the system, illustrating the relationships between the main entities such as employees, clients, managers, attendance, payroll, and schedules. It shows how data is organized, linked, and stored within the database to support accurate timekeeping, payroll computation, and user management. The ERD ensures that each entity is properly connected, allow

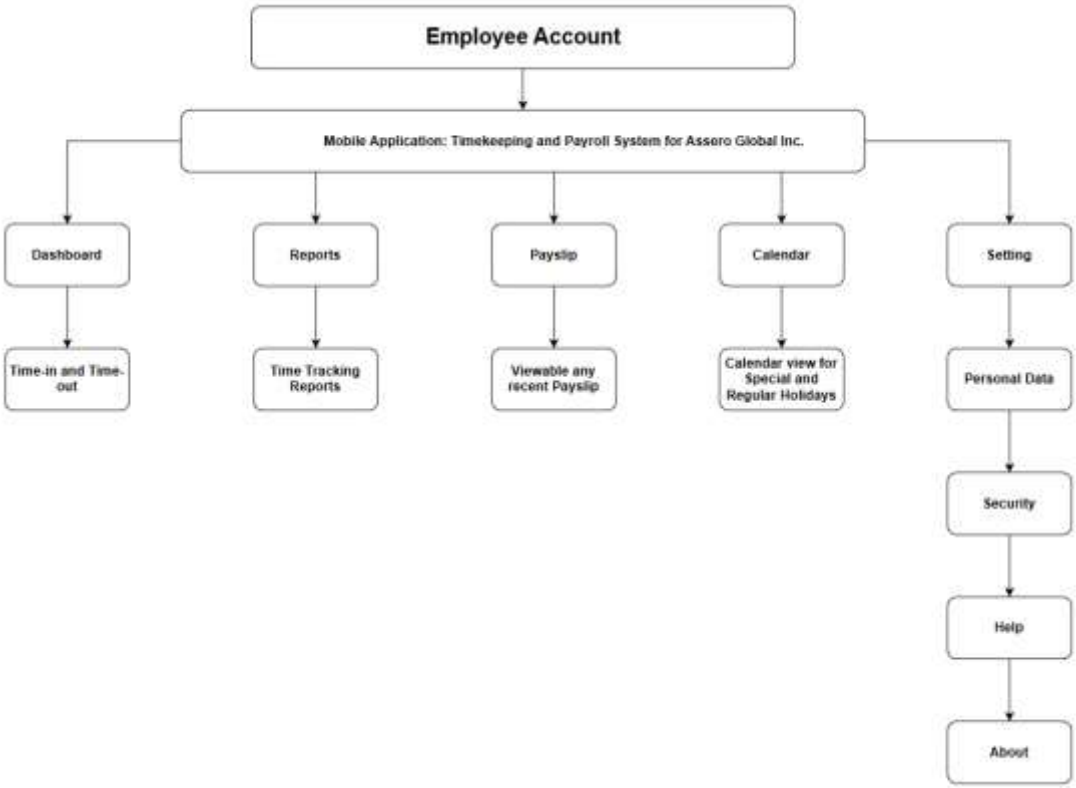


Figure 13 HIPO Chart of Employee/Security Guard Account of a Web - Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

Figure 13 presents the Employee Account structure of the mobile application, outlining the main features accessible to security guards. Through the Dashboard, employees can perform time-in and time-out functions using facial recognition and GPS tracking. The Reports section provides access to detailed time tracking reports that summarize attendance records. The Payslip feature allows employees to view their most recent payslips generated by the system. Meanwhile, the Settings section enables users to manage personal data, adjust security preferences, and access help and about pages. This module ensures employees can conveniently perform attendance logging, view records, and manage their account information through the mobile application.

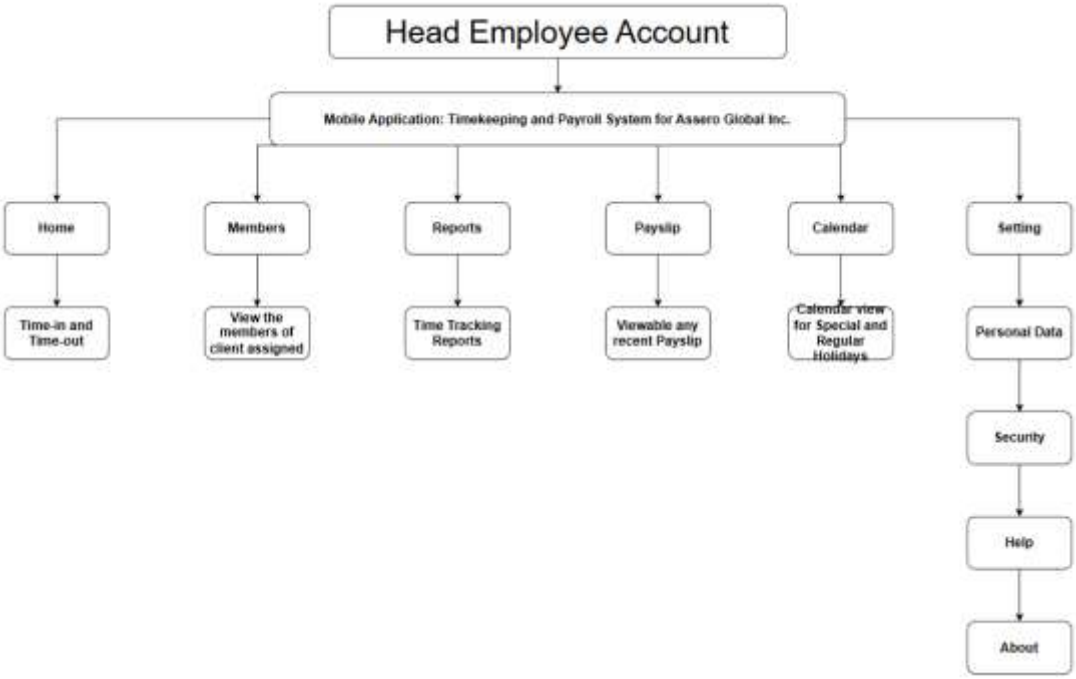


Figure 14 HIPO Chart of Head- Employee Account of a Web - Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

Figure 14 illustrates the Head Employee Account of the mobile application, showing the functions available to users with supervisory roles. The Dashboard enables head employees to perform time-in and time-out actions using facial recognition and GPS. The Members section allows them to view the list of employees assigned under their designated client or location. The Reports feature provides access to time tracking records for monitoring attendance performance, while the Payslip section lets them view their most recent payslip. Additionally, the Settings section allows management of personal data, security preferences, and access to help and pages. This module gives head employees both attendance capabilities and limited oversight over team members assigned to their post.

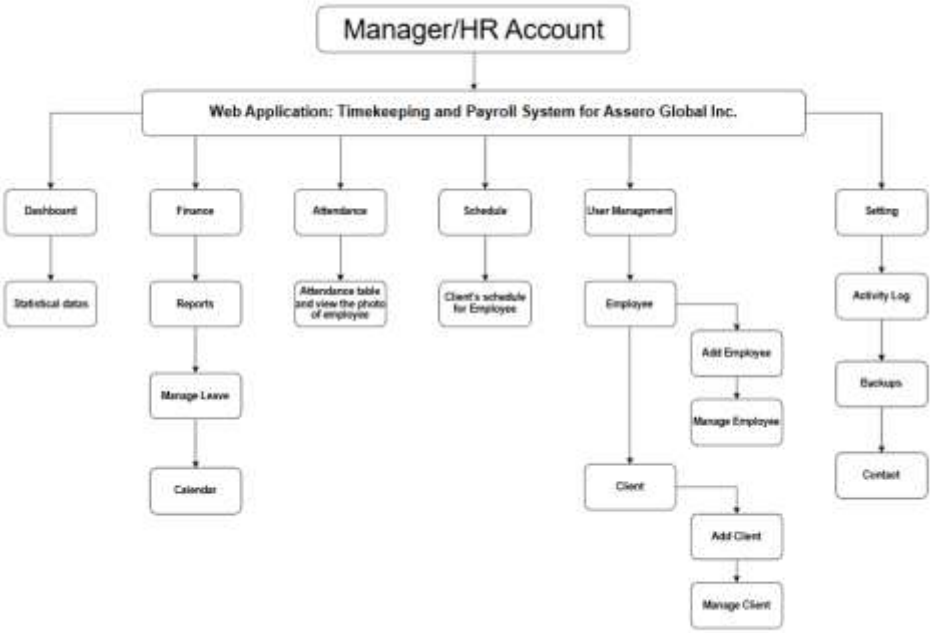


Figure 15 HIPO Chart of Manager Account of a Web - Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

Figure 15 illustrates the Manager Account of the web-based system, which allows managers to oversee and manage key operations within the organization. The Dashboard displays statistical data that provides an overview of system activities. The Finance section includes leave management and calendar functions to monitor employee schedules and events. The Attendance module enables managers to view employee attendance tables and verification photos for monitoring purposes. The User Management section allows the addition and management of employees and clients, ensuring accurate record of maintenance. Meanwhile, the Settings section gives access to the activity log, reports, system backups, and contact information. This module provides managers with the tools to supervise employee performance, maintain organized records, and ensure efficient coordination within the system.

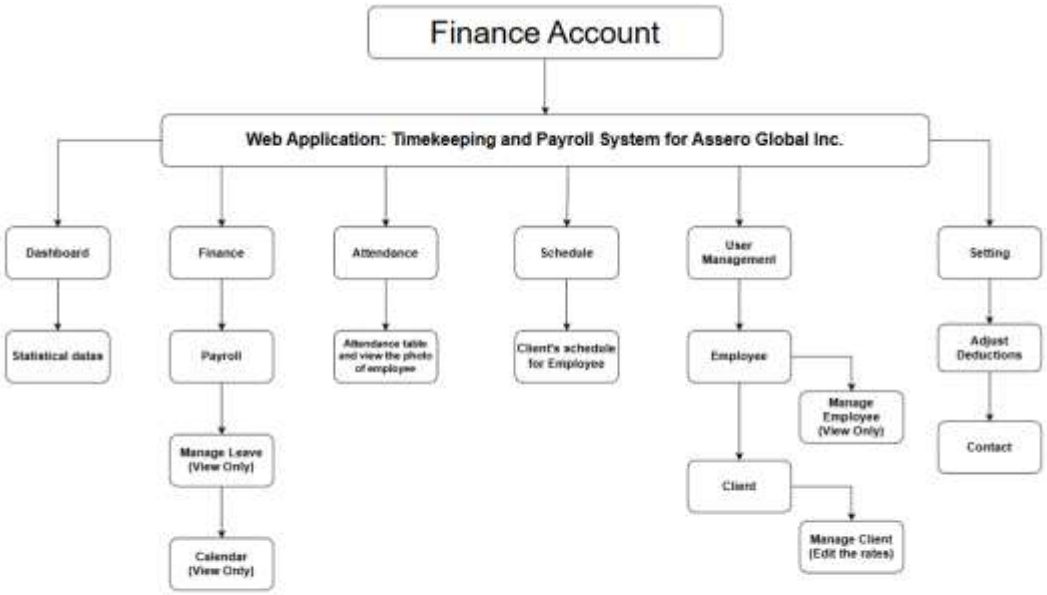


Figure 16 HIPO Chart of Finance Account of a Web - Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

Figure 16 illustrates the Finance Account of the web-based system, which focuses on managing payroll-related tasks and financial records. The Dashboard provides statistical data relevant to financial monitoring. The Finance section handles payroll processes and allows the viewing of leave and calendar details. The Attendance module enables the finance department to review employee attendance tables and verification photos for payroll validation. Through the User Management module, finance users can view employee records and edit client rates when necessary. Lastly, the Settings section allows users to access contact information for support and administrative communication. This module ensures accurate payroll computation, transparent record management, and streamlined coordination within the financial operations of Assero Global Inc.

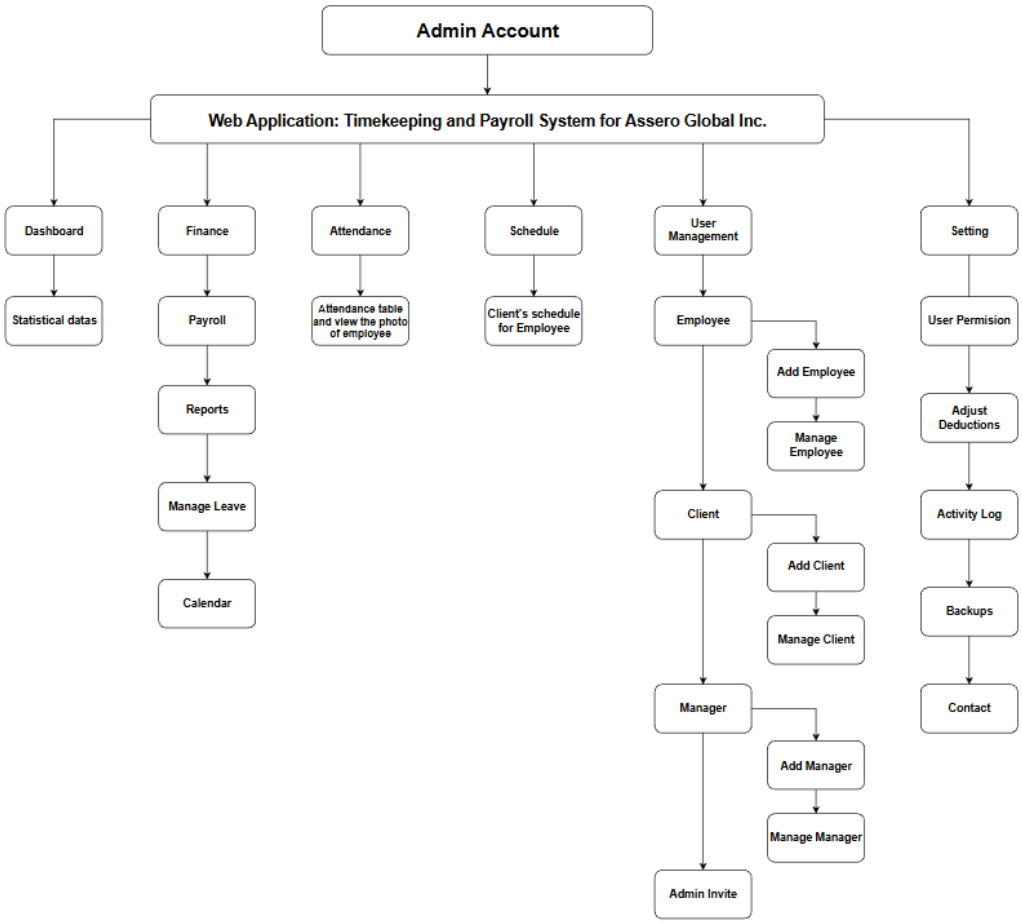


Figure 17 HIPO Chart of Admin Account of a Web - Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

Figure 17 presents the Admin Account, which serves as the core management component of the entire system. The Dashboard displays system-wide statistical data, while the Finance section oversees payroll processing, leave management, and calendar scheduling. The Attendance module enables administrators to monitor employee attendance and verification of photos. The User Management module allows full control over employee, client, and manager accounts—including adding, updating, and managing their records. Meanwhile, the Settings section provides access to the activity log, reports, backups, and contact information, ensuring system integrity and maintenance. Overall, this module allows the administrator to manage all user roles, maintain system data, and oversee operational workflows efficiently.

Payroll Computation Formulas

PROVISION OF SECURITY GUARDS - METRO MANILA (DOLE RATE)

		12 HOURS SHIFT			
		Dayshift		Nightshift	
Daily Wage (W)		PHP	645.00	PHP	645.00
No. of Work Days (Annual Factor-AF)		PHP	394.40	PHP	394.40
	Monthly Pay ((W x AF/12)	PHP	21,199.00	PHP	21,199.00
	13th Month Pay (W x 365/12/12)	PHP	1,634.90	PHP	1,634.90
	5 Days Incentive Leave (W x 5/12)	PHP	268.75	PHP	268.75
	Uniform Allowance	PHP	100.00	PHP	100.00
TOTAL AMOUNT BEFORE OVERTIME		PHP	23,202.65	PHP	23,202.65
	Overtime Pay	PHP	13,496.09	PHP	13,496.09
	Night Differential (10% of Monthly Pay)	PHP	-	PHP	2,119.90
A- TOTAL AMOUNT DIRECTLY TO GUARD		PHP	36,698.74	PHP	38,818.64
	Retirement Pay (W x 22.5 /12)	PHP	1,209.38	PHP	1,209.38
	SSS (Employer's Share)	PHP	2,072.50	PHP	2,072.50
	Philhealth	PHP	529.98	PHP	529.98
	State Insurance/ECC (Employer's Share)	PHP	30.00	PHP	30.00
	Pag-ibig Fund	PHP	200.00	PHP	200.00
B. Total Amount Directly to Government		PHP	4,041.85	PHP	4,041.85
	SUB-TOTAL (A+B)	PHP	40,740.59	PHP	42,860.49
	Admin Overhead (DOLE D.O. 150-16)	PHP	8,148.12	PHP	8,572.10
TOTAL		PHP	48,888.70	PHP	51,432.58
	VAT (10% of the Admin Fee)	PHP	814.81	PHP	857.21
	Add: Additional Allowance	PHP	-	PHP	-
CONTRACT PRICE PER GUARD PER MONTH inclusive of VAT		**PHP**	**49,703.51**	**PHP**	**52,289.79**

TOTAL CONTRACT PRICE PER GUARD PER MONTH (DAYSHIFT)	**PHP**	**49,703.51**
TOTAL CONTRACT PRICE PER GUARD PER MONTH (NIGHTSHIFT)	**PHP**	**52,289.79**

Table 1. Sample Provisions of Security Guards

Daily Wage (W) – The basic daily rate set by DOLE for security guards, which serves as the basis for all computations.

No. of Workdays (Annual Factor - AF) – Represents the total working days in a year used to compute the equivalent monthly rate.

Monthly Pay – Converts the daily wage into its monthly equivalent based on the annual factor divided by 12 months.

$$Monthly\ Pay = \frac{W\ \times AF}{12}$$

13th Month Pay - The prorated amount of the 13th-month benefit spreads over each month.

$$13th\ Month\ Pay = \frac{W\ \times 365}{12^2}$$

5 Days Incentive Leave - The equivalent monthly value of the 5-day service incentive leave benefit.

$$5\ Days\ Incentive\ Leave = \frac{W\ \times 5}{12}$$

Uniform Allowance - A fixed monthly amount provided for uniform expenses.

Overtime Pay - Additional compensation for hours worked beyond the regular 12-hour shift, computed based on the hourly rate multiplier.

Night Differential - Additional pay for night shift employees, equal to 10% of the basic monthly rate.

$$Night\ Differential = 0.10\ \times Monthly\ Pay$$

Total Amount Directly to Guard (A) - The total compensation the guard receives, including all pay and benefits before deductions.

Retirement Pay - The employer's share in the employee's retirement fund contribution, computed monthly.

$$Retirement\ Pay = \frac{W \times 2.25}{12}$$

SSS (Employer's Share) - The employer's monthly contribution to the Social Security System.

PhilHealth - The employer's monthly contribution to the employee's PhilHealth coverage.

State Insurance/ECC (Employer's Share) - Employer's contribution to the Pag-IBIG housing fund.

Total Amount Directly to Government (B) - The total amount employer remits to government agencies for employee benefits.

Admin Overhead - Administrative cost percentage allowed by DOLE to cover management and operational expenses.

$$Admin\ Overhead = (A + B) \times 0.20$$

VAT (Value Added Tax) - The value-added tax applied to the administrative fee portion.

$$VAT = Administrative \times 0.10$$

Contract Price per Guard per Month (Inclusive of VAT) - The final contract rate charged per guard, including VAT, represents the total cost to the client.

Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc.

This survey is part of the process of finishing the IT Capstone Project. The people who will be provided with this survey will be needed to evaluate the proponent's system in making this project a success. Check (/) the box of your choice based on your satisfaction whether it is 5 (Excellent), 4 (Good), 3 (Average), 2 (Fair), or 1 (Poor).

Functionality	1	2	3	4	5
The system allos smooth management of employee information.					
The system accurately displays and updates employee records					
The payroll module accurately computes employee salaries and deductions.					
The payroll process operates smoothly without error.					
The system allows proper management of leaves under the Finance module.					
The calendar page under the Finance module displays schedules accurately.					
The system can generate payroll reports accurately.					
The system can generate attendance reports without issues.					
The facial recognition feature works accurately during attendance.					
The GPS tracking correctly detects the user's assigned location.					
Total Rate:					
Reliability	1	2	3	4	5
The employee data is stored and retrieved without errors.					
The system performs consistently without crashing or data loss.					
The computed payroll results are always consistent and accurate.					
Data in the payroll module remains saved and accessible at all times.					
The Finance module performs tasks without errors or crashes.					
Data recorded in Manage Leave and Calendar pages are accurate and reliable.					
Total Rate:					
Usability					
The interface design of the employee management module is clear and easy to use.					
The system provides user-friendly navigation and understandable labels.					
The payroll page layout is simple and organized.					
Instructions and buttons are clearly labeled and easy to use.					
The report layout is easy to read and well-organized.					
Exporting or printing reports is simple and clear to perform.					
The layout and design of the payslip are readable and organized.					
The navigation to view payslips is simple and user-friendly.					
The attendance page is simple and easy to understand.					
Icons and feedback messages are clear and helpful to users.					
Total Rate:					
Efficiency					
The system loads data and updates information quickly.					
The system reduces the time needed to manage employee records.					
The system computes payroll quickly and without delay.					
Generating payroll reports takes minimal time.					
Total Rate:					
Maintainability					
Employee information errors are easy to correct.					
The system is adaptable to future updates or modifications.					
Payroll errors can be corrected easily.					
The module can easily adjust to updated payroll policies.					
Total Rate:					

Table 2. Survey

Description of the Prototype

Functionality

The prototype of the Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc. was made to help the company manage attendance and payroll automatically. It includes different modules and submodules that work together to make sure the system performs its tasks correctly. The web application is for the admin, manager, and finance staff, while the mobile app is for employees and head employees to log attendance, track their location, and view payslips. The system's performance was checked based on suitability, accuracy, clarity, compliance, and security to ensure it works properly and meets its purpose.

Reliability

Reliability refers to how well the system works without errors and how consistent it is during use. The prototype was tested to make sure it runs smoothly, even if there are small issues or incorrect inputs. The developers allowed users to try the system to see if it can provide correct and steady results. This helps check if the system is stable and ready to be used in real situations without causing problems.

Usability

Usability focuses on how easy it is for users to understand and use the system. Both the web and mobile applications are designed to be simple and easy to navigate. Respondents tested the system to see how quickly they could learn to use it and perform tasks. The system aims to reduce user mistakes and help them recover easily when errors occur. It ensures that the users—employees, managers, or admins—can operate the system comfortably and efficiently.

Efficiency

Efficiency means how well the system performs its functions using only the needed number of resources like memory and time. The prototype was tested to make sure it processes data such as payroll computation and attendance records quickly and accurately. Respondents observed how fast and responsive the system was. The goal is to make sure that important tasks, like creating reports and managing employee data, are done in the shortest time possible without delays.

Maintainability

Maintainability refers to how easy it is to update, improve, or fix the system when needed. The prototype was built with separate parts so that developers can easily make changes or add new features. Respondents and developers checked how well the system can adapt to future updates, such as payroll rule changes or new functions. This ensures that the system can continue working well while keeping maintenance costs and problems low.

Development and Testing

One of the main objectives of this study is to evaluate the quality of the developed system from the point of view of its end users. The survey tool used in this study follows the standards of ISO 9126, which provides a framework for assessing software quality. This approach allows users to share their honest feedback and experiences with the system. The survey method was chosen because it effectively gathers opinions from multiple users, ensuring a fair and accurate evaluation. Respondents were assured that their answers would remain confidential to encourage open participation and honest responses.

Software testing for the Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for the Security Guards of Assero Global Inc. was conducted in accordance with ISO 9126 standards. This standard focuses on six main quality characteristics—functionality, reliability, usability, efficiency, maintainability, and portability. The testing process ensured that each module of the system performed its intended function effectively and that users were satisfied with its performance. The developers also gathered feedback from selected respondents to identify improvements and ensure that the system operates accurately and efficiently. The results of this testing phase helped confirm the dependability of the system and provided valuable insights for future enhancements.

Implementation Plan and Results

The Web-Based Timekeeping and Payroll System with Mobile Solutions was implemented to replace Assero Global Inc.'s manual timekeeping and payroll process. The implementation involved several stages: system development, testing, deployment, and user training.

The mobile application was installed on the security guards' smartphones to enable facial recognition and GPS-based attendance logging. Meanwhile, the web-based platform was deployed for Manager/HR, finance and administrative users to manage employee information, monitor attendance, compute payroll, and generate reports.

Likert Scale

A Likert Scale is a rating scale that is used to assess attitudes or views. Respondents are asked to score items based on their level of agreement using this scale.

Numeric Rating	Remarks
4.51 - 5.00	Excellent
3.51 - 4.50	Good
2.51 - 3.50	Average
1.51 - 2.50	Fair
1.0 - 1.50	Poor

Table 3. Likert Scale Rating

In this study, the Likert Scale was utilized by the developers as one of the criteria for assessing the effectiveness of the Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for Assero Global Inc. The scale served as the basis for measuring the system's functionality, reliability, usability, efficiency, and maintainability.

The performance evaluation was conducted among ten (10) IT professionals and ten (10) end-users or company staff, using a 5-point rating scale ranging from "Excellent" to "Poor." The responses gathered were analyzed to determine the system's overall performance and to identify areas for improvement. Based on the results, the developers formulated conclusions and recommendations that would contribute to enhancing the accuracy, efficiency, and overall performance of the developed system.

Results for Non-IT/Beneficiaries

Criteria	Non-IT/Beneficiaries	Remarks
Functionality	4.36	Good
Reliability	4.41	Good
Usability	4.25	Good
Efficiency	4.42	Good
Maintainability	4.35	Good
Total Average	**4.35**	**Good**

Table 4. Overall Result for Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for Assero Global Inc. of the Non-IT/Beneficiaries Respondents

Results for IT Professionals

Criteria	IT Professionals	Remarks
Functionality	4.01	Good
Reliability	3.95	Good
Usability	4.3	Good
Efficiency	4.27	Good
Maintainability	4.2	Good
Total Average	**4.14**	**Good**

Table 5. Overall Result for Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for Assero Global Inc. of the IT Professionals Respondents

Overall Results of Performance Evaluation

Criteria	IT Professional	Non-IT/Beneficiaries	Mean Rate	Remarks
Functionality	4.01	4.36	4.18	Good
Reliability	3.95	4.41	4.18	Good
Usability	4.3	4.25	4.27	Good
Efficiency	4.27	4.42	4.34	Good
Maintainability	4.2	4.35	4.27	Good
Total Average			4.24	Good

Table 6. Overall Result of Performance Evaluation for Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for Assero Global Inc.

Conclusion and Recommendation

Based on the evaluation results conducted among seven (7) IT professionals and eight (8) non- IT beneficiaries, the Web-Based Timekeeping and Payroll System with Mobile Solutions Utilizing Facial Recognition and GPS Tracking for Assero Global Inc. obtained an overall descriptive rating of "Good." The system met the objectives of automating timekeeping, simplifying payroll computation, and improving accuracy and reliability in managing employee attendance and compensation. Respondents found the system functional, user-friendly, and efficient in reducing manual errors that were present in the company's previous payroll process. The integration of facial recognition and GPS tracking enhanced the security and transparency of attendance monitoring, ensuring that only authorized personnel could record their time logs. Overall, the results indicate that the developed system effectively addressed the organization's operational needs and contributed to a more accurate and efficient payroll management process.

Although the system achieved a "Good" rating and performed satisfactorily, the developers recommend several improvements to further enhance its performance and usability. The system interface may be refined for better accessibility, particularly for mobile users, and optimization should be conducted to ensure faster data processing and report generation. Additional features such as automated tax computation, downloadable payslips, and real-time analytics may also be incorporated to increase functionality and convenience. The developers also recommend gathering continuous user feedback and conducting periodic system updates to maintain efficiency and security. Strengthening data protection measures is advised to ensure compliance with data privacy regulations. Implementing these enhancements will help sustain the effectiveness, reliability, and long-term usability of the system.

APPENDICES

APPENDIX A. REFERENCES

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APPENDIX B. RESOURCES PERSON

RESOURCE PERSONS

Assero Global Inc. Follow-up Interview with Owner, Mr. James Tarroza.

Interviewer: Good Day again Sir James, Again, I'm Lhorea Anne Oliveros from STI College-Munoz – EDSA. So we're here po today para magconfirm ng mga nagather po naming na answers from Ms. Krishna Tacadena yesterday.

Interviewee: Yes, Good morning to you and your group mates, so limited lang yung time ko today so diretso na tayo sa mga questions.

Interviewer: Thank you po, so first question po namin is regarding po sa Cut Off dates po, is it 10th and 25th po?

Interviewee: Yung sahod kasi is every 1st to 15th of the months and 16th to the end of the month. Ayung yung cut off. Yung 10th and 25th of the month, ayun naman yung tinatawag na **Salary Date.** Ayung yung araw kung kalian ire-release yung sahod, kasi halimbawa cut-off one (1) to fifth teen (15), katulad ngayon tapos na yung one (1) to fifthteen (15) bibigyan ngayon yung finance sections para ma-compute yung sahod, mahaba kasi yung proseso nu'n e. Bago kasi yung actual computation may mga submission and validation, back and forth, back and forth na process pa kasi 'yon, Duon sa baba.

So, sa amin, ginawa naming 10 days after the cutoff nandoon yung releasing, kaya yung sahod ditto is every 10th and every 25th of the month.

Interviewer: Mayroon po bang mga incidents po ng pagka-delay ng pagrelease po ng sahod? And ano po yung mga cause ng ganon pong incidents?

Interviewee: So, dito kasi sa amin, mayroon tinatawag na online payroll, which is tumatanggap sila ng sahod thru the bank. Online payroll ang tawag nila 'don. Ngayon, kung sakali man na offline yung bangko, so hindi siya ma-process, that's number 1. Number 2, sa side naman ng company, mayroon diyan tinatawag na approving authority—kumbaga maker and approver, so yung maker, siya yung nag-upload. Sabihan na lang natin may lost of connection yung approver wala siyang signal or internet then hindi siya makapag-approve then dumating sa puntong cut-off time na, alas-otso lang kasi ng gabi yun eh—so na cut off na, hindi siya nakapag-approve, so automatic kinabukasan niya ma-a-approve 'yon, that's a cause of delay. Number 3 naman is lack of funds.

Interviewer: Regarding po sa banks, yung mga personnels po ba ang personal na nag-a-apply ng mga bank account nila, paano po yung process 'non within the company?

Interviewee: Ano, nagbibigay kami ng tintatawag na endorsement, so ibig sabihin pupunta sila doon sa bangko na pinakamalapit sila, magdadala sila ng requirements and yung endorsement. Ang mangyayari is ip-process ka nila, ngayon dahil may dalang endorsement si personnel, ita-tag nila yung account mo na 'yon as payroll account, dito sa company na 'to.

Interviewer: Balikan po natin yung cause of delay ng releasing po ng salary. Ilang beses niyo po ito nae-encounter?

Interviewee: Actually sa tatlong item na 'yon. Number 1, is sa pag-offline ng banko, maybe 2-3 times a year. Second naman is within the company, madalas 'yan nangyayari na yung approving officers is hindi naco-contanct noh, it so happens na maybe twice a week so 8 times a month, so malaking disruption 'yon sa process. Yung pangatlo naman, maybe once or twice every 3 months, but I think its because of the lack of out automated system siyempre may naga-audit pa niyan manually, pero kung mayroon, mas mabilis ang payroll, mas accurate, walang sobra, walang kulang, We won't suffer from insufficiency of funds.

Interviewer: Can you talk more about the deductions of benefits po?

Interviewee: Actually tuwing sahod, may deductions na 'yon. Halimbawa ang, SSS contribution, employee share. Sabihin na lang natin ang supposed to be na bayad niya ay 500 pesos. So ang mangyayari is yung 500 is dahil isang cut off lang so 250 lang yung Employee Share niya. Ayun ay kapag na maximize niya yung kanyang 15 working days.

Interviewer: Next question po, first let us confirm po na, wala po tayong in writing na incident reports regarding the falsification of DTRs? And if you receive a report, ano po yung steps na ginagawa niyo po?

Interviewee: Pag-verbal 'yon, magbibigay agad ng instructions yung head office na imbestigahan yung pangyayari. So yung operations ang magi-investigate. Pero kung kinakailangan gawan ng formal report, ang operations din ang gagawa no'n. So eventually after the report maari siyang masuspinde or ma-terminate.

Interviewer: How many times have you received verbal reports regarding this matter?

Interviewee: Actually there's a thing called "Employee, Employer trust" and its important because that's what builds the company. Integrity in the workplace is important but there are other circumstances where we received this kind of reports. We haven't solve this matter yet, monthly mayroon kaming natatanggap na reports regarding diyan. May be twice or thrice a month, and nakakatakot ito kasi what if gayahin sila ng ibang empleyado, hindi natin ito maiiwasan. But there are of course kabayaran or pataw na parusa kung sakaling mapatunayan ang mga ganito.

This goes against the Section 73 of of our Administrative Policy under Fraudulent Acts. Which is the penalty on the 1st Offense is already Dismissal, but since there's still a lack of employees. Hindi ito nasusunod.

Interviewer: Thank you po for the explanation, next question po naming is regarding po sa types of employee. So naitanong po kasi sa amin ng mga panelist yung different types of employees which are the full time, part time and contractual. As per Ma'am Krishna po, all you have is project-based security personnel?

Interviewee: Let me explain it the other way, Basically, we have two types of guards. Yung mga wala pang 6-months is under probation pag lumagpas na sila ng 6 months automatic regular na sila. So, ngayon yung dalawa na 'yong hindi mo siya pwedeng sabihin contractual or project-based, kasi yung mga security guards na 'yon ay nakadeploy sila sa iba't ibang lugar, kaya lang halimbawa may isang project and it so happen na natapos na, so lilipat nanaman sila ng designation. So basically, may dalawang klase lang ng guard noh, yung mga under probation and regular.

Interviewer: Regarding naman po sa regularization, whether they are under probation and regular employee, they do have the same wage or salary po?

Interviewee: Yes, so lahat naman dito is minimum, mayroon man, sila yung mga may key positions, sila yung mayroon mg aallowances. But the basic salary ng lahat, whether you are under probation or a regular employee it's the same. Ang kinaibahan lang ng regular at under probation kapag nag karoon ng project ang mga priority na makakapasok sa project ang mga regular bale ang mga under probation ang pinapasok kapag kulang ang security personel sa client.

Interviewer: Ano-ano naman yung sakop ng mga allowances?

Interviewee: Mayroon kaming binibigay na officer allowance, ibig sabihin ito yung mga supervisory yung position, it ranges from 1000 pesos to 5000 pesos. Depende duon sa dami ng hawak nilang personnel. Kasi minsan, mayroon lang lima and sampu. Mayroon naman umaabot ng 100 up to 150. Depende sa dami ng hawak na personnel.

Nagibibigay din tayo load allowance, for phone, Fuel allowance, minsan nagbibigay din tayo ng accommodation allowances. Pero yung accommodation hindi napupunta sa mga guard, for example malayo ang designation nila. It is written na free accommodation sila doon, Yung accommodation ang agency nagbabayad so lumalabas yon as accommodation allowance.

Interviewer: Regarding naman po sa contract between client and the agency. Nabanggit po ninyo kasi sa amin last time na may clients po tayo na big companies and sometimes individuals.

Interviewee: Sa amin kasi, may tinatawag tayo na sectors; BPO, Constructions, Schools, Leisure and Hospitality, High Rise/Real Property, Retails, Residential, Logistics, Logistics, Oil and Gas, Mining Sector, Finance Sector, etc… sa amin naman is wala naman kasming client sa lahat ng sector pero mayroon kaming client na one guard only ang naka duty mayroon naming kaming mga clients na up to 100 ang naka duty.

Interviewer: How many would you say is the estimated clients of Assero Global Inc.?

Interviewee: This month na sa 50 plus, kasama na doon yung mga clients na may 100 na security personnel kasama na din don yung small companies and individuals.

Interviewer: Thank you po, next question po regarding sa salary increase, we encounter po kasi a question na what if yung personnel is sobrang tagal niya na sa loob ng company?

Interviewee: Minimum wage lahat e, sumasabay lang tayo sa tinatawag na DOLE wage order, 'di ba naririnig niyo, "Tumaas na yung minimum wage sa Manila…" meaning once lang na nag increase ng minimum wage si DOLE doon lang sila magkakaroon ng increase sa sahod, pero pag matagal na sila sa company hindi tumataas yung sahod, ang tumataas is yung allowance niya.

Interviewer: Let's go back to reports po, sa side naman po ng clients have you received po reports regarding for example po, late po yung ating personnel or absent. Ilang beses po siya nangyayari in a month regarding this incidents.

Interviewee: Yan yung common problem sa industry ng security, Its happening every 12 hours. So sa 100 katao, 50 non and may offense every 12 hours, late. Kasi nagrereport niyan is yung co-security guard nila "Yung kapalitan ko ganyan, late na. Hindi pumasok." Ngayon kung hindi sila makapasok, tatawag yung officer in charge nge reliever. Hindi nagrereport yung client ng ganyang incident ang mahalaga sa client is pag tuminging sila sa post ay mayroon tao doon.

Interviewer: Ano po yugn difference ng Cash Bond and Retirement Pay?

Interviewee: Basically, Cash Bond is ginamit for example may liability yung personnel he will pay it thru cash bond, ngayon pag hindi niya iyon nagalaw pag nag terminate na ng contract ibabalik sa kanya yung cash bond. Insurance siya. 200 per salary per cut off.

Retirement pay naman is For Example, tumagal yung personnel sa client ng 1 year, meron siyang equivalent 1 month pay na matatanggap. Yon yung tinatawag na retirement pay niya, ngayon pag umabot na yung guard ng 60 years old na sa agency, so entitled na siyang makatangap ng retirement pay.

Interviewer: Is there a different approach when it comes to collecting their cash bonds whether they are terminated,fired or resigning?

Interviewee: No, as long as they have their clearance.

Interviewer: Ano naman itong process ng clearance nila?

Interviewee: For example sa post nila, if my liability kailangan nilang mabayaran muna gamit yung cash bonds, need din ng pirma ng OIC and then detachment.

Interviewer: Regarding naman po sa pag keep ng employee records? Saan po ninyo kine-keep yung mga ganitong files for example po yung 201 files? May access pa po ba kayo sa mga lumang records?

Interviewee: Actually lahat ng records nila na meron kami ay ayun nga yung tawag is 201 files. Naka folder siya lahat, it's a physical copy and lahat 'yon ay nakatago sa shelves sa office, all our security guards have a folder so estimated nasa 500 plus folders 'yon.

Interviewer: So that's all our question for today po, Thank you po, Sir.

Interviwee: Thank you din sa inyo, mabuhay kayo.

APPENDIX C. RELEVANT SOURCE CODES

RELEVANT SOURCE CODES

Mobile Application Dependencies & Libraries

```
{
  "name": "ams",
  "version": "1.0.0",
  "main": "expo-router/entry",
  "scripts": {
    "start": "expo start --dev-client",
    "android": "expo run:android",
    "ios": "expo run:ios",
    "web": "expo start --web"
  },
  "dependencies": {
    "@expo/ui": "~0.1.1-alpha.9",
    "@expo/vector-icons": "^14.1.0",
    "@react-native-async-storage/async-
storage": "2.1.2",
    "@react-native-community/netinfo":
"^11.4.1",
    "@react-navigation/drawer": "^7.3.9",
    "@shopify/react-native-skia":
"v2.0.0-next.4",
    "@supabase/supabase-js": "^2.57.4",
    "dayjs": "^1.11.18",
    "expo": "~53.0.23",
    "expo-build-properties": "~0.14.8",
    "expo-constants": "~17.1.6",
    "expo-dev-client": "~5.2.0",
    "expo-font": "~13.3.1",
    "expo-linear-gradient": "~14.1.5",
    "expo-linking": "~7.1.5",
    "expo-local-authentication":
"~16.0.5",
    "expo-localization": "~16.1.5",
    "expo-location": "~18.1.6",
    "expo-network": "~7.1.5",
    "expo-notifications": "^0.32.12",
    "expo-print": "~14.1.4",
    "expo-router": "~5.1.7",
    "expo-sharing": "~13.1.5",
    "expo-splash-screen": "~0.30.10",
    "expo-sqlite": "~15.2.14",
    "expo-status-bar": "~2.2.3",
    "expo-system-ui": "~5.0.11",
    "expo-task-manager": "~13.1.6",
    "expo-updates": "~0.28.17",
    "geolib": "^3.3.4",
    "lodash": "^4.17.21",
    "netinfo": "^0.1.3",
    "react": "19.0.0",
    "react-native": "0.79.5",
    "react-native-bouncy-checkbox":
"^4.1.2",
    "react-native-crypto-js": "^1.0.0",
    "react-native-dotenv": "^3.4.11",
    "react-native-element-dropdown":
"^2.12.4",
    "react-native-gesture-handler":
"~2.24.0",
    "react-native-paper": "^5.14.5",
    "react-native-reanimated": "~3.17.4",
    "react-native-responsive-screen":
"^1.4.2",
    "react-native-safe-area-context":
"5.4.0",
    "react-native-screens": "4.11.1",
    "react-native-svg": "15.11.2",
    "react-native-url-polyfill":
"^2.0.0",
    "react-native-vision-camera":
"^4.7.1",
    "react-native-vision-camera-face-
detector": "^1.8.8",
    "react-native-worklet": "^0.0.0",
    "react-native-worklets-core":
"^1.6.0"
  },
  "devDependencies": {
    "@babel/core": "^7.20.0",
    "@types/react": "~19.0.10",
    "baseline-browser-mapping": "^2.8.30"
  },
  "private": true,
  "expo": {
    "doctor": {
      "reactNativeDirectoryCheck": {
        "listUnknownPackages": false,
        "exclude": [
          "react-native-bouncy-checkbox",
          "react-native-element-
dropdown",
          "react-native-responsive-
screen"
        ]
      }
    }
  }
}
```

Web Application Dependencies & Libraries

```
{
  "name": "my-assero-project",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "@heroicons/react": "^2.2.0",
    "@lordicon/react": "^1.11.0",
    "@supabase/supabase-js": "^2.78.0",
    "@tailwindcss/vite": "^4.1.13",
    "@tensorflow/tfjs": "^4.22.0",
    "@vercel/speed-insights": "^1.2.0",
    "cally": "^0.8.0",
    "chart.js": "^4.5.0",
    "dotenv": "^17.2.3",
    "express": "^5.1.0",
    "face-api.js": "^0.22.2",
    "install": "^0.13.0",
    "leaflet": "^1.9.4",
    "lucide-react": "^0.525.0",
    "moment": "^2.30.1",
    "nodemailer": "^7.0.10",
    "react": "^19.1.0",
    "react-big-calendar": "^1.19.4",
    "react-dom": "^19.1.0",
    "react-leaflet": "^5.0.0",
    "react-router-dom": "^7.6.3",
```

```
    "supabase": "^2.51.0",
    "tailwindcss": "^4.1.13",
    "xlsx": "^0.18.5"
  },
  "devDependencies": {
    "@eslint/js": "^9.29.0",
    "@types/react": "^19.1.8",
    "@types/react-dom": "^19.1.6",
    "@vitejs/plugin-react": "^4.5.2",
    "daisyui": "^5.0.46",
    "eslint": "^9.29.0",
    "eslint-plugin-react-hooks":
"^5.2.0",
    "eslint-plugin-react-refresh":
"^0.4.20",
    "globals": "^16.2.0",
    "vite": "^7.0.0"
  }
}
```

Mobile App Home Page

```
import React, { useState, useEffect,
useRef, useMemo, useCallback } from
'react';
import { ImageBackground, StyleSheet,
Text, View, TouchableOpacity, Dimensions,
Alert } from 'react-native';
import { widthPercentageToDP as wp,
heightPercentageToDP as hp } from 'react-
native-responsive-screen';
import AsyncStorage from '@react-native-
async-storage/async-storage';
import { useRouter, useFocusEffect } from
'expo-router';
import NetInfo from '@react-native-
community/netinfo';
import * as Location from 'expo-
location';
import * as Geolib from 'geolib';
import * as Notifications from 'expo-
notifications';
import { ClockContainer } from
'../../scripts/ClockContainer';
import LocationComponent from
'../../scripts/Location';
import { useAuth } from
'../(auth)/AuthProvider';
import { supabase } from
'../../config/supabaseClient';
import { enqueueOp,
saveFailedPhotoUpload,
getFailedPhotoUploads,
deleteFailedPhotoUploads,
incrementFailedPhotoRetries } from
'../../config/offline.db.js';
import CryptoJS from 'react-native-
crypto-js';
import * as FileSystem from 'expo-file-
system';
import { Buffer } from 'buffer';
import { getManilaDateString,
getTimeData, getManilaTimestamp,
getManilaDayName, convertUTCToManila }
from '../../utils/timeUtils';

const getLeaveAbbreviation = (leaveType)
=> {
    if (!leaveType) return '';
    const type =
leaveType.toLowerCase().trim();
    const standardTypes = ['sick leave',
'emergency leave', 'personal leave',
'vacation leave', 'maternity leave',
'paternity leave', 'bereavement leave',
'sick', 'emergency', 'personal',
'vacation', 'maternity', 'paternity',
'bereavement'];
    if (!standardTypes.includes(type)) {
        return '';
    }
    const abbreviations = {
        'sick leave': 'SL',
        'emergency leave': 'EL',
        'personal leave': 'PL',
        'vacation leave': 'VL',
        'maternity leave': 'ML',
        'paternity leave': 'PL',
        'bereavement leave': 'BL',
        'sick': 'SL',
        'emergency': 'EL',
        'personal': 'PL',
        'vacation': 'VL',
        'maternity': 'ML',
        'paternity': 'PL',
        'bereavement': 'BL',
    };
    return abbreviations[type] || '';
};


const DEBUG = false;

const HomeScreen = () => {
    const router = useRouter();
    const { user, getDisplayName } =
useAuth();
    const BGimg =
require('../../assets/logo/assero-
logo.png');
    const [location, setLocation] =
useState({ street: '...', city: 'Loading'
});
    const [currentCoords,
setCurrentCoords] = useState(null);
    const { height } =
Dimensions.get('window');
    const smallScreen = height < 700;
    const btnFontSizeStyle = { fontSize:
smallScreen ? 20 : 28 };
    const btnWidthStyle = smallScreen ? {
width: wp('60%') } : null;

    const [assignedLocation,
setAssignedLocation] = useState({ street:
'Loading', city: '...' });
    const [assignedTimeRange,
setAssignedTimeRange] = useState('...');
    const [assignedDays, setAssignedDays]
= useState('...');
    const [scheduleStartIso,
setScheduleStartIso] = useState(null);
    const [scheduleEndIso,
setScheduleEndIso] = useState(null);
    const [scheduleDaysArr,
setScheduleDaysArr] = useState([]);
    const destinationLocationRef =
useRef({ latitude: null, longitude: null,
radius: null });
    const [destinationLocationState,
setDestinationLocationState] =
```

```
useState({latitude: null, longitude:
null,  radius: null });
    const [isWithinRadiusState,
setIsWithinRadiusState] =
useState(false);
    const lastRadiusStateRef =
useRef(null);
    const outsideSinceRef = useRef(null);
    const [currentStatus,
setCurrentStatus] = useState('time-in');
    const hasRefreshedAfterLoginRef =
useRef(false);
    const [statusRow, setStatusRow] =
useState(null);
    const [graceWindowCountdown,
setGraceWindowCountdown] =
useState(null);
    const [allowLate, setAllowLate] =
useState(null);
    const [isOnLeave, setIsOnLeave] =
useState(false);
    const [leaveInfo, setLeaveInfo] =
useState(null);
    const reminderScheduledRef =
useRef(false);
    const isLoadingScheduleRef =
useRef(false);
    const processingStatusRef =
useRef(null);
    const retryInProgressRef =
useRef(false);
    const lastRetryAttemptRef =
useRef(0);


    const computeLocationHash =
useCallback((payload) => {
        if (!payload) return null;
        try {
            return
CryptoJS.SHA256(JSON.stringify(payload)).
toString();
        } catch {
            return null;
        }
    }, []);

    const mapDbStatusToUi =
useCallback((dbStatus) => {
        switch ((dbStatus ||
'').toLowerCase()) {
            case 'open':
                return 'time-out';
            case 'completed':
                return 'completed';
            case 'offline_pending':
                return 'processing';
            default:
                return 'time-in';
        }
    }, []);

    const fetchOrCreateStatusRow =
useCallback(async (uid, attendanceDate)
=> {
        if (!uid || !attendanceDate)
return null;
        const { data, error } = await
supabase
            .schema('assero_global')
```

```
.from('attendance_daily_status')
            .select('*')
            .eq('uid', uid)
            .eq('attendance_date',
attendanceDate)
            .maybeSingle();
        if (error && error.code !==
'PGRST116') {
            throw error;
        }
        if (data) {
            return data;
        }
        const { data: inserted, error:
insertErr } = await supabase
            .schema('assero_global')

.from('attendance_daily_status')
            .upsert({ uid,
attendance_date: attendanceDate, status:
'pending', last_action: null }, {
onConflict: 'uid,attendance_date' })
            .select()
            .single();
        if (insertErr) {
            if (insertErr.code ===
'23505') {
                const { data: existing }
= await supabase

.schema('assero_global')

.from('attendance_daily_status')
                    .select('*')
                    .eq('uid', uid)

.eq('attendance_date', attendanceDate)
                    .maybeSingle();
                if (existing) return
existing;
            }
            throw insertErr;
        }
        return inserted;
    }, []);

    const upsertStatusRemote =
useCallback(async (uid, attendanceDate,
patch) => {
        if (!uid || !attendanceDate)
return null;
        const payload = {
            uid,
            attendance_date:
attendanceDate,
            ...patch,
            last_sync_at: new
Date().toISOString()
        };
        const { data, error } = await
supabase
            .schema('assero_global')

.from('attendance_daily_status')
            .upsert(payload, {
onConflict: 'uid,attendance_date' })
            .select()
            .single();
        if (error) {
            throw error;
        }
```

```
        setStatusRow(data);

setCurrentStatus(mapDbStatusToUi(data.sta
tus));
        return data;
    }, [mapDbStatusToUi]);

    const updateLocalStatusRow =
useCallback((patch) => {
        setStatusRow(prev => {
            const base = prev || {
                uid: user?.uid ||
String(user?.id || ''),
                attendance_date:
getManilaDateString(),
                status: 'pending'
            };
            const merged = { ...base,
...patch };
            return merged;
        });
        if (patch?.status) {

setCurrentStatus(mapDbStatusToUi(patch.st
atus));
        }
    }, [user?.uid, user?.id,
mapDbStatusToUi]);

    const flagOfflineProcessing =
useCallback((expectedUiState, lastAction)
=> {

setProcessingExpected(expectedUiState);
        setIsProcessing(true);
        setCurrentStatus('processing');
        updateLocalStatusRow({
            status: 'offline_pending',
            last_action: lastAction
        });
    }, [updateLocalStatusRow]);

    const uploadPhotoToSupabase =
useCallback(async (uid, uri, actionType)
=> {
        try {
            if (!uri) {
                if (DEBUG)
console.error('[Home]
uploadPhotoToSupabase: No URI provided');
                return null;
            }
            const fileUri =
uri.startsWith('file://') ? uri :
`file://${uri}`;
            const fileInfo = await
FileSystem.getInfoAsync(fileUri);
            if (!fileInfo.exists) {
                if (DEBUG)
console.error('[Home]
uploadPhotoToSupabase: File does not
exist:', fileUri);
                return null;
            }
            const base64 = await
FileSystem.readAsStringAsync(fileUri, {
encoding: FileSystem.EncodingType.Base64
});
            if (!base64 || base64.length
=== 0) {
                if (DEBUG)
```

```
console.error('[Home]
uploadPhotoToSupabase: Failed to read
file');
                return null;
            }
            const arrayBuffer =
Buffer.from(base64, 'base64');
            const p = new Date();
            const year = p.getFullYear();
            const month =
String(p.getMonth() + 1).padStart(2,
'0');
            const day =
String(p.getDate()).padStart(2, '0');
            const hour =
String(p.getHours()).padStart(2, '0');
            const minute =
String(p.getMinutes()).padStart(2, '0');
            const second =
String(p.getSeconds()).padStart(2, '0');
            const filePath =
`${uid}/${year}-${month}-${day}/T${hour}-
${minute}-${second}.jpg`;
            const bucket = actionType ===
'time-in' ? 'time_in_url' :
'time_out_url';
            const { error: upErr } =
await
supabase.storage.from(bucket).upload(file
Path, arrayBuffer, {
                cacheControl: '3600',
upsert: true, contentType: 'image/jpeg'
            });
            if (upErr) {
                if (DEBUG)
console.error('[Home]
uploadPhotoToSupabase: Storage upload
error:', upErr);
                return null;
            }
            const { data: pub } =
supabase.storage.from(bucket).getPublicUr
l(filePath);
            const publicUrl =
pub?.publicUrl || null;
            if (!publicUrl) {
                if (DEBUG)
console.error('[Home]
uploadPhotoToSupabase: Failed to get
public URL');
                return null;
            }
            const { error: insertErr } =
await
supabase.from('user_attendance_photo').in
sert({ uid, attendance_log: actionType,
url: publicUrl });
            if (insertErr) {
                if (DEBUG)
console.error('[Home]
uploadPhotoToSupabase: Database insert
error:', insertErr);
                return null;
            }
            if (DEBUG)
console.log('[Home]
uploadPhotoToSupabase: Success:',
publicUrl);
            return publicUrl;
        } catch (error) {
            if (DEBUG)
```

```
console.error('[Home]
uploadPhotoToSupabase: Exception:',
error);
            return null;
        }
    }, []);

    const applyOfflineAttendanceOperation
= useCallback(async (uid, opType,
payload, photos = []) => {
        if (!uid) return false;
        const timestamp =
payload?.timeData?.timestamp ||
getManilaTimestamp();
        const attendanceDate =
getManilaDateString(timestamp);
        const locationText =
payload?.locationData ?
`${payload.locationData.street},
${payload.locationData.city}` : null;
        const locationHash =
payload?.locationHash ||
computeLocationHash(payload?.locationData
);

        let photoUrl = null;
        if (photos && photos.length > 0)
{
            const photo = photos.find(p
=> p.actionType === opType) || photos[0];
            if (photo?.uri) {
                photoUrl = await
uploadPhotoToSupabase(uid, photo.uri,
opType);
                if (!photoUrl) {
                    if (DEBUG)
console.error('[Home]
applyOfflineAttendanceOperation: Photo
upload failed, retrying after delay...');
                    await new
Promise(resolve => setTimeout(resolve,
1000));
                    photoUrl = await
uploadPhotoToSupabase(uid, photo.uri,
opType);
                }
                if (!photoUrl) {
                    if (DEBUG)
console.error('[Home]
applyOfflineAttendanceOperation: Photo
upload failed after retry, saving for
later upload');
                }
            }
        }

        let logId = null;
        if (opType === 'time-in') {
            const { data, error } = await
supabase
                .schema('assero_global')

.from('user_attendance_log')
                .insert([{ uid, time_in:
timestamp, location_in: locationText }])
                .select('id')
                .single();
            if (error || !data) {
                if (DEBUG)
console.error('[Home]
applyOfflineAttendanceOperation: Insert
error:', error);
                throw error || new
Error('Failed to insert time-in log');
            }
            logId = data.id;
            await upsertStatusRemote(uid,
attendanceDate, {
                status: 'open',
                last_action: 'time-in',
                time_in_log_id: data.id,
                location_hash:
locationHash || null
            });
        } else if (opType === 'time-out')
{
            const { data: open } = await
supabase
                .schema('assero_global')

.from('user_attendance_log')
                .select('id')
                .eq('uid', uid)
                .is('time_out', null)
                .order('created_at', {
ascending: false })
                .limit(1);
            if (open && open.length > 0)
{
                logId = open[0].id;
                const { error: updateErr
} = await supabase

.schema('assero_global')

.from('user_attendance_log')
                    .update({ time_out:
timestamp, location_out: locationText })
                    .eq('id', logId);
                if (updateErr) {
                    if (DEBUG)
console.error('[Home]
applyOfflineAttendanceOperation: Update
error:', updateErr);
                    throw updateErr;
                }
            } else {
                const { data: inserted,
error: insertErr } = await supabase

.schema('assero_global')

.from('user_attendance_log')
                    .insert([{ uid,
time_out: timestamp, location_out:
locationText }])
                    .select('id')
                    .single();
                if (insertErr ||
!inserted) {
                    if (DEBUG)
console.error('[Home]
applyOfflineAttendanceOperation: Insert
error:', insertErr);
                    throw insertErr ||
new Error('Failed to insert time-out
log');
                }
                logId = inserted.id;
            }
            await upsertStatusRemote(uid,
attendanceDate, {
```

```
                status: 'completed',
                last_action: 'time-out',
                time_out_log_id: logId,
                location_hash:
locationHash || null
            });
        }

        if (!photoUrl && photos &&
photos.length > 0) {
            const photo = photos.find(p
=> p.actionType === opType) || photos[0];
            if (photo?.uri) {
                try {
                    await
saveFailedPhotoUpload(uid, photo.uri,
opType, attendanceDate, logId);
                    if (DEBUG)
console.log('[Home] Saved failed photo
upload for retry');
                } catch (err) {
                    if (DEBUG)
console.error('[Home] Error saving failed
photo upload:', err);
                }
            }
        }

        return true;
    }, [computeLocationHash,
uploadPhotoToSupabase,
upsertStatusRemote]);

    const retryFailedPhotoUploads =
useCallback(async () => {
        const now = Date.now();
        if (retryInProgressRef.current) {
            if (DEBUG)
console.log('[Home] Retry already in
progress, skipping');
            return;
        }
        if (now -
lastRetryAttemptRef.current < 5000) {
            if (DEBUG)
console.log('[Home] Retry attempted too
soon, skipping');
            return;
        }
        retryInProgressRef.current =
true;
        lastRetryAttemptRef.current =
now;
        try {
            const uid = user?.uid ||
String(user?.id || '');
            if (!uid) {

retryInProgressRef.current = false;
                return;
            }
            const failedPhotos = await
getFailedPhotoUploads(uid, 20);
            if (!failedPhotos ||
failedPhotos.length === 0) {

retryInProgressRef.current = false;
                return;
            }
            if (DEBUG)
console.log('[Home] Retrying failed photo
uploads:', failedPhotos.length);
            const succeeded = [];
            for (let i = 0; i <
failedPhotos.length; i++) {
                const failedPhoto =
failedPhotos[i];
                try {
                    const fileInfo =
await
FileSystem.getInfoAsync(failedPhoto.photo
_uri);
                    if (!fileInfo.exists)
{
                        if (DEBUG)
console.error('[Home] Failed photo file
no longer exists:',
failedPhoto.photo_uri);

succeeded.push(failedPhoto.id);
                        continue;
                    }
                    const photoUrl =
await
uploadPhotoToSupabase(failedPhoto.uid,
failedPhoto.photo_uri,
failedPhoto.action_type);
                    if (photoUrl) {
                        if (DEBUG)
console.log('[Home] Successfully retried
photo upload:', failedPhoto.id);

succeeded.push(failedPhoto.id);
                    } else {
                        await
incrementFailedPhotoRetries(failedPhoto.i
d);
                        if
(failedPhoto.retries >= 5) {
                            if (DEBUG)
console.error('[Home] Photo upload
exceeded max retries, removing:',
failedPhoto.id);

succeeded.push(failedPhoto.id);
                        }
                    }
                    if (i <
failedPhotos.length - 1) {
                        await new
Promise(resolve => setTimeout(resolve,
500));
                    }
                } catch (error) {
                    if (DEBUG)
console.error('[Home] Error retrying
photo upload:', error);
                    await
incrementFailedPhotoRetries(failedPhoto.i
d);
                }
            }
            if (succeeded.length > 0) {
                await
deleteFailedPhotoUploads(succeeded);
                if (DEBUG)
console.log('[Home] Cleaned up',
succeeded.length, 'failed photo
uploads');
            }
        } catch (error) {
            if (DEBUG)
```

```
console.error('[Home] Error in
retryFailedPhotoUploads:', error);
        } finally {
            retryInProgressRef.current =
false;
        }
    }, [user?.uid, user?.id,
uploadPhotoToSupabase]);

    const queueScheduleReload =
useCallback(() => {
        if (DEBUG)
console.log('[Schedule] Reloading
schedule data due to change');
        loadScheduleData();
    }, [loadScheduleData]);

    const refreshAfterLogin =
useCallback(async () => {
        if (DEBUG)
console.log('Refreshing after login...');
        await loadScheduleData();
        await checkCurrentStatus();
        updateRadiusStatus();
        hasRefreshedAfterLoginRef.current
= true;
    }, [checkCurrentStatus,
updateRadiusStatus, loadScheduleData]);
    const [isProcessing, setIsProcessing]
= useState(false);
    const [processingExpected,
setProcessingExpected] = useState(null);

    // Realtime subscription for schedule
changes
    useEffect(() => {
        if (DEBUG) console.log('Schedule
subscription effect - user?.uid:',
user?.uid);
        if (!user?.uid) {
            if (DEBUG) console.log('No
user UID, skipping schedule
subscription');
            return;
        }

        if (DEBUG) console.log('Setting
up schedule subscription for UID:',
user.uid);
        const channel = supabase

.channel('mobile_schedule_changes')
            .on('postgres_changes',
                {
                    event: '*',
                    schema:
'assero_global',
                    table: 'schedule',
                    filter:
`employee_uid=eq.${user.uid}`
                },
                (payload) => {
                    if (DEBUG)
console.log('Schedule change detected:',
payload);
                    // Debounced reload
to ensure we pick up the latest committed
change

queueScheduleReload();
                }
            )
            .on('postgres_changes',
                {
                    event: '*',
                    schema:
'assero_global',
                    table:
'client_schedule',
                    filter:
user?.client_code ?
`client_code=eq.${user.client_code}` :
undefined
                },
                (payload) => {
                    if (DEBUG)
console.log('Client schedule change
detected:', payload);

queueScheduleReload();
                }
            )
            .subscribe();

        return () => {
            if (DEBUG)
console.log('Cleaning up schedule
subscription');

supabase.removeChannel(channel);
        };
    }, [user?.uid, user?.client_code,
queueScheduleReload]);

    const startProcessingPoll =
useCallback(async () => {
        const uid = user?.uid ||
String(user?.id || 'unknown');
        if (!uid) return;
        try {
            const flag = await
AsyncStorage.getItem(`attendance_processi
ng_${uid}`);
            if (flag) {
                setIsProcessing(true);
                let expected = 'time-
out';
                try {
                    const parsed =
JSON.parse(flag);
                    expected =
parsed?.action === 'time-in' ? 'time-out'
: 'completed';

setProcessingExpected(expected);

processingStatusRef.current = expected;
                } catch {

setProcessingExpected('time-out');

processingStatusRef.current = 'time-out';
                    expected = 'time-
out';
                }

                await
checkCurrentStatus();
                // Use a fresh fetch to
avoid stale state closure
                const currentUserUid =
user?.uid || String(user?.id);
```

```
                const
currentAttendanceDate =
getManilaDateString();
                const freshStatusRow =
currentUserUid && currentAttendanceDate ?
await
fetchOrCreateStatusRow(currentUserUid,
currentAttendanceDate) : null;
                const initialDbStatus =
freshStatusRow?.status ?
mapDbStatusToUi(freshStatusRow.status) :
null;
                if (initialDbStatus) {

setCurrentStatus(initialDbStatus);

processingStatusRef.current =
initialDbStatus;
                } else {

processingStatusRef.current = expected;
                }

                try { await
AsyncStorage.removeItem(`attendance_proce
ssing_${uid}`); } catch { }

isLoadingScheduleRef.current = false;
                await loadScheduleData();

                if (DEBUG)
console.log('[ProcessingPoll] After
loadScheduleData, checking status
again...');
                await
checkCurrentStatus();

                const userUid = user?.uid
|| String(user?.id);
                const attendanceDate =
getManilaDateString();
                if (userUid &&
attendanceDate) {
                    let retries = 5;
                    let finalRow = null;
                    const
expectedDbStatus = expected === 'time-
out' ? 'open' : expected === 'completed'
? 'completed' : null;

                    while (retries > 0) {
                        await new
Promise(resolve => setTimeout(resolve,
500));
                        const freshRow =
await fetchOrCreateStatusRow(userUid,
attendanceDate);
                        if (freshRow) {
                            if (DEBUG)
console.log('[ProcessingPoll] Retry
check:', { retry: 6 - retries, dbStatus:
freshRow.status, expectedDbStatus,
expected });
                            if
(expectedDbStatus) {
                                if
(freshRow.status === expectedDbStatus) {

finalRow = freshRow;

break;
```

```
                                }
                            } else {
                                if
(freshRow.status !== 'pending') {

finalRow = freshRow;

break;
                                }
                            }
                        }
                        retries--;
                    }

                    if (finalRow &&
finalRow.status === expectedDbStatus) {

setStatusRow(finalRow);
                        const finalStatus
= mapDbStatusToUi(finalRow.status);
                        if (DEBUG)
console.log('[ProcessingPoll] Final
status check (fresh read):', { dbStatus:
finalRow.status, uiStatus: finalStatus
});

setCurrentStatus(finalStatus);
                    } else {
                        const lastCheck =
await fetchOrCreateStatusRow(userUid,
attendanceDate);
                        if (lastCheck) {

setStatusRow(lastCheck);
                            const
finalStatus =
mapDbStatusToUi(lastCheck.status);
                            if (DEBUG)
console.log('[ProcessingPoll] Final
status (last check):', { dbStatus:
lastCheck.status, uiStatus: finalStatus,
expectedDbStatus, matches:
lastCheck.status === expectedDbStatus });
                            if
(expectedDbStatus && lastCheck.status ===
expectedDbStatus) {

setCurrentStatus(finalStatus);
                            } else if
(expectedDbStatus) {
                                if
(DEBUG) console.log('[ProcessingPoll]
Status mismatch, scheduling delayed re-
check');
                                const
delayedTimeoutId = setTimeout(async () =>
{
                                    const
delayedCheck = await
fetchOrCreateStatusRow(userUid,
attendanceDate);
                                    if
(delayedCheck && delayedCheck.status ===
expectedDbStatus) {

setStatusRow(delayedCheck);

const delayedStatus =
mapDbStatusToUi(delayedCheck.status);

if (DEBUG) console.log('[ProcessingPoll]
```

```
Delayed check found correct status:', {
dbStatus: delayedCheck.status, uiStatus:
delayedStatus });

setCurrentStatus(delayedStatus);

setProcessingExpected(null);

processingStatusRef.current = null;

setIsProcessing(false);
                                    }
else if (delayedCheck) {

if (DEBUG) console.log('[ProcessingPoll]
Delayed check still mismatch:', {
dbStatus: delayedCheck.status,
expectedDbStatus });

setCurrentStatus(mapDbStatusToUi(delayedC
heck.status));

setProcessingExpected(null);

processingStatusRef.current = null;

setIsProcessing(false);
                                    }
                                }, 2000);

processingStatusRef.current = {
timeoutId: delayedTimeoutId };
                                return;
                            } else {

setCurrentStatus(finalStatus);
                            }
                        } else {
                            if (DEBUG)
console.log('[ProcessingPoll] No status
row found after processing');
                        }
                    }
                }


setProcessingExpected(null);
                if
(processingStatusRef.current?.timeoutId)
{

clearTimeout(processingStatusRef.current.
timeoutId);
                }

processingStatusRef.current = null;
                setIsProcessing(false);
            } else {
                setIsProcessing(false);

setProcessingExpected(null);

processingStatusRef.current = null;
                await
checkCurrentStatus();
            }
        } catch (error) {
            if (DEBUG)
console.error('[ProcessingPoll] Error:',
error);
            setIsProcessing(false);
            setProcessingExpected(null);
            if
(processingStatusRef.current?.timeoutId)
{

clearTimeout(processingStatusRef.current.
timeoutId);
            }
            processingStatusRef.current =
null;
        }
    }, [user?.uid, user?.id,
checkCurrentStatus, loadScheduleData,
mapDbStatusToUi,
fetchOrCreateStatusRow]);

    const isWithinRadius =
useCallback((currentCoords) => {
        if (!currentCoords?.latitude ||
!currentCoords?.longitude) {
            if (DEBUG)
console.log('[Radius] No current
coords:', currentCoords);
            return { distance: null,
withinRadius: false };
        }

        const dest =
destinationLocationRef.current;
        if (dest.latitude == null ||
dest.longitude == null || dest.radius ==
null) {
            if (DEBUG)
console.log('[Radius] No destination:',
dest);
            return { distance: null,
withinRadius: false };
        }

        const distance =
Geolib.getDistance(
            { latitude:
currentCoords.latitude, longitude:
currentCoords.longitude },
            { latitude: dest.latitude,
longitude: dest.longitude }
        );
        const buffer = 5;
        const effectiveRadius =
dest.radius + buffer;
        const withinRadius = distance <=
effectiveRadius;

        if (DEBUG) console.log('[Radius]
Calculation:', {
            currentCoords: { lat:
currentCoords.latitude, lng:
currentCoords.longitude },
            destination: { lat:
dest.latitude, lng: dest.longitude },
            radius: dest.radius,
            buffer: buffer,
            effectiveRadius:
effectiveRadius,
            distance: distance,
            withinRadius: withinRadius
        });

        return { distance: distance,
withinRadius: withinRadius };
    }, []);
```

```
    const updateRadiusStatus =
useCallback(() => {
        const result =
isWithinRadius(currentCoords);
        if (lastRadiusStateRef.current
!== result.withinRadius) {
            if (DEBUG)
console.log('[Radius] State change:', {
                previous:
lastRadiusStateRef.current,
                new: result.withinRadius,
                distance: result.distance
            });
            lastRadiusStateRef.current =
result.withinRadius;

setIsWithinRadiusState(result.withinRadiu
s);
        }
    }, [currentCoords, isWithinRadius]);


    const checkCurrentStatus =
useCallback(async () => {
        try {
            const userUid = user?.uid ||
String(user?.id);
            if (!userUid) {
                setCurrentStatus('time-
in');
                return;
            }
            const attendanceDate =
getManilaDateString();
            const row = await
fetchOrCreateStatusRow(userUid,
attendanceDate);
            if (row) {
                setStatusRow(row);
                const uiStatus =
mapDbStatusToUi(row.status);
                if (DEBUG && row.status
!== 'pending') {

console.log('[StatusCheck] DB status:',
row.status, '-> UI status:', uiStatus);
                }

setCurrentStatus(uiStatus);
            } else {
                setStatusRow(null);
                setCurrentStatus('time-
in');
            }
        } catch (error) {
            if (DEBUG)
console.error('[StatusCheck] Error:',
error);
            setStatusRow(null);
            setCurrentStatus('time-in');
        }
    }, [user?.uid, user?.id,
fetchOrCreateStatusRow,
mapDbStatusToUi]);

    const handleTimeIn =
useCallback(async () => {
        if (DEBUG) console.log('[TimeIn]
handleTimeIn called', {
            isWithinRadiusState,
            currentStatus,
            hasCurrentCoords:
!!currentCoords,
            canTimeInNow
        });

        if (!isWithinRadiusState) {
            if (DEBUG)
console.log('[TimeIn] Blocked - not
within radius');
            Alert.alert('Location Error',
'You must be within the designated radius
to time in/out.');
            return;
        }

        if (currentStatus ===
'completed') {
            if (DEBUG)
console.log('[TimeIn] Blocked - already
completed');
            Alert.alert(
                'Daily Time Tracking
Complete',
                'You have already
completed both time-in and time-out for
today. The system will reset tomorrow.',
                [{ text: 'OK', style:
'default' }]
            );
            return;
        }

        if (isOnLeave && currentStatus
=== 'time-in') {
            if (DEBUG)
console.log('[TimeIn] Blocked - user is
on leave');
            const leaveAbbr =
leaveInfo?.leave_type ?
getLeaveAbbreviation(leaveInfo.leave_type
) : '';
            const leaveTypeText =
leaveAbbr ? ` (${leaveAbbr})` :
(leaveInfo?.leave_type ? `
(${leaveInfo.leave_type})` : '');
            Alert.alert(
                'Leave Status',
                `You are currently on
leave${leaveTypeText} and cannot time
in.`,
                [{ text: 'OK', style:
'default' }]
            );
            return;
        }

        if (currentStatus === 'time-in'
&& !canTimeInNow) {
            if (DEBUG)
console.log('[TimeIn] Blocked - outside
schedule');
            Alert.alert(
                'Schedule Error',
                'You cannot time in
outside your scheduled day and time.
Please check your schedule.',
                [{ text: 'OK', style:
'default' }]
            );
            return;
```

```
        }

        if (DEBUG) console.log('[TimeIn]
Validating reverse geocode...');
        await
validateReverseGeocode(currentCoords,
assignedLocation);
        if (DEBUG) console.log('[TimeIn]
Reverse geocode validated');

        const timeDataObj =
getTimeData();
        const locationData = {
            street: location?.street ||
'Unknown Street',
            city: location?.city ||
'Unknown City',
            coordinates: currentCoords
        };
        const timeData = {
            time: timeDataObj.time,
            date: timeDataObj.date,
            timestamp:
timeDataObj.timestamp
        };
        const locationHash =
computeLocationHash(locationData);

        if (currentStatus === 'time-out')
{
            Alert.alert(
                'Confirm Time Out',
                'Are you sure you want to
time out? You will need to authenticate
with your face.',
                [
                    { text: 'Cancel',
style: 'cancel' },
                    {
                        text: 'Time Out',
                        style:
'destructive',
                        onPress: async ()
=> {
                            const net =
await NetInfo.fetch();
                            if
(net.isConnected) {

router.navigate(`../(auth)/FaceAuth?actio
n=time-
out&location=${encodeURIComponent(JSON.st
ringify(locationData))}&timeData=${encode
URIComponent(JSON.stringify(timeData))}`)
;
                            } else {
                                await
enqueueOp('time-out', { locationData,
timeData, locationHash });

flagOfflineProcessing('completed', 'time-
out');

Alert.alert('Queued Offline', 'Your time-
out was saved and will sync when
online.');
                            }
                        }
                    }
                ]
            );
```

```
        } else {
            if (DEBUG)
console.log('[TimeIn] Processing time-
in...');
            const net = await
NetInfo.fetch();
            if (DEBUG)
console.log('[TimeIn] Network status:',
net.isConnected);
            if (net.isConnected) {
                if (DEBUG)
console.log('[TimeIn] Navigating to
FaceAuth');

router.navigate(`../(auth)/FaceAuth?actio
n=time-
in&location=${encodeURIComponent(JSON.str
ingify(locationData))}&timeData=${encodeU
RIComponent(JSON.stringify(timeData))}`);
            } else {
                if (DEBUG)
console.log('[TimeIn] Offline - queuing
operation');
                await enqueueOp('time-
in', { locationData, timeData,
locationHash });

flagOfflineProcessing('time-out', 'time-
in');
                Alert.alert('Queued
Offline', 'Your time-in was saved and
will sync when online.');
            }
        }
    }, [isWithinRadiusState,
currentStatus, currentCoords, location,
computeLocationHash, router,
canTimeInNow, isOnLeave, leaveInfo,
assignedLocation, validateReverseGeocode,
flagOfflineProcessing]);


    const loadScheduleData =
useCallback(async () => {
        if (isLoadingScheduleRef.current)
{
            if (DEBUG)
console.log('loadScheduleData already in
progress, skipping');
            return;
        }
        if (DEBUG)
console.log('loadScheduleData called -
user:', user ? { uid: user.uid, id:
user.id } : 'null');
        isLoadingScheduleRef.current =
true;
        try {
            const userUid = user?.uid ||
String(user?.id);
            if (DEBUG)
console.log('loadScheduleData -
userUid:', userUid);
            if (!userUid) {
                if (DEBUG)
console.log('No userUid, returning
early');
                return;
            }

            // Fetch user with
```

```
client_code
            if (DEBUG)
console.log('Fetching user data for
UID:', userUid);
            const { data: userRows } =
await supabase
                .schema('assero_global')
                .from('user')
                .select('uid,
client_code, firstname, lastname')
                .eq('uid', userUid)
                .limit(1);
            const userRow = (userRows &&
userRows[0]) || null;
            if (DEBUG) console.log('User
row found:', userRow ? { uid:
userRow.uid, client_code:
userRow.client_code } : 'null');

            // Fetch client's allow_late
value
            if (userRow?.client_code) {
                const { data: clientData
} = await supabase

.schema('assero_global')
                    .from('client')
                    .select('allow_late')
                    .eq('client_code',
userRow.client_code)
                    .maybeSingle();
                if
(clientData?.allow_late !== undefined &&
clientData.allow_late !== null) {
                    if (DEBUG)
console.log('[GraceWindow] Client
allow_late:', clientData.allow_late,
'minutes');

setAllowLate(clientData.allow_late);
                } else {
                    if (DEBUG)
console.log('[GraceWindow] No allow_late
value found for client:',
userRow.client_code);
                    setAllowLate(null);
                }
            } else {
                if (DEBUG)
console.log('[GraceWindow] No client_code
found for user');
                setAllowLate(null);
            }

            // Fetch schedules (include
updated_at like web app)
            if (DEBUG)
console.log('Fetching schedules for
employee UID:', userUid);
            const [{ data: userSched }, {
data: clientSched }] = await
Promise.all([
                supabase

.schema('assero_global')
                    .from('schedule')
                    .select('start_at,
end_at, location, created_at, updated_at,
radius_m, days')
                    .eq('employee_uid',
userUid),
                    userRow?.client_code
                        ? supabase

.schema('assero_global')

.from('client_schedule')

.select('start_at, end_at, location,
created_at, updated_at, radius_m, days')

.eq('client_code', userRow.client_code)
                        : Promise.resolve({
data: [] })
            ]);
            if (DEBUG) console.log('User
schedules found:', userSched?.length ||
0);
            if (DEBUG)
console.log('Client schedules found:',
clientSched?.length || 0);

            // Helper sort like web:
updated_at desc, then created_at desc,
then start_at desc
            const sortLikeWeb = (a, b) =>
{
                const au = a?.updated_at
? new Date(a.updated_at).getTime() : 0;
                const bu = b?.updated_at
? new Date(b.updated_at).getTime() : 0;
                if (bu !== au) return bu
- au;
                const ac = a?.created_at
? new Date(a.created_at).getTime() : 0;
                const bc = b?.created_at
? new Date(b.created_at).getTime() : 0;
                if (bc !== ac) return bc
- ac;
                const as = a?.start_at ?
new Date(a.start_at).getTime() : 0;
                const bs = b?.start_at ?
new Date(b.start_at).getTime() : 0;
                return bs - as;
            };

            // Choose latest non-
placeholder employee schedule
            const nonPlaceholders =
(userSched || []).filter(r =>
!(r.start_at == null && r.end_at == null
&& r.location == null));
            let chosen =
nonPlaceholders.length > 0 ?
[...nonPlaceholders].sort(sortLikeWeb)[0]
: null;

            // Find placeholder row for
overlaying radius later
            const placeholderRow =
(userSched || []).find(r => r.start_at ==
null && r.end_at == null && r.location ==
null) || null;

            // Compute latest client
schedule
            let latestClientSched = null;
            if (!chosen &&
Array.isArray(clientSched) &&
clientSched.length > 0) {
                latestClientSched =
[...clientSched].sort(sortLikeWeb)[0];
```

```
                chosen =
latestClientSched ? {
...latestClientSched } : null;
            }

            // Resolve days: prefer
employee's own days if provided; else
client days
            const resolvedDays =
(Array.isArray(chosen?.days) &&
chosen.days.length > 0)
                ? chosen.days
                :
(Array.isArray(latestClientSched?.days) ?
latestClientSched.days : []);
            if (chosen) chosen.days =
resolvedDays;

            // Resolve location: prefer
employee schedule location, else client
location
            if (chosen) chosen.location =
chosen.location ||
latestClientSched?.location || null;

            // Resolve time range: if
employee schedule lacks times, inherit
from client
            if (chosen &&
latestClientSched && (!chosen.start_at ||
!chosen.end_at)) {
                chosen.start_at =
chosen.start_at ||
latestClientSched.start_at || null;
                chosen.end_at =
chosen.end_at || latestClientSched.end_at
|| null;
            }

            // Resolve radius: prefer
employee radius_m; else client radius_m
            if (chosen &&
(chosen.radius_m == null ||
!Number.isFinite(chosen.radius_m))) {
                if (latestClientSched &&
Number.isFinite(latestClientSched.radius_
m)) {
                    chosen.radius_m =
latestClientSched.radius_m;
                }
            }

            // If employee only has
placeholder, overlay its radius onto
chosen
            if (!nonPlaceholders ||
nonPlaceholders.length === 0) {
                if (placeholderRow &&
chosen &&
Number.isFinite(placeholderRow.radius_m))
{
                    chosen.radius_m =
placeholderRow.radius_m;
                }
            }

            // Location display text
(reverse geocode) handled below
            let coords = null;
            if (chosen?.location &&
typeof chosen.location === 'string') {
                const m =
chosen.location.trim().match(/^\s*([+-
]?\d{1,2}(?:\.\d+)?)\s*,\s*([+-
]?\d{1,3}(?:\.\d+)?)\s*$/);
                if (m) {
                    const lat =
parseFloat(m[1]);
                    const lng =
parseFloat(m[2]);
                    if
(Number.isFinite(lat) &&
Number.isFinite(lng)) {
                        coords = {
latitude: lat, longitude: lng };
                    }
                }
            }

            if (coords) {
                try {
                    const reverseGeocode
= await
Location.reverseGeocodeAsync(coords);
                    if (reverseGeocode &&
reverseGeocode.length > 0) {
                        const g =
reverseGeocode[0];

setAssignedLocation({
                            street:
g.street || g.name || 'Unknown Street',
                            city: g.city
|| g.subregion || '...'
                        });
                    }
                } catch {
                    setAssignedLocation({
street: 'Assigned Location', city: '' });
                }
            } else {
                setAssignedLocation({
street: 'No assigned location', city: ''
});
            }

            // Time range display
            const fmt = (iso) => iso ?
new Date(iso) : null;
            const s =
fmt(chosen?.start_at);
            const e =
fmt(chosen?.end_at);
            const toHuman = (d) => d ?
d.toLocaleTimeString('en-PH', { hour: '2-
digit', minute: '2-digit', hour12: true,
timeZone: 'Asia/Manila' }) : null;
            const sTxt = toHuman(s);
            const eTxt = toHuman(e);
            setAssignedTimeRange((sTxt &&
eTxt) ? `${sTxt} - ${eTxt}` : 'No time
assigned');

setScheduleStartIso(chosen?.start_at ||
null);

setScheduleEndIso(chosen?.end_at ||
null);

            // Days text
            const days = chosen?.days;
            const normalizeDays = (arr)
```

```
=> {
                if (!Array.isArray(arr))
return [];
                const map = {
                    'M': 'Monday', 'Mon':
'Monday', 'Monday': 'Monday',
                    'T': 'Tuesday',
'Tue': 'Tuesday', 'Tuesday': 'Tuesday',
                    'W': 'Wednesday',
'Wed': 'Wednesday', 'Wednesday':
'Wednesday',
                    'Th': 'Thursday',
'Thu': 'Thursday', 'Thursday':
'Thursday',
                    'F': 'Friday', 'Fri':
'Friday', 'Friday': 'Friday',
                    'S': 'Saturday',
'Sat': 'Saturday', 'Saturday':
'Saturday',
                    'Su': 'Sunday',
'Sun': 'Sunday', 'Sunday': 'Sunday'
                };
                return arr.map(d =>
map[String(d).trim()] ||
String(d).trim());
            };
            const toDayLetters = (arr) =>
{
                if (!Array.isArray(arr)
|| arr.length === 0) return 'No days
assigned';
                return arr.map((d) => {
                    const dayStr =
String(d).trim();
                    if
(dayStr.startsWith('Thu')) return 'Th';
                    if
(dayStr.startsWith('Sun')) return 'Su';
                    return
dayStr.charAt(0).toUpperCase();
                }).join(',');
            };
            const normalized =
normalizeDays(days);

setAssignedDays(toDayLetters(normalized))
;

setScheduleDaysArr(normalized);

            // Radius
            const chosenRadius =
Number.isFinite(chosen?.radius_m) ?
chosen.radius_m : null;
            const oldRadius =
destinationLocationRef.current?.radius;
            const oldLat =
destinationLocationRef.current?.latitude;
            const oldLng =
destinationLocationRef.current?.longitude
;
            const newDestLocation =
coords
                ? { latitude:
coords.latitude, longitude:
coords.longitude, radius: chosenRadius }
                : { latitude: null,
longitude: null, radius: chosenRadius };
            const locationChanged =
oldLat !== newDestLocation.latitude ||
oldLng !== newDestLocation.longitude ||
oldRadius !== chosenRadius;
            if (DEBUG)
console.log('[Schedule] Updating
destination location:', {
                radius: chosenRadius,
                oldRadius: oldRadius,
                locationChanged:
locationChanged,
                coords: coords ? { lat:
coords.latitude, lng: coords.longitude }
: null,
                oldCoords: oldLat !==
null ? { lat: oldLat, lng: oldLng } :
null
            });

destinationLocationRef.current =
newDestLocation;
            if (locationChanged) {

setDestinationLocationState({
                    latitude:
newDestLocation.latitude,
                    longitude:
newDestLocation.longitude,
                    radius:
newDestLocation.radius,
                    updatedAt: Date.now()
                });
                if (DEBUG)
console.log('[Schedule]
destinationLocationState updated,
triggering radius check');
            } else {
                if (DEBUG)
console.log('[Schedule] Location
unchanged, skipping state update');
            }
        } catch (error) {
            if (DEBUG)
console.error('loadScheduleData error:',
error);
            setAssignedLocation({ street:
'Error loading location', city: '' });
            setAssignedTimeRange('...');
            setAssignedDays('...');
            const errorDestLocation = {
latitude: null, longitude: null, radius:
null };

destinationLocationRef.current =
errorDestLocation;
            setDestinationLocationState({
...errorDestLocation });
        } finally {
            isLoadingScheduleRef.current
= false;
        }
    }, [user?.uid, user?.id,
user?.client_code]);

    const checkLeaveStatus =
useCallback(async () => {
        if (!user?.uid) {
            setIsOnLeave(false);
            setLeaveInfo(null);
            return;
        }
        try {
            const today =
getManilaDateString();
```

```
            const { data } = await
supabase
                .schema('assero_global')
                .from('leave')
                .select('uid, leave_type,
status, date_from, date_to')
                .eq('uid', user.uid)
                .eq('status', 'Approved')
                .lte('date_from', today)
                .gte('date_to', today)
                .maybeSingle();
            if (data) {
                setIsOnLeave(true);
                setLeaveInfo(data);
                if (DEBUG)
console.log('[Leave] User is on leave
today:', data);
            } else {
                setIsOnLeave(false);
                setLeaveInfo(null);
            }
        } catch (error) {
            if (DEBUG)
console.error('[Leave] Error checking
leave status:', error);
            setIsOnLeave(false);
            setLeaveInfo(null);
        }
    }, [user?.uid]);

    useEffect(() => {
        if (DEBUG)
console.log('loadScheduleData effect
triggered - user:', user ? { uid:
user.uid, id: user.id } : 'null');
        // Reset refresh flag when user
changes
        hasRefreshedAfterLoginRef.current
= false;
        loadScheduleData();
        checkLeaveStatus();
    }, [user?.uid, user?.client_code,
loadScheduleData, checkLeaveStatus]);

    useEffect(() => {
        if (!isProcessing && user?.uid) {
            checkCurrentStatus();
        }
    }, [isProcessing, user?.uid,
checkCurrentStatus]);

    useEffect(() => {
        if (currentCoords &&
destinationLocationState &&
destinationLocationState.radius !== null)
{
            updateRadiusStatus();
        } else if (!currentCoords ||
!destinationLocationState ||
destinationLocationState.radius === null)
{
            if
(lastRadiusStateRef.current !== false) {

lastRadiusStateRef.current = false;

setIsWithinRadiusState(false);
            }
        }
    }, [currentCoords,
destinationLocationState?.latitude,
destinationLocationState?.longitude,
destinationLocationState?.radius,
updateRadiusStatus]);

    const validateReverseGeocode =
useCallback(async (coords,
expectedLocation) => {
        if (!coords?.latitude ||
!coords?.longitude) return true;
        if (!expectedLocation?.city ||
expectedLocation.city === '...' ||
expectedLocation.city === 'Loading')
return true;
        try {
            const reverse = await
Location.reverseGeocodeAsync(coords);
            if (!reverse ||
reverse.length === 0) return true;
            const actual = reverse[0];
            const actualCity =
(actual.city || actual.subregion ||
actual.district ||
'').toLowerCase().trim();
            const expectedCity =
(expectedLocation?.city ||
'').toLowerCase().trim();
            if (!expectedCity ||
!actualCity) return true;
            if
(actualCity.includes(expectedCity) ||
expectedCity.includes(actualCity)) return
true;
            if (DEBUG) console.log('[GPS]
Reverse geocode mismatch (non-
blocking):', { actualCity, expectedCity,
actual: reverse[0] });
            return true;
        } catch (error) {
            if (DEBUG) console.log('[GPS]
Reverse geocode error (non-blocking):',
error);
            return true;
        }
    }, []);

    const setupGraceWindowCountdown =
useCallback(() => {
        if (!scheduleStartIso ||
currentStatus !== 'time-in' || allowLate
=== null || allowLate === undefined ||
allowLate === 0) {
            if (DEBUG && scheduleStartIso
&& currentStatus === 'time-in') {

console.log('[GraceWindow] Not showing -
allowLate:', allowLate);
            }

setGraceWindowCountdown(null);
            return;
        }
        const now = new Date();
        const manilaTime = new
Date(now.getTime() + (8 * 3600000));
        const start = new
Date(scheduleStartIso);
        const sManila = new
Date(start.getTime() + (8 * 3600000));
        const toMinutes = (d) =>
d.getHours() * 60 + d.getMinutes();
        const startMin =
```

```
toMinutes(sManila);
        const nowMin =
toMinutes(manilaTime);
        const graceMinutes = allowLate;

        const startDateStr =
sManila.toISOString().split('T')[0];
        const nowDateStr =
manilaTime.toISOString().split('T')[0];
        const isSameDay = startDateStr
=== nowDateStr;

        if (!isSameDay) {
            if (DEBUG)
console.log('[GraceWindow] Not same day
as schedule start:', {
                scheduleStartDate:
startDateStr,
                currentDate: nowDateStr,
                scheduleStart:
sManila.toISOString(),
                currentTime:
manilaTime.toISOString()
            });

setGraceWindowCountdown(null);
            return;
        }

        const windowEndMin = startMin +
graceMinutes;
        if (nowMin >= startMin && nowMin
< windowEndMin) {
            const remainingMs =
(windowEndMin - nowMin) * 60 * 1000;
            const remainingMinutes =
Math.ceil(remainingMs / 1000 / 60);
            if (DEBUG)
console.log('[GraceWindow] Active:', {
                allowLate: graceMinutes,
                scheduleStart:
sManila.toISOString(),
                currentTime:
manilaTime.toISOString(),
                startMin,
                nowMin,
                windowEndMin,
                remainingMinutes
            });

setGraceWindowCountdown(remainingMinutes)
;
        } else {
            if (DEBUG)
console.log('[GraceWindow] Expired or not
started:', {
                allowLate: graceMinutes,
                scheduleStart:
sManila.toISOString(),
                currentTime:
manilaTime.toISOString(),
                startMin,
                nowMin,
                windowEndMin
            });

setGraceWindowCountdown(null);
        }
    }, [scheduleStartIso, currentStatus,
allowLate]);
```

```
    const scheduleTimeInReminder =
useCallback(async () => {
        if (reminderScheduledRef.current
|| !scheduleStartIso || currentStatus !==
'time-in') return;
        try {
            const { status } = await
Notifications.requestPermissionsAsync();
            if (status !== 'granted')
return;
            const now = new Date();
            const manilaTime = new
Date(now.getTime() + (8 * 3600000));
            const start = new
Date(scheduleStartIso);
            const sManila = new
Date(start.getTime() + (8 * 3600000));
            const toMinutes = (d) =>
d.getHours() * 60 + d.getMinutes();
            const startMin =
toMinutes(sManila);
            const nowMin =
toMinutes(manilaTime);
            if (nowMin < startMin) {
                const delayMs = (startMin
- nowMin) * 60 * 1000;
                await
Notifications.scheduleNotificationAsync({
                    content: {
                        title: 'Time In
Reminder',
                        body: 'Your
scheduled time-in window is starting now.
Please time in.',
                        sound: true,
                    },
                    trigger: { seconds:
Math.floor(delayMs / 1000) },
                });

reminderScheduledRef.current = true;
            }
        } catch {
        }
    }, [scheduleStartIso,
currentStatus]);

    useEffect(() => {
        const interval = setInterval(()
=> {
            setupGraceWindowCountdown();
        }, 60000);
        setupGraceWindowCountdown();
        scheduleTimeInReminder();
        return () =>
clearInterval(interval);
    }, [scheduleStartIso, currentStatus,
allowLate, setupGraceWindowCountdown,
scheduleTimeInReminder]);

    useEffect(() => {
        if (isWithinRadiusState) {
            outsideSinceRef.current =
null;
            try {

AsyncStorage.removeItem('outside_radius_s
ince');
            } catch { }
            return;
        }
```

```
        (async () => {
            try {
                if
(!outsideSinceRef.current) {
                    const existing =
await
AsyncStorage.getItem('outside_radius_sinc
e');
                    if (existing) {

outsideSinceRef.current =
Number(existing) || Date.now();
                    } else {

outsideSinceRef.current = Date.now();
                        await
AsyncStorage.setItem('outside_radius_sinc
e', String(outsideSinceRef.current));
                    }
                }
            } catch { }
        })();
    }, [isWithinRadiusState]);

    useEffect(() => {
        if (!user?.uid) return;
        const retryTimeoutRef = {
current: null };
        const unsubscribe =
NetInfo.addEventListener(state => {
            if (state.isConnected &&
state.isInternetReachable) {
                if
(retryTimeoutRef.current) {

clearTimeout(retryTimeoutRef.current);
                }
                retryTimeoutRef.current =
setTimeout(() => {
                    if (DEBUG)
console.log('[Home] Network connected,
retrying failed photo uploads');

retryFailedPhotoUploads();
                }, 3000);
            } else {
                if
(retryTimeoutRef.current) {

clearTimeout(retryTimeoutRef.current);

retryTimeoutRef.current = null;
                }
            }
        });
        return () => {
            if (retryTimeoutRef.current)
{

clearTimeout(retryTimeoutRef.current);
            }
            unsubscribe();
        };
    }, [user?.uid,
retryFailedPhotoUploads]);


    useFocusEffect(
        React.useCallback(() => {
            let retryTimeoutId = null;
            const refreshStatus = async
```

```
() => {
                await
checkCurrentStatus();
                await
startProcessingPoll();
                await
checkCurrentStatus();
                if (DEBUG)
console.log('[FocusEffect] Final status
after refresh:', {
                    currentStatus,
                    statusRowStatus:
statusRow?.status,
                    displayStatus:
isProcessing && processingExpected ?
processingExpected : currentStatus
                });
            };
            refreshStatus();

            // Refresh once after login
if not already done
            if (user?.uid &&
!hasRefreshedAfterLoginRef.current) {
                refreshAfterLogin();
            }

            (async () => {
                const net = await
NetInfo.fetch();
                if (net.isConnected) {
                    const { popNextOps }
= await
import('../../config/offline.db.js');
                    const ops = await
popNextOps(20);
                    if (ops && ops.length
> 0) {
                        const succeeded =
[];
                        for (const op of
ops) {
                            try {
                                const
payload = JSON.parse(op.payload || '{}');
                                const uid
= user?.uid || String(user?.id ||
'unknown');
                                const ok
= await
applyOfflineAttendanceOperation(uid,
op.op_type, payload, op.photos || []);
                                if (ok)
succeeded.push(op.id);
                            } catch (err)
{
                                if
(DEBUG) console.error('Offline queue
processing failed:', err);
                            }
                        }
                        if
(succeeded.length > 0) {
                            const {
deleteOps } = await
import('../../config/offline.db.js');
                            await
deleteOps(succeeded);
                        }
                    }
```

```
setIsProcessing(false);

setProcessingExpected(null);
                    const netCheck =
await NetInfo.fetch();
                    if
(netCheck.isConnected &&
netCheck.isInternetReachable) {
                        retryTimeoutId =
setTimeout(() => {

retryFailedPhotoUploads();
                        }, 2000);
                    }
                }
            })();

            return () => {
                if (retryTimeoutId) {

clearTimeout(retryTimeoutId);
                }
            };
        }, [user?.uid,
retryFailedPhotoUploads,
checkCurrentStatus, startProcessingPoll,
refreshAfterLogin,
applyOfflineAttendanceOperation])
    );

    const displayStatus = useMemo(() => {
        const status = isProcessing &&
processingExpected ? processingExpected :
currentStatus;
        if (DEBUG && status !== 'time-
in') {
            console.log('[DisplayStatus]
Current:', {
                isProcessing,
                processingExpected,
                currentStatus,
                displayStatus: status,
                statusRowStatus:
statusRow?.status
            });
        }
        return status;
    }, [isProcessing, processingExpected,
currentStatus, statusRow?.status]);

    const canTimeInNow = useMemo(() => {
        try {
            if (DEBUG)
console.log('canTimeInNow check -
scheduleStartIso:', scheduleStartIso,
'scheduleEndIso:', scheduleEndIso);
            if (!scheduleStartIso ||
!scheduleEndIso) {
                if (DEBUG)
console.log('canTimeInNow: Missing
schedule times');
                return false;
            }

            const todayName =
getManilaDayName();
            const now = new Date();
            const manilaNow =
convertUTCToManila(now.toISOString());
            const yesterdayManila = new
Date(manilaNow.getTime() - 24 * 60 * 60 *
1000);
            const yesterdayName =
getManilaDayName(yesterdayManila.toISOStr
ing());

            if (DEBUG)
console.log('Simple timezone calculation
debug:', {
                'now': now.toISOString(),
                'manilaNow':
manilaNow.toISOString(),
                'todayName': todayName,
                'yesterdayName':
yesterdayName,
                'scheduleDaysArr':
scheduleDaysArr
            });

            const sManila =
convertUTCToManila(scheduleStartIso);
            const eManila =
convertUTCToManila(scheduleEndIso);

            const toMinutes = (d) =>
d.getUTCHours() * 60 + d.getUTCMinutes();
            const startMin =
toMinutes(sManila);
            const endMin =
toMinutes(eManila);
            const nowMin =
toMinutes(manilaNow);

            const hasDays =
Array.isArray(scheduleDaysArr) &&
scheduleDaysArr.length > 0;
            const overnight = endMin <=
startMin;

            if (DEBUG)
console.log('canTimeInNow details:', {
                manilaNow:
manilaNow.toISOString(),
                todayName,
                yesterdayName,
                scheduleDaysArr,
                startMin,
                endMin,
                nowMin,
                hasDays,
                overnight
            });

            if (!overnight) {
                if (!hasDays) {
                    if (DEBUG)
console.log('canTimeInNow: No days
assigned (day shift)');
                    return false;
                }

                // Normalize day names to
handle potential format differences
                const normalizedToday =
todayName.toLowerCase();
                const
normalizedScheduleDays =
scheduleDaysArr.map(day =>
day.toLowerCase());

                if (DEBUG)
console.log('Day matching debug:', {
```

```
                    'normalizedToday':
normalizedToday,

'normalizedScheduleDays':
normalizedScheduleDays,
                    'includes check':
normalizedScheduleDays.includes(normalize
dToday)
                });

                if
(!normalizedScheduleDays.includes(normali
zedToday)) {
                    if (DEBUG)
console.log('canTimeInNow: Not assigned
day (day shift)');
                    return false;
                }
                // Early time-in allowed
anytime before end time
                const result = nowMin <=
endMin;
                if (DEBUG)
console.log('canTimeInNow day shift
result:', result, 'nowMin <= endMin:',
nowMin, '<=', endMin);
                return result;
            }

            // Overnight: require days to
be assigned
            if (!hasDays) {
                if (DEBUG)
console.log('canTimeInNow: No days
assigned (overnight shift)');
                return false;
            }

            // Overnight: allow time-in
on same day from start to 23:59, and next
day until end
            const normalizedToday =
todayName.toLowerCase();
            const normalizedYesterday =
yesterdayName.toLowerCase();
            const normalizedScheduleDays
= scheduleDaysArr.map(day =>
day.toLowerCase());

            // For overnight shifts,
check if current time is after start time
on assigned day
            // OR if current time is
before end time on the day after assigned
day
            const sameDayOk =
normalizedScheduleDays.includes(normalize
dToday) && nowMin >= startMin;
            const morningOk =
normalizedScheduleDays.includes(normalize
dYesterday) && nowMin <= endMin;

            // Additional check: if it's
the assigned day and we're in the early
morning hours (after midnight but before
end time)
            const earlyMorningOk =
normalizedScheduleDays.includes(normalize
dToday) && nowMin <= endMin && nowMin <
720; // 720 = 12:00 PM
```

```
            const result = sameDayOk ||
morningOk || earlyMorningOk;

            if (DEBUG)
console.log('canTimeInNow overnight
details:', {
                sameDayOk,
                morningOk,
                earlyMorningOk,
                result,
                'sameDayOk check':
`scheduleDaysArr.includes(todayName) &&
nowMin >= startMin`,
                'morningOk check':
`scheduleDaysArr.includes(yesterdayName)
&& nowMin <= endMin`,
                'earlyMorningOk check':
`scheduleDaysArr.includes(todayName) &&
nowMin <= endMin && nowMin < 720`
            });

            return result;
        } catch (error) {
            if (DEBUG)
console.log('canTimeInNow error:',
error);
            return false;
        }
    }, [scheduleStartIso, scheduleEndIso,
scheduleDaysArr]);

    useEffect(() => {
        if (DEBUG && (displayStatus !==
'time-in' || isProcessing)) {
            console.log('[HomeScreen]
State update:', {
                displayStatus,
                isWithinRadiusState,
                isProcessing,
                currentStatus,
                statusRowStatus:
statusRow?.status
            });
        }
    }, [displayStatus,
isWithinRadiusState, isProcessing,
currentStatus, statusRow?.status]);

    return (
        <ImageBackground source={BGimg}
imageStyle={{ opacity: 0.2 }}
style={styles.container}>
            <LocationComponent

onLocationChange={setLocation}
                onCoordsChange={(coords)
=> {

setCurrentCoords(coords);
                }}
            />
            <View
style={styles.assignedLTD}>
                <Text
style={styles.companyLabel}>{assignedLoca
tion.street}{assignedLocation.city ? `,
${assignedLocation.city}` : ''}</Text>
                <Text
style={styles.companyLabel}>{assignedTime
Range}</Text>
                <Text
```

```
style={styles.companyLabel}>{assignedDays
}</Text>
                <Text
style={styles.companyLabel}>{(destination
LocationRef.current?.radius)}m
radius</Text>
                {graceWindowCountdown !=
null && graceWindowCountdown > 0 && (
                    <Text
style={[styles.companyLabel, { color:
'#FF5722', fontWeight: 'bold' }]}>
                        Grace window:
{graceWindowCountdown} min remaining
                    </Text>
                )}
            </View>
            <View
style={styles.clockHandler}>
                <ClockContainer />
            </View>

            <View style={styles.NTL}>
                <Text
style={styles.userName}>

{getDisplayName(user)}
                </Text>
                <Text
style={styles.userId}>
                    ID: {user?.uid ||
user?.id || 'N/A'}
                </Text>
                <Text
style={styles.userRole}>
                    Role: {user?.role ||
'Employee'}
                </Text>
                {useMemo(() => {
                    const buttonText =
isProcessing ? 'PROCESSING' :
displayStatus === 'completed' ?
'COMPLETED' :
                        displayStatus ===
'time-out' ? 'TIME OUT' : 'TIME IN';
                    const buttonDisabled
= isProcessing || !isWithinRadiusState ||
displayStatus === 'completed' ||
                        (displayStatus
=== 'time-in' && !canTimeInNow);

                    if (DEBUG &&
(isProcessing || displayStatus !== 'time-
in' || !isWithinRadiusState)) {

console.log('[Button] State:', {
                            buttonText,

buttonDisabled,

displayStatus,
                            isProcessing,

isWithinRadiusState
                        });
                    }

                    return (
                        <TouchableOpacity

disabled={buttonDisabled}
                            style={[
```

```
styles.btnCard,

isProcessing ? { width: wp('60%') } :
null,

btnWidthStyle,

isProcessing ? styles.inactiveButton :
displayStatus === 'completed' ?
styles.completedButton :

displayStatus === 'time-out' ?
styles.timeOutButton :

(isWithinRadiusState && (displayStatus
!== 'time-in' || canTimeInNow) ?
styles.activeButton :
styles.inactiveButton)
                            ]}

activeOpacity={(!isProcessing &&
isWithinRadiusState && displayStatus !==
'completed' && (displayStatus !== 'time-
in' || canTimeInNow)) ? 0.8 : 1}

onPress={handleTimeIn}
                        >
                            <Text
numberOfLines={1} ellipsizeMode='clip'
style={[

styles.btnText,

btnFontSizeStyle,

isProcessing ? styles.inactiveBtnText :
displayStatus === 'completed' ?
styles.completedBtnText :

displayStatus === 'time-out' ?
styles.timeOutBtnText :

(isWithinRadiusState && (displayStatus
!== 'time-in' || canTimeInNow) ?
styles.activeBtnText :
styles.inactiveBtnText)
                            ]}>

{buttonText}
                            </Text>

</TouchableOpacity>
                        );
                    }, [isProcessing,
isWithinRadiusState, displayStatus,
currentStatus, processingExpected,
statusRow?.status, canTimeInNow,
btnWidthStyle, btnFontSizeStyle,
handleTimeIn])}
                <Text
style={[styles.locationText, { fontSize:
height < 750 ? 20 : 28 }]}>
                    Location:
{location?.street === null ? '...' :
location?.street + ','}
                </Text>
                <Text
style={[styles.locationText, { fontSize:
height < 750 ? 20 : 28 }]}>
                    {location?.city}
```

```
                </Text>
            </View>
        </ImageBackground>
    );
}

const styles = StyleSheet.create({
    container: {
        flex: 1,
        flexDirection: 'column',
        justifyContent: 'flex-start',
        backgroundColor: 'white'
    },
    assignedLTD: {
        flex: hp('10%'),
        alignItems: 'flex-end',
        padding: wp('2%'),
    },
    clockHandler: {
        flex: hp('40%'),
        justifyContent: 'center',
        alignItems: 'center',
    },
    companyLabel: {
        fontFamily: 'InterMedium',
        color: '#000000',
    },
    NTL: {
        flex: hp('50%'),
        justifyContent: 'center',
        alignItems: 'center',
    },
    userName: {
        fontFamily: 'InterMedium',
        fontSize: 24,
        color: '#000000'
    },
    userId: {
        fontFamily: 'OpenSansLight',
        fontSize: 16,
        fontWeight: 'bold',
        color: '#666666'
    },
    userRole: {
        fontFamily: 'OpenSansLight',
        fontSize: 14,
        marginBottom: hp('1%'),
        color: '#000000'
    },
    btnCard: {
        height: hp('10%'),
        width: wp('50%'),
        padding: wp('2%'),
        alignItems: 'center',
        justifyContent: 'center',
        marginBottom: hp('2%'),
        borderRadius: 20
    },
    btnText: {
        fontFamily: 'ArialBlack',
        fontSize: 28
    },
    activeButton: {
        backgroundColor: '#84C986',
        elevation: 5,
        shadowColor: '#000',
        shadowOffset: {
            width: 0,
            height: 2,
        },
        shadowOpacity: 0.25,
        shadowRadius: 3.84,
    },
    inactiveButton: {
        backgroundColor: '#ccc',
        opacity: 0.6,
    },
    activeBtnText: {
        color: '#E9E9E9'
    },
    inactiveBtnText: {
        color: '#999'
    },
    timeOutButton: {
        backgroundColor: '#FF5722',
        elevation: 5,
        shadowColor: '#000',
        shadowOffset: {
            width: 0,
            height: 2,
        },
        shadowOpacity: 0.25,
        shadowRadius: 3.84,
    },
    timeOutBtnText: {
        color: '#E9E9E9'
    },
    locationText: {
        fontFamily: 'InterMedium',
        color: '#000000',
        fontWeight: 'bold',
        textAlign: 'center'
    },
    completedButton: {
        backgroundColor: '#999',
        opacity: 0.7,
    },
    completedBtnText: {
        color: '#E9E9E9',
        fontWeight: 'bold',
    }
});

export default HomeScreen;
```

Web App Dashboard

```
import React, { useEffect, useState }
from 'react'
import supabase from
'../../config/supabaseClient'
import UserDonut from
'../../components/Dashboard/UserDonut'
import ActivityLog from
'../../components/Dashboard/ActivityLog'
import SkeletonLoader from
'../../components/Common/SkeletonLoader'
import { useDashboardData } from
'../../hooks/Dashboard/useDashboardData'
import KPICards from
'../../components/Dashboard/KPICards'
import AttendanceChart from
'../../components/Dashboard/AttendanceCha
rt'
import PendingApprovals from
'../../components/Dashboard/PendingApprov
als'
import ClientDistribution from
'../../components/Dashboard/ClientDistrib
ution'
```

```
import CalendarEvents from
'../../components/Dashboard/CalendarEvent
s'

const DEBUG = false

const Dashboard = () => {
    const {
        chartData,
        kpis,
        attendance7d,
        approvals,
        clientDistribution,
        timeKpis,
        prevKpis,
        isLoading
    } = useDashboardData()

    const [events, setEvents] =
useState([])
    const [currentTime, setCurrentTime] =
useState(new Date())

    const fetchCalendarEvents = async ()
=> {
        try {
            const today = new Date()
            const startOfWeek = new
Date(today.getTime() - 7 * 24 * 60 * 60 *
1000)
            const endOfWeek = new
Date(today.getTime() + 7 * 24 * 60 * 60 *
1000)
            const { data: weekEvents,
error } = await supabase
                .schema('assero_global')
                .from('calendar')
                .select('*')
                .gte('date',
startOfWeek.toISOString().split('T')[0])
                .lte('date',
endOfWeek.toISOString().split('T')[0])
                .order('date', {
ascending: true })

            if (error) {
                if (DEBUG)
console.error('Error fetching calendar
events:', error)
                return
            }

            const transformedEvents =
(weekEvents || []).map(event => {
                const eventDate = new
Date(event.date)

                let startTime, endTime

                if (event.is_long_event
&& event.from_date && event.to_date) {
                    startTime = new
Date(event.from_date)
                    endTime = new
Date(event.to_date)
                    endTime.setHours(23,
59, 59, 999)
                } else if
(event.start_time && event.end_time) {
                    const [startHour,
startMin] = event.start_time.split(':')
                    const [endHour,
endMin] = event.end_time.split(':')
                    startTime = new
Date(eventDate)

startTime.setHours(parseInt(startHour),
parseInt(startMin), 0, 0)
                    endTime = new
Date(eventDate)

endTime.setHours(parseInt(endHour),
parseInt(endMin), 0, 0)
                } else {
                    startTime = new
Date(eventDate)
                    startTime.setHours(0,
0, 0, 0)
                    endTime = new
Date(eventDate)
                    endTime.setHours(23,
59, 59, 999)
                }

                return {
                    id: event.id,
                    title:
event.holiday_name,
                    description:
event.description || '',
                    startTime,
                    endTime,
                    type:
event.holiday_type,
                    isPhilippineHoliday:
event.is_philippine_holiday,
                    isLongEvent:
event.is_long_event,
                    date: event.date
                }
            })

            setEvents(transformedEvents)
        } catch (error) {
            if (DEBUG)
console.error('Error fetching calendar
events:', error)
        }
    }

    useEffect(() => {
        fetchCalendarEvents()

        const timer = setInterval(() => {
            setCurrentTime(new Date())
        }, 60000)

        const calendarChannel = supabase

.channel('dashboard_calendar_events_chang
es')
            .on('postgres_changes',
                {
                    event: '*',
                    schema:
'assero_global',
                    table: 'calendar'
                },
                () => {
                    if (DEBUG)
console.log('[Dashboard] Calendar events
change detected, refreshing...')
```

```
                    fetchCalendarEvents()
                }
            )
            .subscribe()

        return () => {
            calendarChannel.unsubscribe()
            clearInterval(timer)
        }
    }, [])

    if (isLoading) {
        return <SkeletonLoader
type="dashboard" />
    }

    return (
        <div>
            <div className='shadow-lg bg-
gray-200 dark:bg-gray-900 dark:text-white
mt-5 mx-6 mb-1.5 sm:mb-0 rounded-lg h-
[70px] p-2 flex flex-col justify-center
items-start '>
                <div className='ml-2
sm:ml-5'>
                    <h1 className='text-
xl font-bold sm:text-2xl'>Dashboard</h1>
                    <p className='text-
gray-500 text-sm dark:text-gray-300
sm:text-md'>Welcome to Dashboard</p>
                </div>
            </div>

            <div className='grid grid-
cols-1 gap-3 p-3'>
                <KPICards kpis={kpis}
timeKpis={timeKpis} prevKpis={prevKpis}
/>

                <div className='grid
grid-cols-1 lg:grid-cols-3 gap-3 items-
start'>
                    <div
className='lg:col-span-2'>
                        <AttendanceChart
attendance7d={attendance7d}
isLoading={isLoading} />
                    </div>
                    <div className='flex
flex-col space-y-2 pt-3'>
                        <UserDonut
total={chartData.total}
cHead={chartData.cHead}
cEmp={chartData.cEmp}
cFinance={chartData.cFinance} />
                        <PendingApprovals
approvals={approvals} />
                    </div>
                </div>

                <div className='grid
grid-cols-1 lg:grid-cols-3 gap-3 items-
start'>
                    <ClientDistribution
clientDistribution={clientDistribution}
/>
                    <CalendarEvents
events={events} currentTime={currentTime}
/>
                    <div>
                        <ActivityLog />
                    </div>
                </div>
            </div>
        </div>
    )
}

export default Dashboard
```

Web App Landing Page

```
import React, { useCallback, useEffect,
useRef, useState } from 'react'
import supabase from
'../config/supabaseClient'
import logo from '../assets/logo.png'
import heroImage from
'../assets/images/1.jpg'
import aboutImage from
'../assets/images/2.jpg'
import contactImage from
'../assets/images/3.jpg'
import sectorsImage from
'../assets/images/4.jpg'
import inquireImage from
'../assets/images/5.jpg'
import servicesImage from
'../assets/images/6.jpg'

const sections = [
    { id: 'home', label: 'Home' },
    { id: 'about', label: 'About' },
    { id: 'services', label: 'Services'
},
    { id: 'inquire', label: 'Inquire' },
    { id: 'contact', label: 'Contact' }
]

const parallaxStyles = (image) => ({
    backgroundImage: `linear-
gradient(0deg, rgba(7,11,25,0.65),
rgba(7,11,25,0.65)), url(${image})`,
    backgroundSize: 'cover',
    backgroundPosition: 'center',
    backgroundAttachment: 'fixed'
})

const Landing = () => {
    const [contactInfo, setContactInfo] =
useState(null)
    const [contactLoading,
setContactLoading] = useState(true)
    const [inquiry, setInquiry] =
useState({ name: '', email: '', company:
'', location: '', message: '', honeypot:
'' })
    const [inquiryStatus,
setInquiryStatus] = useState(null)
    const [inquirySubmitting,
setInquirySubmitting] = useState(false)
    const [activeSection,
setActiveSection] = useState('home')
    const statusTimeoutRef = useRef(null)
    const sectionRefs = useRef({})

    const loadContactInfo =
useCallback(async () => {
        try {
            const res = await
```

```
fetch('/secret_API/site-contact')
            const payload = await
res.json().catch(() => ({}))
            if (res.ok && payload?.data)
{
                setContactInfo({
                    title:
payload.data.title || '',
                    location:
payload.data.location || '',
                    phone:
payload.data.phone || '',
                    email:
payload.data.email || ''
                })
            } else {
                setContactInfo(null)
            }
        } catch {
            setContactInfo(null)
        } finally {
            setContactLoading(false)
        }
    }, [])

    useEffect(() => {
        loadContactInfo()
        const handler = () =>
loadContactInfo()

window.addEventListener('contactInfoUpdat
ed', handler)

        const channel =
supabase.channel('site-contact-listener')
            .on('postgres_changes', {
event: '*', schema: 'assero_global',
table: 'site_contact' }, () => {
                loadContactInfo()
            })
            .subscribe()

        return () => {

window.removeEventListener('contactInfoUp
dated', handler)

supabase.removeChannel(channel)
        }
    }, [loadContactInfo])

    useEffect(() => {
        const observerOptions = {
            root: null,
            rootMargin: '-20% 0px -60%
0px',
            threshold: 0
        }

        const observerCallback =
(entries) => {
            entries.forEach((entry) => {
                if (entry.isIntersecting)
{
                    const sectionId =
entry.target.id

setActiveSection(sectionId)
                }
            })
        }
```

```
        const observer = new
IntersectionObserver(observerCallback,
observerOptions)

        sections.forEach((section) => {
            const element =
document.getElementById(section.id)
            if (element) {
                observer.observe(element)

sectionRefs.current[section.id] = element
            }
        })

        return () => {

Object.values(sectionRefs.current).forEac
h((element) => {
                if (element) {

observer.unobserve(element)
                }
            })
        }
    }, [])

    const updateInquiry = (field, value)
=> {
        setInquiry(prev => ({ ...prev,
[field]: value }))
    }

    const handleInquirySubmit = async (e)
=> {
        e.preventDefault()
        if (inquirySubmitting) return
        setInquiryStatus(null)
        if (statusTimeoutRef.current) {

clearTimeout(statusTimeoutRef.current)
            statusTimeoutRef.current =
null
        }
        setInquirySubmitting(true)
        try {
            const res = await
fetch('/secret_API/send-inquiry', {
                method: 'POST',
                headers: { 'Content-
Type': 'application/json' },
                body:
JSON.stringify(inquiry)
            })
            const payload = await
res.json().catch(() => ({}))
            if (!res.ok) {
                throw new
Error(payload?.error || 'Failed to send
inquiry')
            }
            setInquiryStatus({ type:
'success', message: 'Inquiry sent
successfully. Our team will reach out
soon.' })
            statusTimeoutRef.current =
setTimeout(() => setInquiryStatus(null),
5000)
            setInquiry({ name: '', email:
'', company: '', location: '', message:
'', honeypot: '' })
```

```
        } catch (error) {
            setInquiryStatus({ type:
'error', message: error?.message ||
'Failed to send inquiry.' })
            statusTimeoutRef.current =
setTimeout(() => setInquiryStatus(null),
5000)
        } finally {
            setInquirySubmitting(false)
        }
    }

    return (
        <div className="min-h-screen w-
full bg-slate-50 text-slate-900">
            <header className="fixed top-
0 left-0 right-0 z-20 bg-white/80
backdrop-blur border-b border-slate-200">
                <div className="mx-auto
flex max-w-6xl items-center justify-
between px-6 py-3">
                    <div className="flex
items-center gap-3">
                        <img src={logo}
alt="Assero Global logo" className="h-15
w-auto" />
                        <div>
                            <p
className="text-xs uppercase tracking-
widest text-slate-500">Assero Global</p>
                            <p
className="text-sm font-semibold text-
slate-900">AGISSS Security Services
Inc.</p>
                        </div>
                    </div>
                    <nav
className="hidden gap-6 text-sm font-
medium md:flex">

{sections.map((section) => (
                            <a

key={section.id}

href={`#${section.id}`}

className={`transition ${activeSection
=== section.id
                                    ?
'text-red-700'
                                    :
'text-slate-700 hover:text-red-700'
                                    }`}
                            >

{section.label}
                            </a>
                        ))}
                    </nav>
                </div>
            </header>

            <section id="home"
className="flex min-h-screen items-
center"
style={parallaxStyles(heroImage)}>
                <div className="mx-auto
mt-20 max-w-4xl px-6 py-16 text-center
text-white">
                    <p className="text-sm
font-semibold uppercase tracking-[0.3em]
text-blue-100">Securing the Philippines
since 2010</p>
                    <h1 className="mt-6
text-4xl font-semibold leading-tight
md:text-5xl">
                        Delivering
professional security and risk management
nationwide
                    </h1>
                    <p className="mt-6
text-lg text-blue-100">
                        Close protection,
static and mobile security, training, and
operational consultancy built for
Philippine realities.
                    </p>
                    <div className="mt-10
flex flex-col items-center justify-center
gap-4 sm:flex-row">
                        <a
href="#inquire" className="rounded-full
bg-blue-600 px-8 py-3 text-sm font-
semibold uppercase tracking-wide text-
white transition hover:bg-blue-500">
                            Inquire Today
                        </a>
                        <a href="#about"
className="rounded-full border border-
white/60 px-8 py-3 text-sm font-semibold
uppercase tracking-wide text-white
transition hover:bg-white/10">
                            Learn More
                        </a>
                    </div>
                </div>
            </section>

            <section id="about"
className="bg-white py-20">
                <div className="mx-auto
flex max-w-6xl flex-col gap-10 px-6
md:flex-row">
                    <div className="md:w-
1/2">
                        <h2
className="text-sm font-semibold
uppercase tracking-[0.3em] text-blue-
600">About</h2>
                        <p className="mt-
4 text-3xl font-semibold text-slate-
900">Operational risk management rooted
in Philippine realities.</p>
                        <p className="mt-
6 text-base leading-relaxed text-slate-
600">
                            Established
in 2010, Assero Global Philippines has
enabled multinational corporations, NGOs,
industrial operators, and private
principals to execute mission-critical
work securely. We craft regulatory-
compliant SOPs, train site teams, and
embed supervisors who understand the
terrain, communities, and regulatory
frameworks across the archipelago.
                        </p>
                        <p className="mt-
4 text-base leading-relaxed text-slate-
600">
                            Our
```

```
specialists blend global standards with
the on-the-ground perspective required to
anticipate threats, coordinate with local
authorities, and keep operations
resilient.
                        </p>
                    </div>
                    <div className="md:w-
1/2">
                        <div
className="rounded-3xl shadow-xl"
style={parallaxStyles(aboutImage)}>
                            <div
className="rounded-3xl bg-slate-900/70 p-
10 text-white backdrop-blur">
                                <p
className="text-sm uppercase tracking-
[0.3em] text-blue-200">What we do</p>
                                <p
className="mt-5 text-lg leading-relaxed
text-blue-50">

Recruitment, vetting, training,
deployment, and active supervision of
licensed security personnel, with
embedded quality assurance and rapid
incident response.
                                </p>
                            </div>
                        </div>
                    </div>
                </div>
            </section>

            <section id="services"
className="py-20 text-white"
style={parallaxStyles(servicesImage)}>
                <div className="mx-auto
max-w-6xl px-6">
                    <div className="text-
center">
                        <h2
className="text-sm font-semibold
uppercase tracking-[0.3em] text-blue-
100">Services</h2>
                        <p className="mt-
4 text-3xl font-semibold">Full-spectrum
protective solutions</p>
                    </div>
                    <div className="mt-12
grid gap-8 md:grid-cols-3">
                        {[
                            { title:
'Static & Mobile Security', detail:
'Licensed guard operations with dedicated
site supervisors, mobile patrols, quick-
response teams, and 24/7 command
oversight.' },
                            { title:
'Close Protection & Specialized Teams',
detail: 'Executive protection, secure
transport, and tailored detail
composition for principals, visiting
delegations, and high-profile events.' },
                            { title:
'Training & Consultancy', detail:
'Regulatory-compliant training, risk
assessments, SOP development, incident
response planning, and operational
audits.' }
                        ].map((service)
=> (
                            <div
key={service.title} className="rounded-
2xl border border-white/30 bg-white/5 p-6
shadow-lg backdrop-blur">
                                <p
className="text-lg font-
semibold">{service.title}</p>
                                <p
className="mt-3 text-sm leading-relaxed
text-blue-100">{service.detail}</p>
                            </div>
                        ))}
                    </div>
                    <section
className="mt-16 rounded-3xl border
border-white/30 bg-white/5 p-10 text-
white backdrop-blur">
                        <div
className="text-center">
                            <p
className="text-sm font-semibold
uppercase tracking-[0.3em] text-blue-
100">Sectors</p>
                            <p
className="mt-3 text-xl font-medium text-
blue-50">Supporting clients across
industries</p>
                        </div>
                        <div
className="mt-8 grid gap-6 md:grid-cols-
3">
                            {[

'Corporate headquarters & commercial
estates',

'Industrial plants, logistics hubs, and
special economic zones',

'Residential communities and mixed-use
developments',
                                'NGOs and
development projects operating
nationwide',

'Government institutions and diplomatic
engagements',
                                'Private
principals, visiting delegations, and
events'
                            ].map((item)
=> (
                                <div
key={item} className="rounded-2xl border
border-white/20 bg-white/5 p-4 text-sm
leading-relaxed text-blue-100">

{item}
                                </div>
                            ))}
                        </div>
                    </section>
                </div>
            </section>

            <section id="inquire"
className="py-20"
style={parallaxStyles(inquireImage)}>
                <div className="mx-auto
max-w-5xl px-6 text-center text-white">
```

```
                    <h2 className="text-
sm font-semibold uppercase tracking-[1em]
text-blue-600">Inquire</h2>
                    <p className="mt-4
text-3xl font-semibold text-slate-
200">Tell us about your site, mission, or
project.</p>
                    <p className="mt-4
text-base leading-relaxed text-slate-
400">
                        We build
pragmatic security frameworks—guarding
strength, supervision, and training—
aligned with your operational tempo.
Share your requirements and our risk
advisors will respond within one business
day.
                    </p>
                    <form
                        className="mt-10
space-y-4 text-left rounded-3xl bg-
white/10 p-6 shadow-2xl backdrop-blur"

onSubmit={handleInquirySubmit}

action="javascript:void(0);"

autoComplete="off"
                    >
                        <input
                            type="text"

value={inquiry.honeypot}
                            onChange={(e)
=> updateInquiry('honeypot',
e.target.value)}

className="hidden"
                            tabIndex={-1}

autoComplete="off"
                            aria-
hidden="true"
                        />
                        <div
className="grid gap-6 md:grid-cols-2">
                            <input
value={inquiry.name} onChange={(e) =>
updateInquiry('name', e.target.value)}
type="text" placeholder="Full Name"
className="rounded-2xl border border-
slate-300 bg-white/30 px-4 py-3 text-sm
text-slate-900 outline-none focus:border-
blue-500 focus:ring-1 focus:ring-blue-
500" required />
                            <input
value={inquiry.email} onChange={(e) =>
updateInquiry('email', e.target.value)}
type="email" placeholder="Email Address"
className="rounded-2xl border border-
slate-300 bg-white/30 px-4 py-3 text-sm
text-slate-900 outline-none focus:border-
blue-500 focus:ring-1 focus:ring-blue-
500" required />
                            <input
value={inquiry.company} onChange={(e) =>
updateInquiry('company', e.target.value)}
type="text" placeholder="Company /
Organization" className="rounded-2xl
border border-slate-300 bg-white/30 px-4
py-3 text-sm text-slate-900 outline-none
focus:border-blue-500 focus:ring-1
focus:ring-blue-500" required />
                            <input
value={inquiry.location} onChange={(e) =>
updateInquiry('location',
e.target.value)} type="text"
placeholder="Location / Site"
className="rounded-2xl border border-
slate-300 bg-white/30 px-4 py-3 text-sm
text-slate-900 outline-none focus:border-
blue-500 focus:ring-1 focus:ring-blue-
500" required />
                        </div>
                        <textarea
value={inquiry.message} onChange={(e) =>
updateInquiry('message', e.target.value)}
placeholder="Tell us about your security
objectives..." rows={4} className="w-full
rounded-2xl border border-slate-300 bg-
white/30 px-4 py-3 text-sm text-slate-900
outline-none focus:border-blue-500
focus:ring-1 focus:ring-blue-500"
required></textarea>
                        {inquiryStatus &&
(
                            <div

className={`rounded-2xl px-4 py-2 text-xs
font-semibold uppercase tracking-wide
${inquiryStatus.type === 'success'
                                ?
'bg-green-50 text-green-700'
                                :
'bg-red-50 text-red-700'
                                }`}
                            >

{inquiryStatus.message}
                            </div>
                        )}
                        <button
                            type="submit"

disabled={inquirySubmitting}

className="mt-4 rounded-full bg-blue-600
px-10 py-3 text-sm font-semibold
uppercase tracking-wide text-white
transition disabled:cursor-not-allowed
disabled:bg-blue-300 hover:bg-blue-500"
                        >

{inquirySubmitting ? 'Sending…' : 'Submit
Inquiry'}
                        </button>
                    </form>
                </div>
            </section>

            <section
                id="contact"
                className="relative
overflow-hidden bg-slate-900 py-20 text-
white"
                style={{
                    backgroundImage:
`linear-gradient(0deg,
rgba(7,11,25,0.85), rgba(7,11,25,0.85)),
url(${contactImage})`,
                    backgroundSize:
'cover',
```

```
                    backgroundPosition:
'center',
                    backgroundAttachment:
'fixed'
                }}
            >
                <div className="absolute
inset-0 bg-slate-900/50"></div>
                <div className="relative
mx-auto grid max-w-5xl gap-10 px-6
md:grid-cols-2">
                    <div>
                        <h2
className="text-sm font-semibold
uppercase tracking-[0.3em] text-blue-
300">Contact</h2>
                        <p className="mt-
4 text-3xl font-semibold">

{contactLoading ? 'Loading…' :
(contactInfo?.title || 'Contact
unavailable')}
                        </p>
                        <p className="mt-
4 text-sm uppercase tracking-[0.3em]
text-blue-200">Office Location</p>
                        <p className="mt-
2 text-base leading-relaxed text-blue-
100">

{contactLoading ? 'Loading…' :
(contactInfo?.location || 'No address on
file')}
                        </p>
                        <p className="mt-
4 text-lg font-semibold text-
white">{contactLoading ? 'Loading…' :
(contactInfo?.phone || 'No contact
number')}</p>
                        <p className="mt-
1 text-sm text-blue-100">{contactLoading
? 'Loading…' : (contactInfo?.email || 'No
email configured')}</p>
                        <div
className="mt-8 flex flex-wrap gap-4">

{contactInfo?.phone ? (
                                <a
href={`tel:${contactInfo.phone.replace(/[
^0-9+]/g, '')}`} className="rounded-full
border border-white/30 px-6 py-2 text-xs
font-semibold uppercase tracking-wide
text-white transition hover:border-white
hover:bg-white/10">
                                    Call
Us
                                </a>
                            ) : null}
                            <a
href="#inquire" className="rounded-full
bg-blue-500 px-6 py-2 text-xs font-
semibold uppercase tracking-wide text-
white transition hover:bg-blue-400">
                                Request
Proposal
                            </a>
                        </div>
                    </div>
                    <div
className="rounded-3xl border border-
white/10 bg-white/5 p-8 backdrop-blur">
                        <p
className="text-sm uppercase tracking-
[0.3em] text-blue-200">Managed
Services</p>
                        <p className="mt-
4 text-base leading-relaxed text-blue-
100">
                            Nationwide
coverage with dedicated supervisors, QA
audits, and command-center support. We
design site-specific plans, training
cycles, and reporting frameworks so
clients operate with confidence.
                        </p>
                        <div
className="mt-6 grid gap-4 text-sm text-
blue-100">
                            <p>Static and
mobile guard deployments</p>
                            <p>Close
protection and secure transport</p>
                            <p>Security
training and competency validation</p>
                            <p>Risk
assessments and incident response
planning</p>
                            <p>On-site
supervision, QA, and compliance
reporting</p>
                        </div>
                    </div>
                </div>
                <div className="relative
mt-12 border-t border-white/10 pt-6 text-
center text-xs uppercase tracking-[0.4em]
text-blue-300">
                    © {new
Date().getFullYear()} Assero Global
Philippines
                </div>
            </section>
        </div>
    )
}

export default Landing
```

Database Connection - Supabase

```
import { createClient } from
'@supabase/supabase-js';
import { EXPO_PUBLIC_SUPABASE_URL,
EXPO_PUBLIC_SUPABASE_KEY } from '@env';

export const supabase =
createClient(EXPO_PUBLIC_SUPABASE_URL,
EXPO_PUBLIC_SUPABASE_KEY, {
  auth: {
    autoRefreshToken: true,
    persistSession: true,
    detectSessionInUrl: false,
  },
  db: {
    schema: '[REDACTED]'
  }
});

export default supabase;
```

Database Connection - SQLite

```
import * as SQLite from 'expo-sqlite';

let db = null;
export function getDb() {
    if (!db) {
        db =
SQLite.openDatabase('assero_offline_v1.db
');
        db.exec([{ sql: 'PRAGMA
journal_mode = WAL;', args: [] }], false,
() => {});
        db.transaction(tx => {
            tx.executeSql(`create table
if not exists offline_ops (
                id integer primary key
autoincrement,
                created_at integer not
null,
                op_type text not null,
                payload text not null,
                retries integer not null
default 0
            );`);
            tx.executeSql(`create index
if not exists idx_ops_created on
offline_ops(created_at);`);
            tx.executeSql(`create table
if not exists offline_photos (
                id integer primary key
autoincrement,
                op_id integer not null,
                photo_uri text not null,
                action_type text not
null,
                created_at integer not
null,
                foreign key (op_id)
references offline_ops(id) on delete
cascade
            );`);
            tx.executeSql(`create index
if not exists idx_photos_op on
offline_photos(op_id);`);
            tx.executeSql(`create table
if not exists cache_kv (
                key text primary key,
                value text not null,
                updated_at integer not
null
            );`);
            tx.executeSql(`create table
if not exists failed_photo_uploads (
                id integer primary key
autoincrement,
                uid text not null,
                photo_uri text not null,
                action_type text not
null,
                attendance_date text,
                log_id integer,
                created_at integer not
null,
                retries integer not null
default 0
            );`);
            tx.executeSql(`create index
if not exists idx_failed_photos_uid on
failed_photo_uploads(uid);`);
            tx.executeSql(`create index
if not exists idx_failed_photos_created
on failed_photo_uploads(created_at);`);
        });
    }
    return db;
}

export function enqueueOp(opType,
payloadObj) {
    return new Promise((resolve, reject)
=> {
        const payload =
JSON.stringify(payloadObj || {});
        const now = Date.now();
        getDb().transaction(tx => {
            tx.executeSql(
                'insert into offline_ops
(created_at, op_type, payload) values (?,
?, ?);',
                [now, opType, payload],
                (_, res) => {
                    const opId =
res.insertId;
                    if
(payloadObj?.photoUri) {
                        tx.executeSql(
                            'insert into
offline_photos (op_id, photo_uri,
action_type, created_at) values (?, ?, ?,
?);',
                            [opId,
payloadObj.photoUri, opType, now],
                            () =>
resolve(opId),
                            (_, err) => {
reject(err); return false; }
                        );
                    } else {
                        resolve(opId);
                    }
                },
                (_, err) => {
reject(err); return false; }
            );
        });
    });
}

export function popNextOps(limit = 20) {
    return new Promise((resolve, reject)
=> {
        getDb().transaction(tx => {
            tx.executeSql('select id,
op_type, payload, retries from
offline_ops order by created_at asc limit
?;', [limit], (_, { rows }) => {
                const ops = rows._array
|| [];
                if (ops.length === 0) {
                    resolve([]);
                    return;
                }
                const opIds = ops.map(o
=> o.id);
                const placeholders =
opIds.map(() => '?').join(',');
                tx.executeSql(`select
op_id, photo_uri, action_type from
offline_photos where op_id in
(${placeholders});`, opIds, (_,photoRows)
```

```
=> {
                    const photosByOpId =
{};

(photoRows.rows._array || []).forEach(p
=> {
                        if
(!photosByOpId[p.op_id])
photosByOpId[p.op_id] = [];

photosByOpId[p.op_id].push({ uri:
p.photo_uri, actionType: p.action_type
});
                    });
                    ops.forEach(op => {
                        op.photos =
photosByOpId[op.id] || [];
                    });
                    resolve(ops);
                }, (_, err) => {
reject(err); return false; });
            }, (_, err) => { reject(err);
return false; });
        });
    });
}

export function deleteOps(ids) {
    if (!ids || ids.length === 0) return
Promise.resolve();
    const placeholders = ids.map(() =>
'?').join(',');
    return new Promise((resolve, reject)
=> {
        getDb().transaction(tx => {
            tx.executeSql(`delete from
offline_photos where op_id in
(${placeholders});`, ids, () => {
                tx.executeSql(`delete
from offline_ops where id in
(${placeholders});`, ids, () =>
resolve(), (_, err) => { reject(err);
return false; });
            }, (_, err) => { reject(err);
return false; });
        });
    });
}

export function bumpRetries(id) {
    return new Promise((resolve, reject)
=> {
        getDb().transaction(tx => {
            tx.executeSql('update
offline_ops set retries = retries + 1
where id = ?;', [id], () => resolve(),
(_, err) => { reject(err); return false;
});
        });
    });
}

export function cacheSet(key, valueObj) {
    return new Promise((resolve, reject)
=> {
        const now = Date.now();
        const value =
JSON.stringify(valueObj || {});
        getDb().transaction(tx => {
            tx.executeSql('insert into
cache_kv (key, value, updated_at) values
(?, ?, ?) on conflict(key) do update set
value=excluded.value,
updated_at=excluded.updated_at;', [key,
value, now], () => resolve(), (_, err) =>
{ reject(err); return false; });
        });
    });
}

export function cacheGet(key) {
    return new Promise((resolve, reject)
=> {
        getDb().transaction(tx => {
            tx.executeSql('select value
from cache_kv where key = ?;', [key], (_,
{ rows }) => {
                if (rows.length > 0) {
                    try {
resolve(JSON.parse(rows.item(0).value));
} catch { resolve(null); }
                } else {
                    resolve(null);
                }
            }, (_, err) => { reject(err);
return false; });
        });
    });
}

export async function
processOfflineQueue(processor) {
    const ops = await popNextOps(20);
    if (!ops || ops.length === 0) return
0;
    const succeeded = [];
    for (const op of ops) {
        try {
            const payload =
JSON.parse(op.payload || '{}');
            const ok = await
processor(op.op_type, payload);
            if (ok)
succeeded.push(op.id);
            else await
bumpRetries(op.id);
        } catch {
            await bumpRetries(op.id);
        }
    }
    if (succeeded.length) await
deleteOps(succeeded);
    return succeeded.length;
}

export function
saveFailedPhotoUpload(uid, photoUri,
actionType, attendanceDate = null, logId
= null) {
    return new Promise((resolve, reject)
=> {
        const now = Date.now();
        getDb().transaction(tx => {
            tx.executeSql(
                'insert into
failed_photo_uploads (uid, photo_uri,
action_type, attendance_date, log_id,
created_at) values (?, ?, ?, ?, ?, ?);',
                [uid, photoUri,
actionType, attendanceDate, logId, now],
                (_, res) =>
resolve(res.insertId),
```

```
                (_, err) => {
reject(err); return false; }
            );
        });
    });
}

export function getFailedPhotoUploads(uid
= null, limit = 50) {
    return new Promise((resolve, reject)
=> {
        getDb().transaction(tx => {
            const query = uid
                ? 'select id, uid,
photo_uri, action_type, attendance_date,
log_id, retries from failed_photo_uploads
where uid = ? order by created_at asc
limit ?;'
                : 'select id, uid,
photo_uri, action_type, attendance_date,
log_id, retries from failed_photo_uploads
order by created_at asc limit ?;';
            const params = uid ? [uid,
limit] : [limit];
            tx.executeSql(query, params,
(_, { rows }) => {
                resolve(rows._array ||
[]);
            }, (_, err) => { reject(err);
return false; });
        });
    });
}

export function
deleteFailedPhotoUpload(id) {
    return new Promise((resolve, reject)
=> {
        getDb().transaction(tx => {
            tx.executeSql('delete from
failed_photo_uploads where id = ?;',
[id], () => resolve(), (_, err) => {
reject(err); return false; });
        });
    });
}

export function
deleteFailedPhotoUploads(ids) {
    if (!ids || ids.length === 0) return
Promise.resolve();
    const placeholders = ids.map(() =>
'?').join(',');
    return new Promise((resolve, reject)
=> {
        getDb().transaction(tx => {
            tx.executeSql(`delete from
failed_photo_uploads where id in
(${placeholders});`, ids, () =>
resolve(), (_, err) => { reject(err);
return false; });
        });
    });
}

export function
incrementFailedPhotoRetries(id) {
    return new Promise((resolve, reject)
=> {
        getDb().transaction(tx => {
            tx.executeSql('update
failed_photo_uploads set retries =
retries + 1 where id = ?;', [id], () =>
resolve(), (_, err) => { reject(err);
return false; });
        });
    });
}
```

APPENDIX D. EVIDENCES

EVIDENCES

2025 1st 2025 2nd 2025 3rd 2025 4th 2025 BRGY. 2025 Sanitary 2025

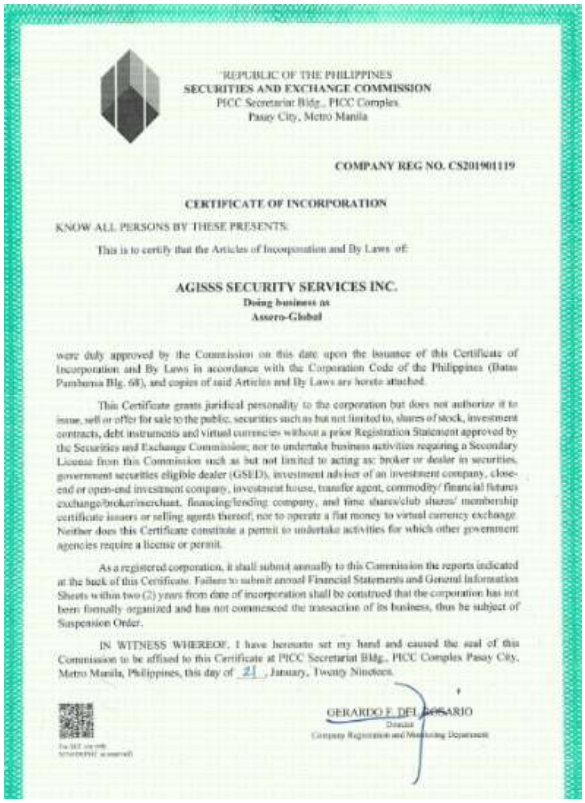
CITY OF TAGUIG
BUSINESS PERMIT

NAME OF CORPORATION/ENTITY: AGISSS SECURITY SERVICES INC
TRADE NAME: ASSERO GLOBAL
BUSINESS LOCATION: NO. 67 UNIT 808 AMBER PLACE BUILDING BAYANI ROAD WESTERN BICUTAN CITY OF TAGUIG
BUSINESS OWNER: AGISSS SECURITY SERVICES INC
OWNERSHIP TYPE: CORPORATION
STATUS: RENEWAL
NATURE OF BUSINESS: SECURITY SERVICES
LCN NO.: 20-028483
AREA (SQM): 28
DATE ISSUED: 01/01/2025
VALID UNTIL: 12/31/2025
O.R. NUMBER: A-08475476
APPROVED BY: MARIA THERESA S. VELOSO, HEAD, BUSINESS PERMITS AND LICENSING OFFICE

TAX DESCRIPTION	TAX BASE	AMOUNT DUE
CONTRACTS/SECURITY SERVICES	[illegible]	[illegible]
MAYORS FEE - SECURITY SERVICES		[illegible]
ENVIRONMENTAL IMPACT FEE		[illegible]
SANITARY INSPECTION FEE		[illegible]
BUILDING INSPECTION FEE		[illegible]
ELECTRICAL INSPECTION FEE		[illegible]
PLUMBING INSPECTION FEE		[illegible]
MECHANICAL INSPECTION FEE		[illegible]
MEDICAL/HEALTH FEE		[illegible]
FIRE PERMIT FEE		[illegible]
BUSINESS PLATE (STICKER)		[illegible]
FORM FEE		[illegible]
SIGNBOARD FEE		[illegible]
FIRE CODE RA 9514		[illegible]
CEDULA CORPORATION		[illegible]
BARANGAY FEE		[illegible]
TOTAL		207,820.25

This permit may be revoked for violation of applicable laws or regulations or for non-payment of taxes and fees. Please notify BPLO at (02) 7795-8874 and (02) 7795-9999 loc. 104 or bplo@taguig.gov.ph for any concerns about this permit. | ERASURES AND/OR ALTERATIONS WILL INVALIDATE THIS PERMIT

www.taguig.gov.ph /taguigcity @taguigcity @iloveTaguig!



REPUBLIC OF THE PHILIPPINES
SECURITIES AND EXCHANGE COMMISSION
PICC Secretariat Bldg., PICC Complex
Pasay City, Metro Manila

COMPANY REG NO. CS201901119

CERTIFICATE OF INCORPORATION

KNOW ALL PERSONS BY THESE PRESENTS:

This is to certify that the Articles of Incorporation and By Laws of:

AGISSS SECURITY SERVICES INC.
Doing business as
Assero-Global

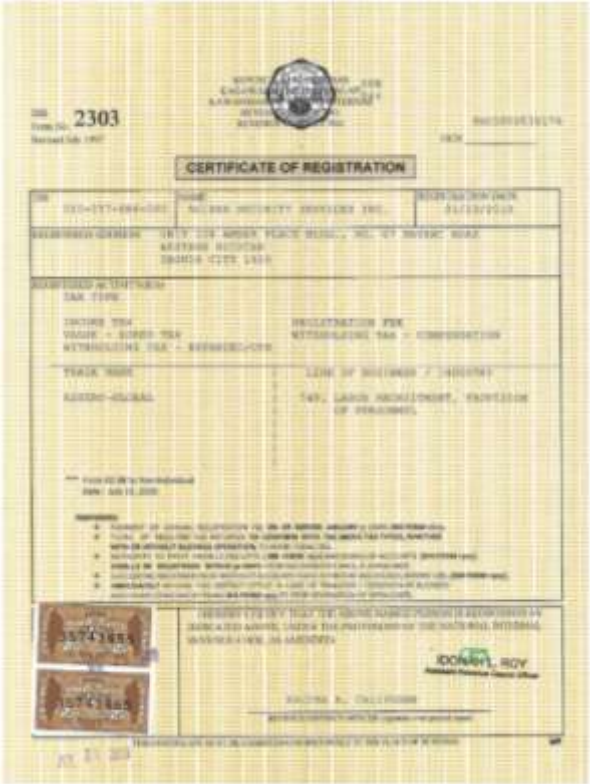
were duly approved by the Commission on this date upon the Issuance of this Certificate of Incorporation and By Laws in accordance with the Corporation Code of the Philippines (Batas Pambansa Blg. 68), and copies of said Articles and By Laws are hereto attached.

This Certificate grants juridical personality to the corporation but does not authorize it to issue, sell or offer for sale to the public, securities such as but not limited to, shares of stock, investment contracts, debt instruments and virtual currencies without a prior Registration Statement approved by the Securities and Exchange Commission; nor to undertake business activities requiring a Secondary License from this Commission such as but not limited to acting as: broker or dealer in securities, government securities eligible dealer (GSED), investment adviser of an investment company, close-end or open-end investment company, investment house, transfer agent, commodity/ financial futures exchange/broker/merchant, financing/lending company, and time shares/club shares/ membership certificate issuers or selling agents thereof; nor to operate a fiat money to virtual currency exchange. Neither does this Certificate constitute a permit to undertake activities for which other government agencies require a license or permit.

As a registered corporation, it shall submit annually to this Commission the reports indicated at the back of this Certificate. Failure to submit annual Financial Statements and General Information Sheets within two (2) years from date of incorporation shall be construed that the corporation has not been formally organized and has not commenced the transaction of its business, thus be subject of Suspension Order.

IN WITNESS WHEREOF, I have hereunto set my hand and caused the seal of this Commission to be affixed to this Certificate at PICC Secretariat Bldg., PICC Complex Pasay City, Metro Manila, Philippines, this day of 21, January, Twenty Nineteen.

GERARDO F. DEL ROSARIO
Director
Company Registration and Monitoring Department



2303

CERTIFICATE OF REGISTRATION

SAMPLES OF BUDGET PROPOSAL FOR CLIENTS

PROVISION OF SECURITY GUARDS - METRO MANILA (DOLE RATE)

		12 HOURS SHIFT			
		Dayshift		Nightshift	
Daily Wage (W)		PHP	645.00	PHP	645.00
No. of Work Days (Annual Factor-AF)		PHP	394.40	PHP	394.40
	Monthly Pay ((W x AF/12)	PHP	21,199.00	PHP	21,199.00
	13th Month Pay (W x 365/12/12)	PHP	1,634.90	PHP	1,634.90
	5 Days Incentive Leave (W x 5/12)	PHP	268.75	PHP	268.75
	Uniform Allowance	PHP	100.00	PHP	100.00
TOTAL AMOUNT BEFORE OVERTIME		PHP	23,202.65	PHP	23,202.65
	Overtime Pay	PHP	13,496.09	PHP	13,496.09
	Night Differential (10% of Monthly Pay)	PHP	-	PHP	2,119.90
A- TOTAL AMOUNT DIRECTLY TO GUARD		PHP	36,698.74	PHP	38,818.64
	Retirement Pay (W x 22.5 /12)	PHP	1,209.38	PHP	1,209.38
	SSS (Employer's Share)	PHP	2,072.50	PHP	2,072.50
	Philhealth	PHP	529.98	PHP	529.98
	State Insurance/ECC (Employer's Share)	PHP	30.00	PHP	30.00
	Pag-ibig Fund	PHP	200.00	PHP	200.00
B. Total Amount Directly to Government		PHP	4,041.85	PHP	4,041.85
	SUB-TOTAL (A+B)	PHP	40,740.59	PHP	42,860.49
	Admin Overhead (DOLE D.O. 150-16)	PHP	8,148.12	PHP	8,572.10
TOTAL		PHP	48,888.70	PHP	51,432.58
	VAT (10% of the Admin Fee)	PHP	814.81	PHP	857.21
	Add: Additional Allowance	PHP	-	PHP	-
CONTRACT PRICE PER GUARD PER MONTH inclusive of VAT		**PHP**	**49,703.51**	**PHP**	**52,289.79**

TOTAL CONTRACT PRICE PER GUARD PER MONTH (DAYSHIFT)	**PHP**	**49,703.51**
TOTAL CONTRACT PRICE PER GUARD PER MONTH (NIGHTSHIFT)	**PHP**	**52,289.79**

PROVISION OF SECURITY GUARDS - ZAMBOANGA CITY (DOLE Rate)

		12 HOURS SHIFT			
		Dayshift		Nightshift	
Daily Wage (W)		PHP	443.00	PHP	443.00
No. of Work Days (Annual Factor-AF)		PHP	394.40	PHP	394.40
	Monthly Pay ((W x AF/12)	PHP	14,559.93	PHP	14,559.93
	13th Month Pay (W x 365/12/12)	PHP	1,122.88	PHP	1,122.88
	5 Days Incentive Leave (W x 5/12)	PHP	184.58	PHP	184.58
	Uniform Allowance	PHP	100.00	PHP	100.00
TOTAL AMOUNT BEFORE OVERTIME		PHP	15,967.40	PHP	15,967.40
	Overtime Pay	PHP	8,616.89	PHP	8,616.89
	Night Differential (10% of Monthly Pay)	PHP	728.00	PHP	728.00
A- TOTAL AMOUNT DIRECTLY TO GUARD		PHP	25,312.29	PHP	25,312.29
	Retirement Pay (W x 22.5 /12)	PHP	830.63	PHP	830.63
	SSS Premium (Jan 2023)	PHP	1,900.00	PHP	1,900.00
	SSS WISP	PHP	285.00	PHP	285.00
	Philhealth	PHP	364.00	PHP	364.00
	State Insurance/ECC (Employer's Share)	PHP	30.00	PHP	30.00
	Pag-ibig Fund	PHP	200.00	PHP	200.00
B. Total Amount Directly to Government		PHP	3,609.62	PHP	3,609.62
	SUB-TOTAL (A+B)	PHP	28,921.91	PHP	28,921.91
	Admin Overhead (DOLE DO 150-16)	PHP	5,784.38	PHP	5,784.38
TOTAL	**VAT (10% of the Admin Fee)**	PHP	2,892.19	PHP	2,892.19
	VAT (12% of the Admin Fee)	PHP	694.13	PHP	694.13
	Add: Additional Allowance	PHP	-	PHP	-
CONTRACT PRICE PER GUARD PER MONTH inclusive of VAT		**PHP**	**3,586.32**	**PHP**	**3,586.32**

TOTAL CONTRACT PRICE PER GUARD PER MONTH (DAYSHIFT)	**PHP**	**3,586.32**
TOTAL CONTRACT PRICE PER GUARD PER MONTH (NIGHTSHIFT)	**PHP**	**3,586.32**

SAMPLE OF DTR

AGISSS SECURITY SERVICES INC.
U108 Amber Place, Bayani Road, Fort Bonifacio Brgy. Western Bicutan Taguig City
Tel. No. (632)8511-1658

TIMESHEET SUMMARY

Date Covered: JANUARY 16 TO 31, 2024
Client: DMCI DARONG FUEL DEPOT PROJECT
Location: Brgy. Darong Purok Rambutan, Sta. Cruz Davao del Sur

NO.	EMPLOYEE NUMBER	EMPLOYEE NAME	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15		TOTAL HOURS	No. of Days	Regular Hours	OT Hours	Night Duty	Reg Hol	Spec Hol	SIGNATURE
			16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31								
1	102045	Andagan, Arthur G.	12	12	12	12	12	12		12	12	12	12	12	12		12	12	168	14			14			
2	101976	Jagape, Alfredo Sr. C.	12		12	12	12	12	12	12		12	12	12	12	12	12		156	13						
3	102075	Opong, Regiemel M.		12	12	12	12	12	12		12	12	12	12	12	12		12	156	13			13			
4	102114	Rellanos, Kenneth V.	12	12		12	12	12	12	12	12		12	12	12	12	12	12	168	14			14			
5	102248	Rocho, Rodrigo Sr. C.	12	12	12	12	12	12		12	12	12	12	12	12		12	12	168	14						
7	101921	Senoy, Ruel A.		12	12	12	12	12	12		12	12	12	12	12	12		12	156	13			10			
8	101739	Largado, Kelvin G.	12	12	12	12	12	12		12	12	12	12	12	12		12	12	168	14						
9	102297	Abejuela , Jerrol H.		12	12	12	12	12	12		12	12	12	12	12	12		12	156	13						
10	202439	Gilbolingo, Sulpicio Jr. L.	12	12	12	12		12	12	12	12	12	12		12	12	12	12	168	14			14			
11																			0	0						
12																			0	0						
13																			0	0						
																			0	0						
																			0	0						
TOTAL HOURS			84	108	108	120	96	120	84	84	108	108	120	96	120	84	84	108	1,632	136						
TOTAL PERSONNEL			7	9	9	10	8	10	7	7	9	9	10	8	10	7	7	9								

REMARKS:

Prepared by:

SO SENOY, RUEL A.
Signature over Printed Name

Approved By

Signature over Printed Name and Designation

SAMPLE PAYSLIP

AGISSS *Security Services Inc.*

Employee's Name:	[illegible]	
Employee Number:	103101	
Pay Period:	**March 01 to 15, 2025**	
Email Address:	manuelabain@gmail.com	
EARNINGS		***AMOUNT***
Regular Hours	9,150.00	
Gross Payment		**9,150.00**
DEDUCTIONS		***AMOUNT***
Cash Bond	200.00	
SSS Contribution	250.00	
Philhealth Contribution	250.00	
HDMF Contribution	200.00	
Total Deductions		**900.00**
NET PAYMENT		***8,250.00***

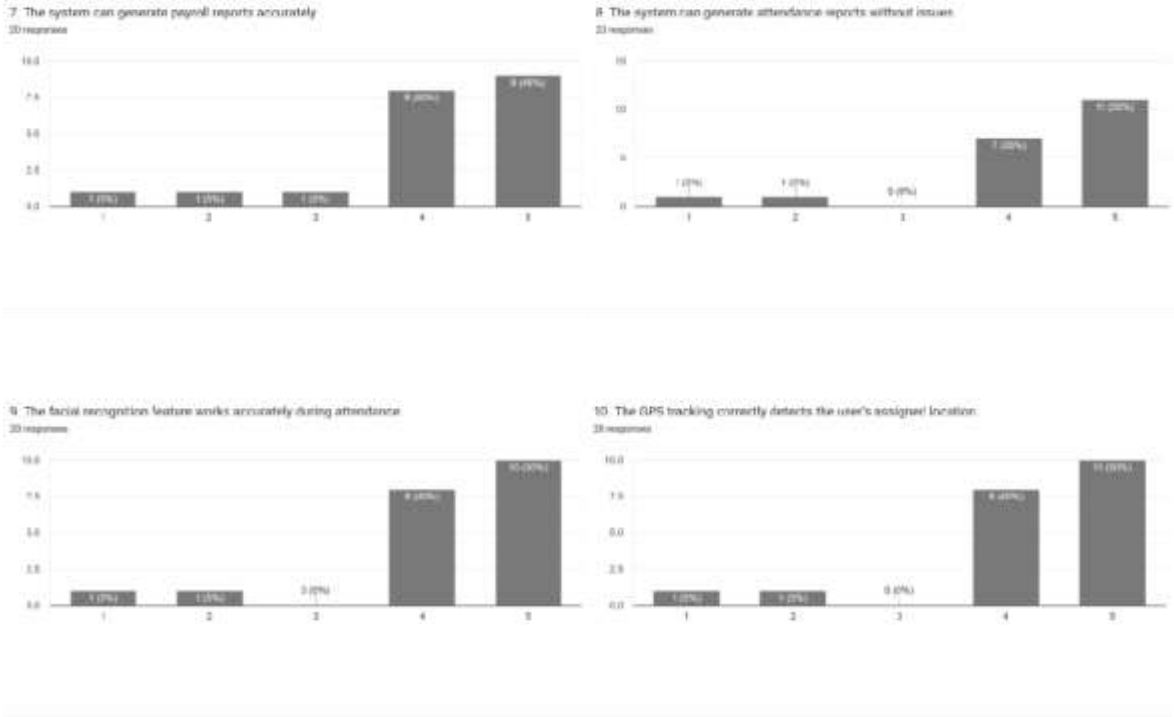
AGISSS *Strategic Security Services Inc.*

Employee's Name:	[illegible]	
Pay Period:	**March 01 to 15, 2025**	
OTHER EARNINGS	***AMOUNT***	
NS Allowance	741.00	
Allowance	4,743.75	
Total		***5,484.75***

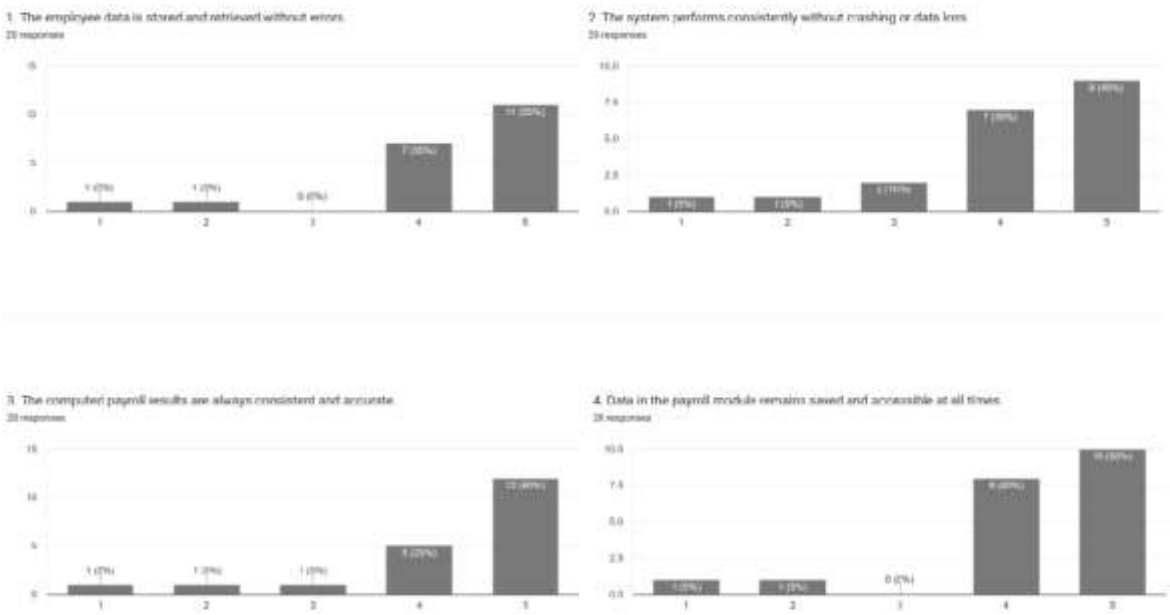
Days Work	15.00
Total Amt	***13,734.75***

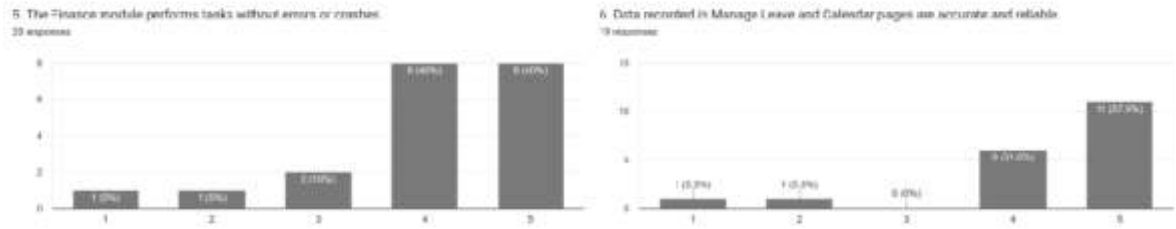
APPENDIX E. EVALUATION TOOLS/TEST DOCUMENTS

| Time Stamp | | First Name | Last Name | It always appears accurate when person worldwide need access | | |

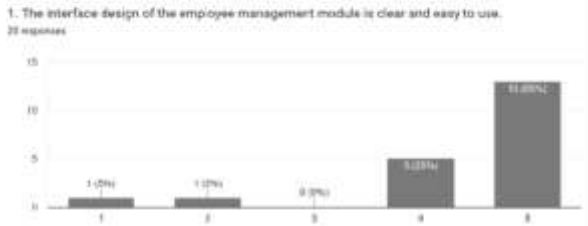


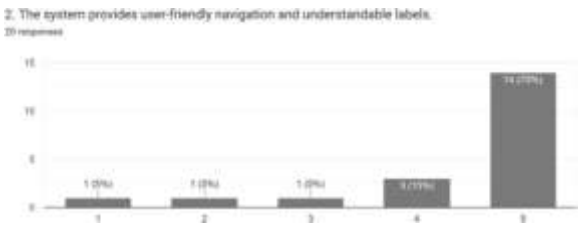
Survey Results for Functionality

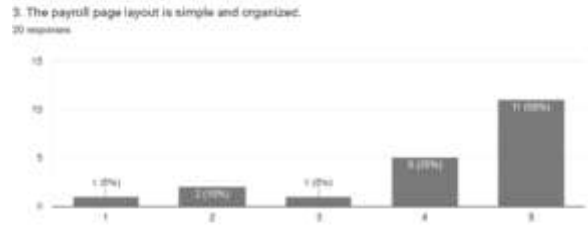


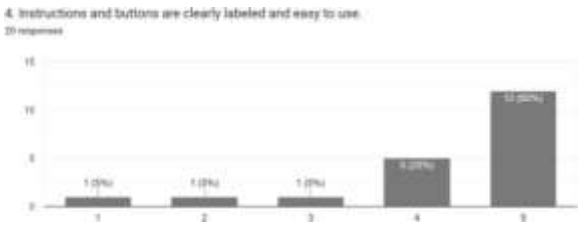


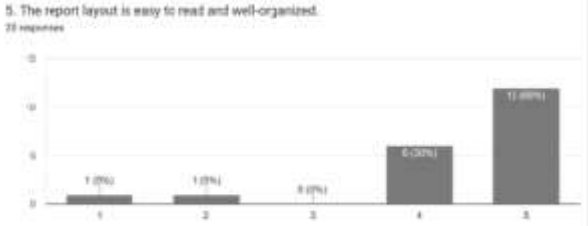
Survey for Reliability

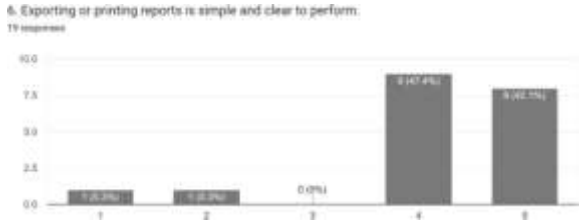


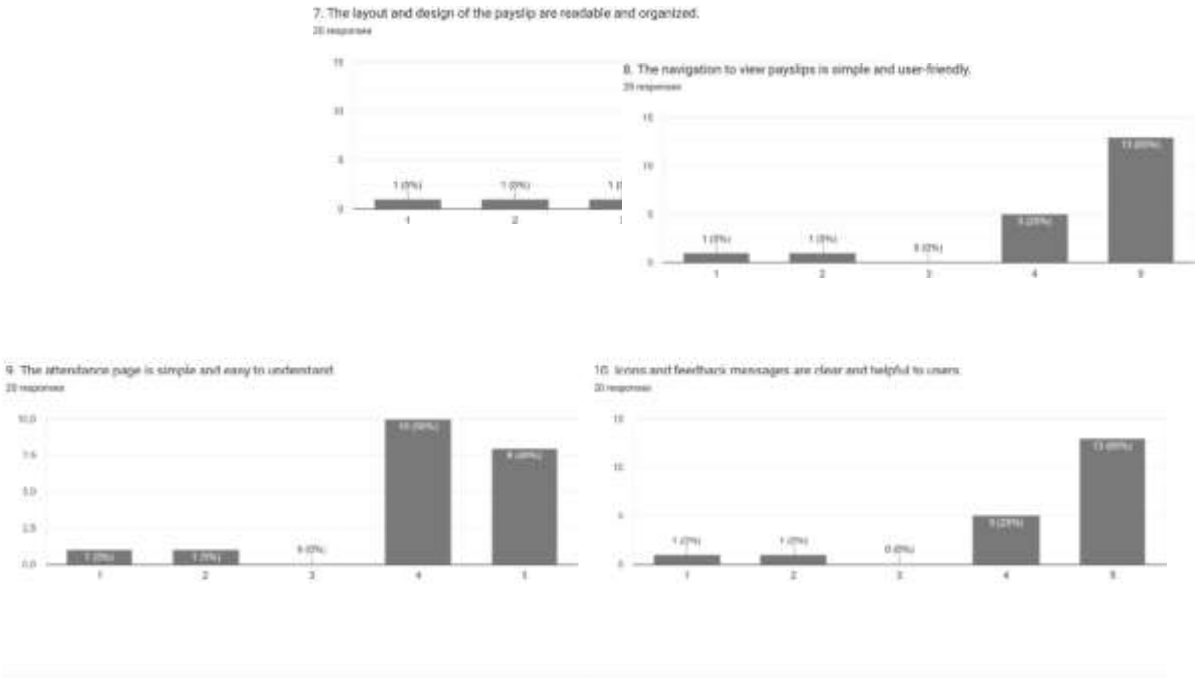




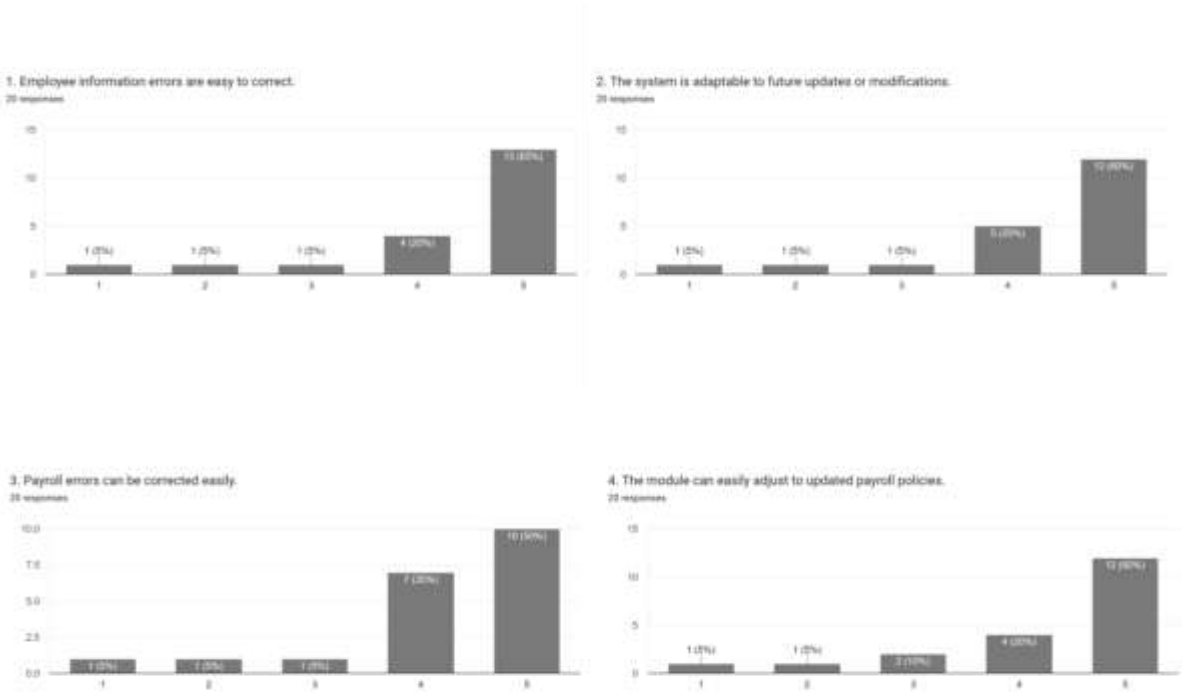




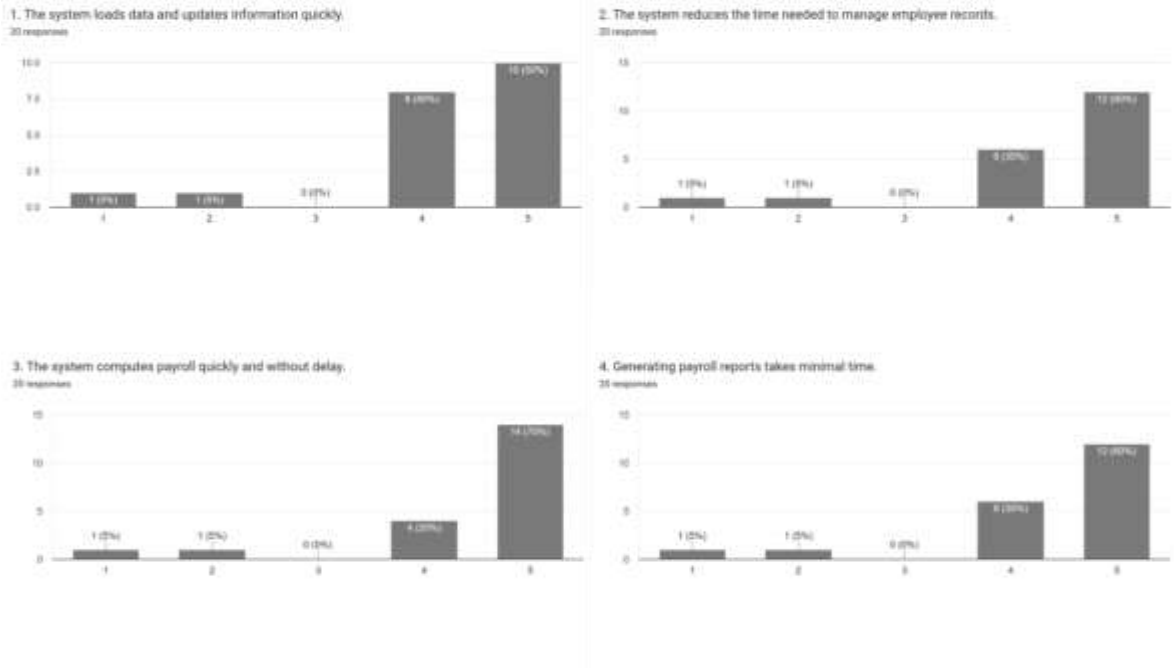




Survey Results for Usability



Survey Results for Efficiency



Survey Results for Maintainability

APPENDIX F.SAMPLE INPUT/OUTPUT/REPORTS

SAMPLE INPUT/OUTPUT/REPORTS

SAMPLE GENERATED REPORT/PAYSLIP

Assero Global - Strategic Security Solutions

PAYSLIP

Pay Period: 2025-10-16 to 2025-10-31

EMPLOYEE INFORMATION

Employee ID: 2025100054
Name: Juhorman Masukat Kuday
Client: Jollibee - Taguig City (Night shift)
Client Code: JTC20251292

PERIOD DETAILS

Period Start: 2025-10-16
Period End: 2025-10-31
Working Days: 15
Status: Issued

EARNINGS

Basic Salary (128 hrs × ₱71.25) ₱9,120.00
Overtime Pay (60 hrs × ₱89.06) ₱5,343.75
Night Differential ₱855.00
Uniform Allowance ₱52.60

DEDUCTIONS

SSS Total ₱462.50
PhilHealth ₱228.00
Pag-IBIG ₱100.00
Cash Bond ₱200.00

Total Deductions **₱990.50**

GROSS PAY **₱15,371.35**

NET PAY **₱14,380.85**

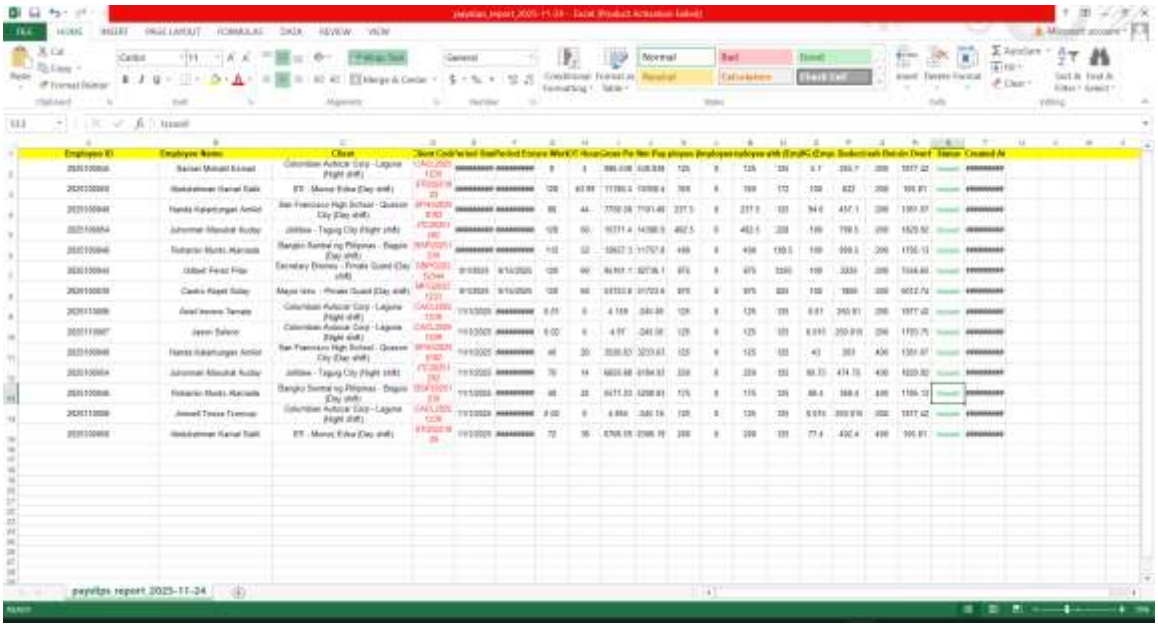
DATES WORKED

2025-10-16	2025-10-17	2025-10-18	2025-10-19	2025-10-20	2025-10-21	2025-10-22	2025-10-23
2025-10-24	2025-10-25	2025-10-26	2025-10-27	2025-10-28	2025-10-29	2025-10-30	

This is a system-generated payslip. No signature required.

Generated on 2025-11-25

SAMPLE EXPORTED CSV FILE



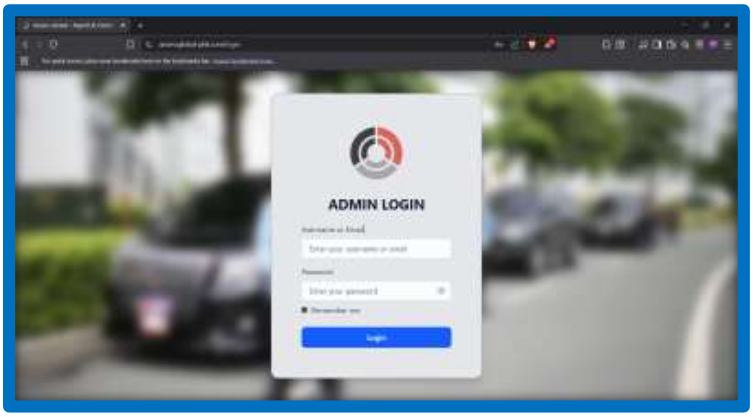
APPENDIX G. USER'S GUIDE MANUAL

APPENDIX G. USER GUIDE MANUAL

Assero Global Inc.'s Web Application

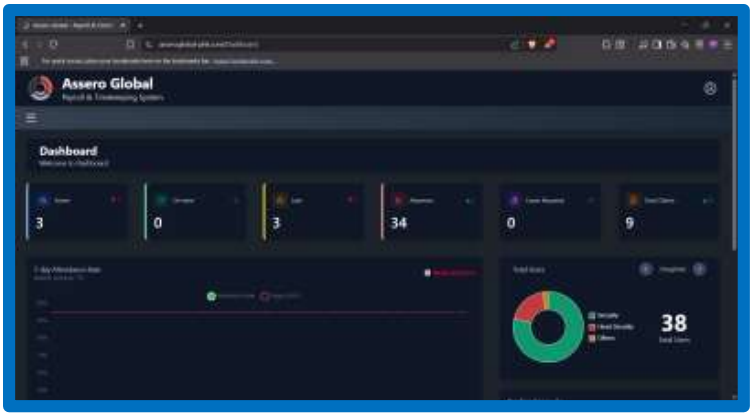
Open your browser (Chrome, Edge, Firefox).

- In the address bar, type the system URL (https://asseroglobalphil.com/asserologin).
- Enter Your Account Credentials and press Login.



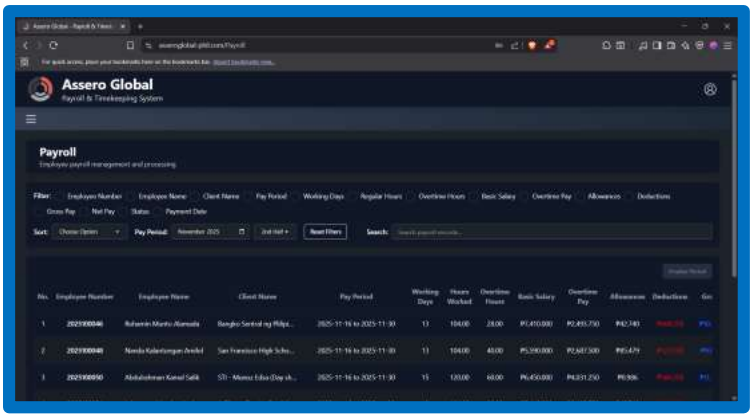
You will be directed to the Dashboard where you can view statistics such as;

- Total employees
- Attenadnce summaries
- Pending leave requests
- Payroll overview



- Click any card or statistic to navigate to its corresponding module.
- Click on the Navigation Drawer.

➢ **Navigate to Finance → Payroll**



➢ Select:

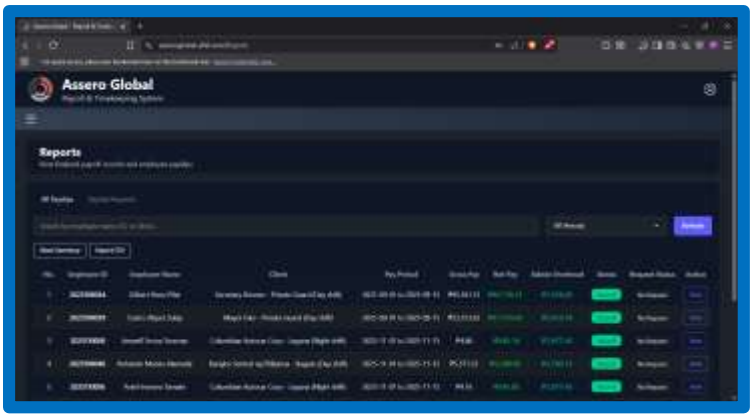
- Employee
- Cutoff period
- Month

➢ Review auto-generated computation:

- Regular hours
- Overtime
- Night differential
- Allowances
- Deductions (SSS, PhilHealth, Pag-IBIG)

After a 15-day payroll, you may click Finalize Period which will be directed to Reports.

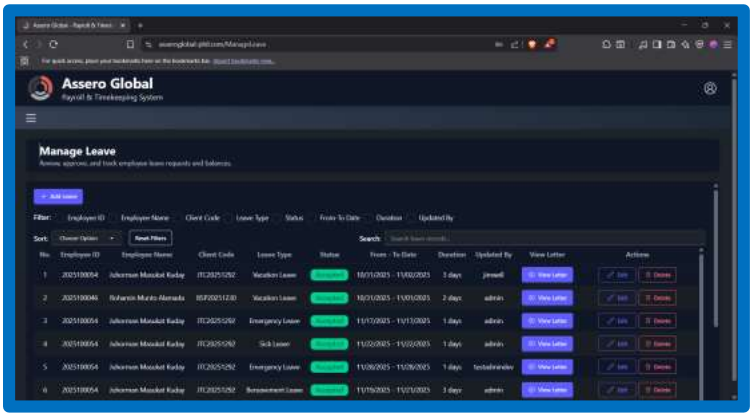
➢ **Navigate to Finance → Reports**



➢ In this module you can;

- View and Print Employee Payslips
- Approve Payslip Requests

➢ **Go to Finance → Leave Management**



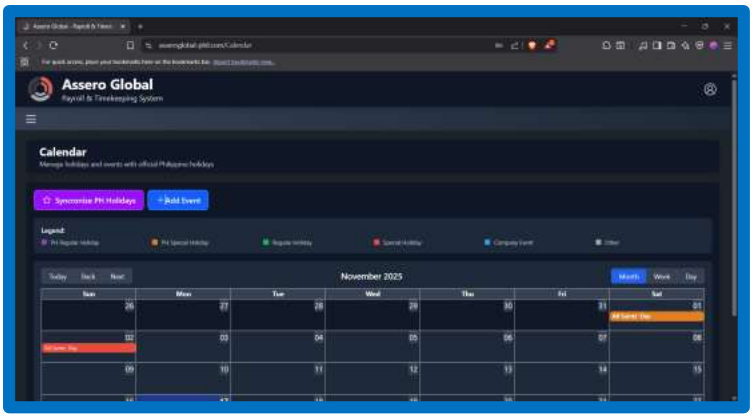
➢ View all pending or approved leave requests.

➢ Select a request and click:

- Approve
- Reject

Approved leave is automatically reflected in the payroll.

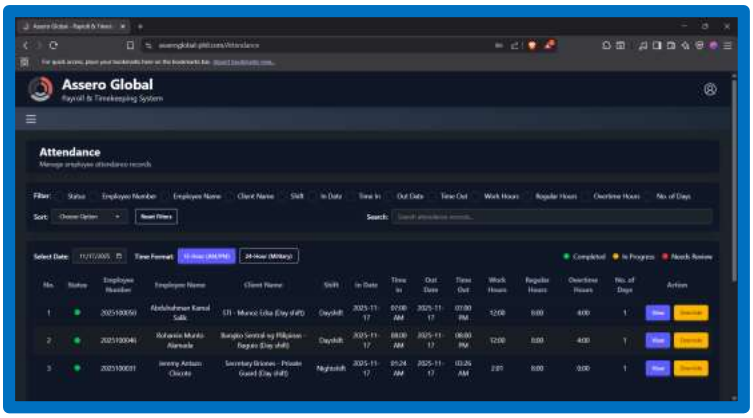
Go to Finance → Calendar



- Navigate using:
  - Month
  - Week
  - Day view

- Click any date to open details such as:
  - Payroll cutoff
  - Events
  - Leave entries

➢ **Open Attendance from the sidebar.**



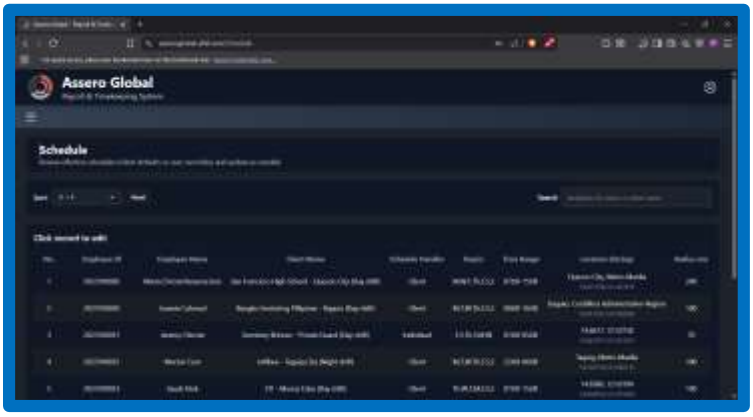
➢ Search by:

- Employee name
- Date and Location

➢ View:

- Time-in and Time-out
- GPS-verified coordinates
- Facial recognition images

➢ **Navigate to Schedule.**



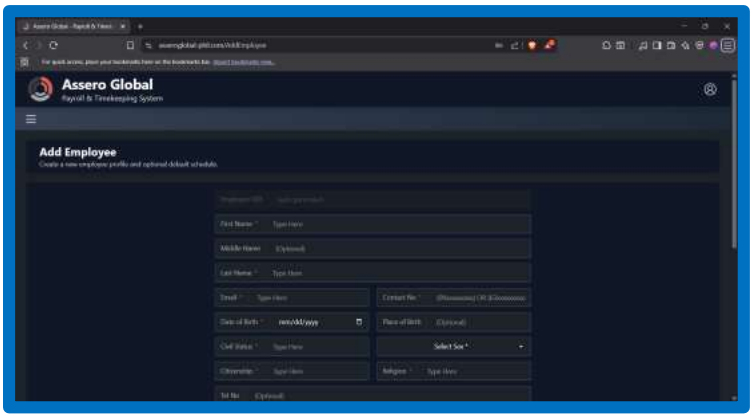
➢ Select Add Schedule and Input the following:

- Employee, Client & Location
- Time-in / Time-out
- Rest days and ***Click Save.***

The schedule automatically synchronizes to the mobile app.

How to Add an Employee?

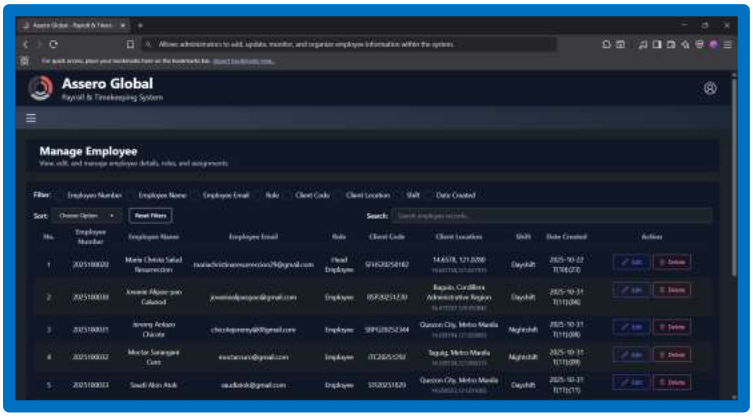
- **Go to User Management → Add Employee.**



- Encode required information:
  - Full name
  - Contact number
  - Address
  - Rate
  - Schedule
- Upload ID photos or required files and ***Click Save.***

How to Edit Employee Information?

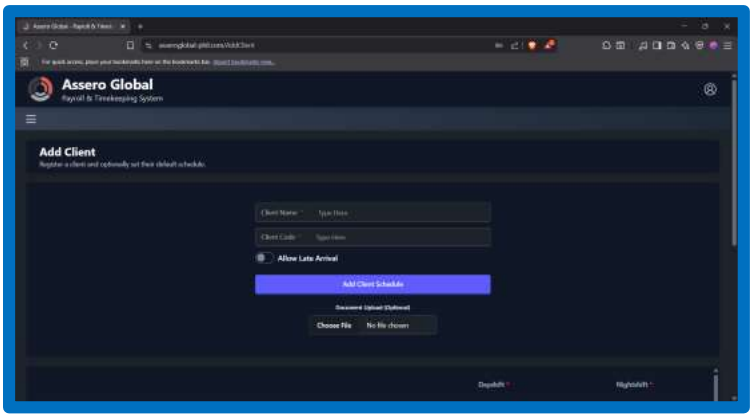
- **Go to Manage Employee**



- Search for employee names.
- Click Edit
- Update the details and ***Save Changes.***

How to Add a Client?

➢ **Go to User Management → Add Client.**

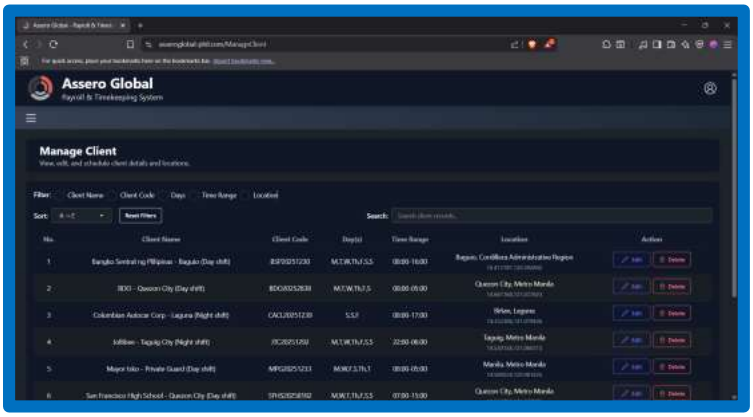


➢ Fill in:

- Client Name
- Client Code
- Contact Person
- Add "Provisions of Security Guard" Computations of Payroll and ***Click Save.***

How to Manage Client Information?

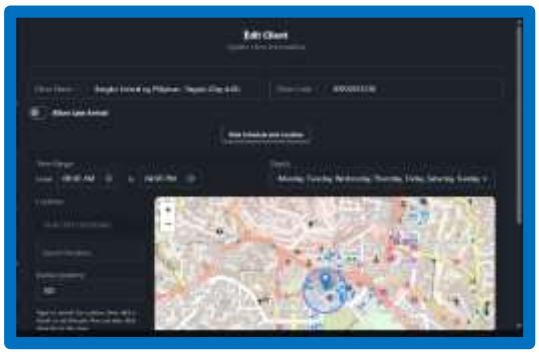
➢ **Go to Manage Client.**



➢ Edit or delete client information as needed.

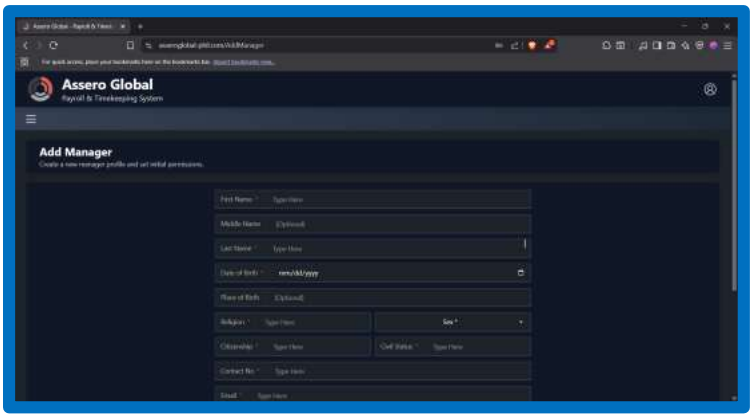
➢ Edit information such as;

- Client Name and Code
- Time of Duty and Location
- Rates



How to Add Managers of the Web Application?

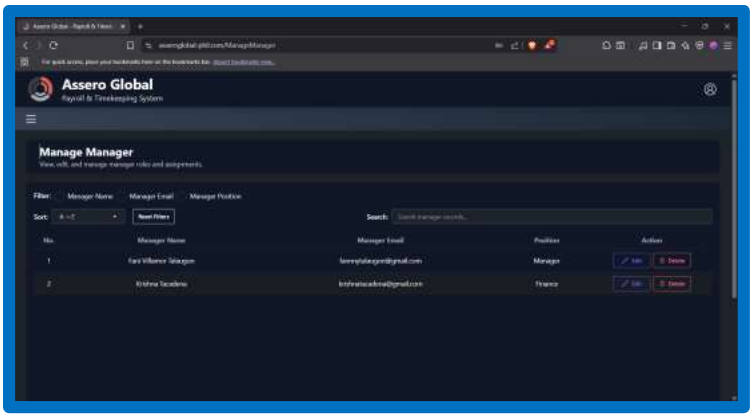
- **Go to User Management → Add Manager.**



- Encode required details.
- Assign permissions.
- Get their Credentials to access the Web.
- Save.

How to Manage Managers?

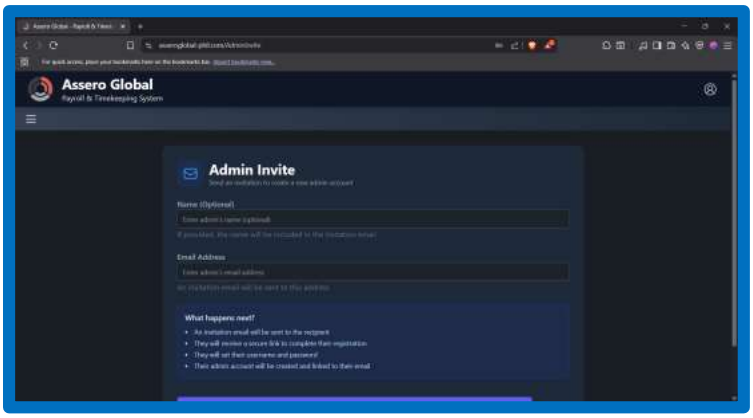
- **Go to Manage Manager.**



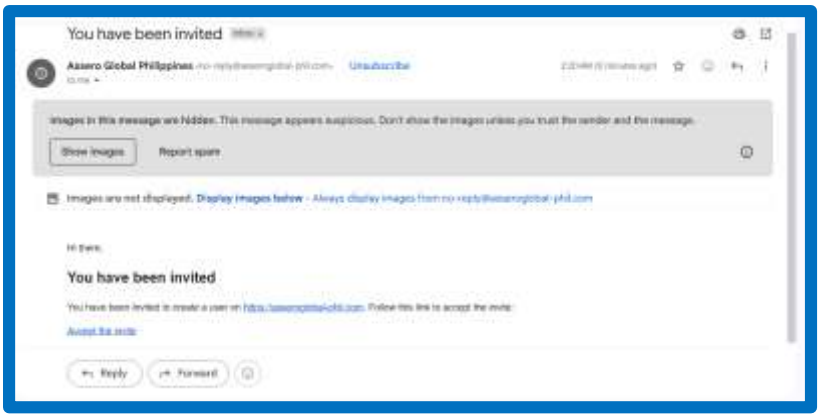
- Update or deactivate manager credentials.

How to Invite an Administrator?

- **Go to Admin Invite.**

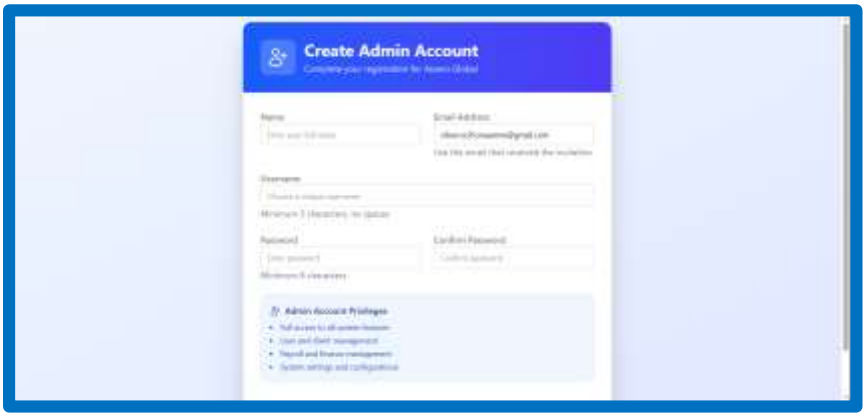


- Input the email address of the new admin and click Send Invite.
- The invited user receives an email with:

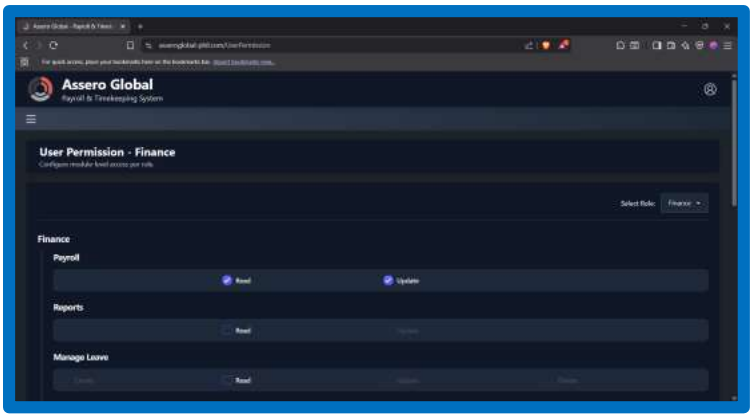


  - Registration link

- Create Access Credentials



➢ **Go to Settings → User Permissions.**



➢ Select user role:

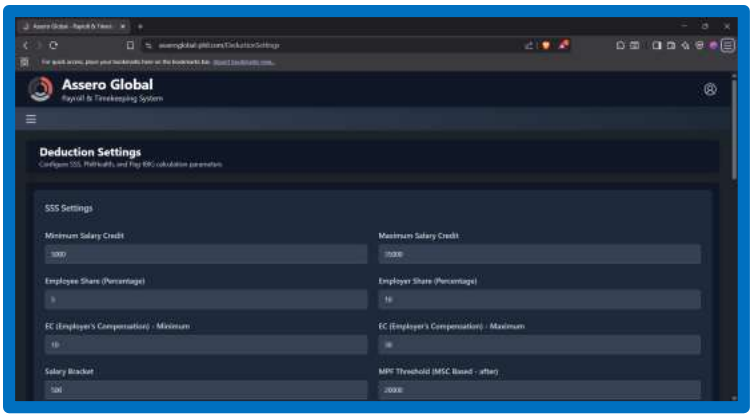
- Admin
- Manager
- Finance

➢ Select user role:

➢ Toggle which modules the role can access.

➢ Click Save Permissions.

➢ **Go to Settings → Adjust Deductions.**

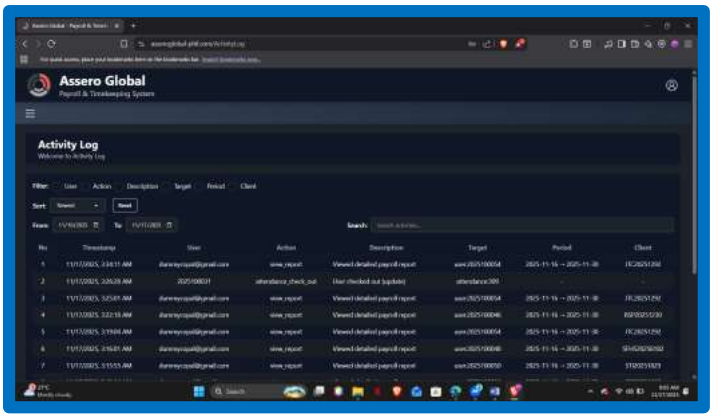


➢ Input:

- SSS employee + employer rate
- PhilHealth contribution brackets
- Pag-IBIG contribution

➢ Click Update.

➢ **Open Settings → Activity Log.**

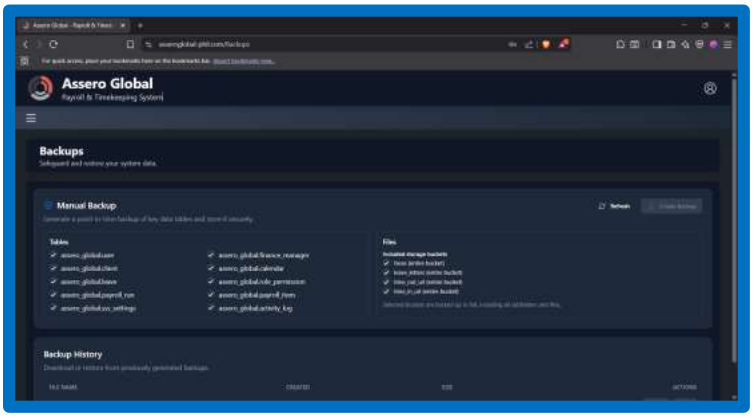


➢ View logs such as:

- Login attempts
- Data changes
- Payroll generation

➢ Filter logs by user or date.

➢ **Go to Settings → Backups.**



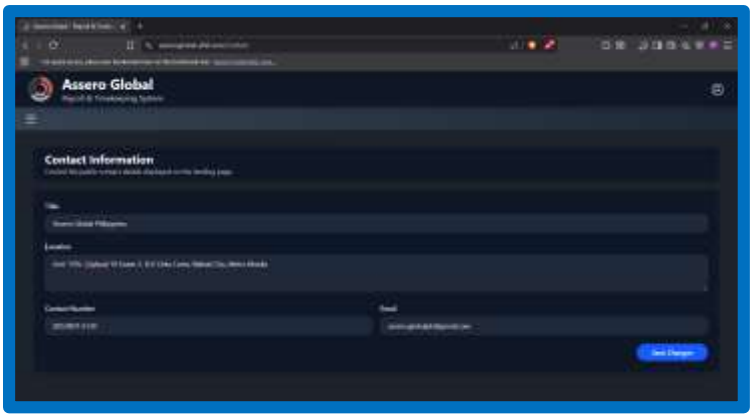
➢ Click Create Backup.

➢ Download the backup file for offline security.

➢ To restore:

- Upload backup file and Click Restore.

➢ **Open Settings → Contacts.**



➢ Edit Title, Location, Contact Number and Email. All will be visible on the Web Application's Landing Page.

Assero Global Inc.'s Mobile Application

➢ **Open the mobile app.**

➢ Enter:

- Username
- Password

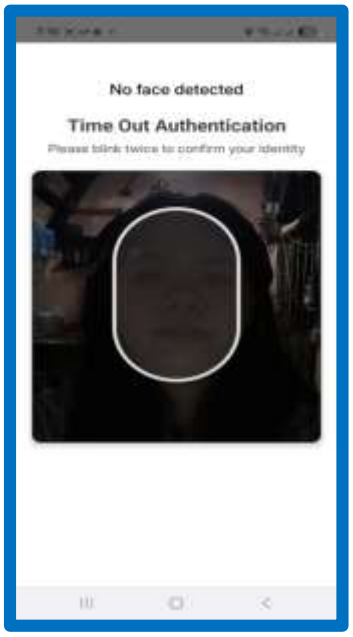
➢ **Tap Login**

You will be directed to the Home Page.

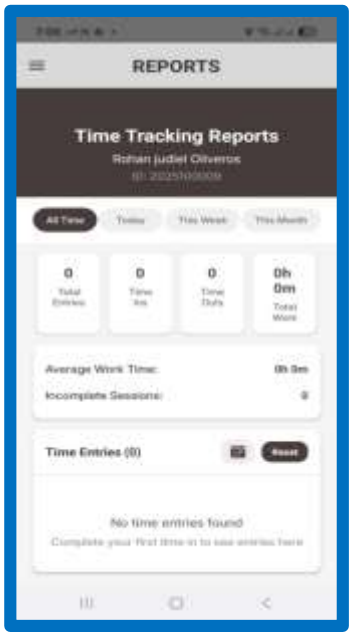
➢ Features:

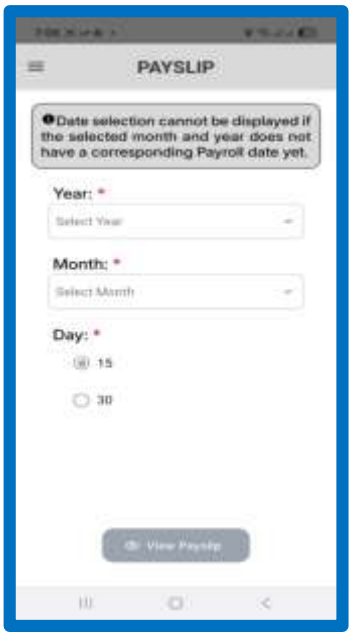
- Time-in and Time-out buttons
- Navigation menu
- GPS indicator

After Clicking Time in or out you will be directed to the Facial Recognition Module where you should;



- ➢ Hold the phone up to the face.
- ➢ Ensure:
  - Good lighting
  - Face within frame
- ➢ System automatically:
  - Matches face
  - Confirms GPS location
  - Records attendance
- ➢ **Tap Reports**
- ➢ Choose:
  - Date range
  - Specific month
- ➢ View:
  - Total work hours
  - Number of entries
  - Time-in/out logs





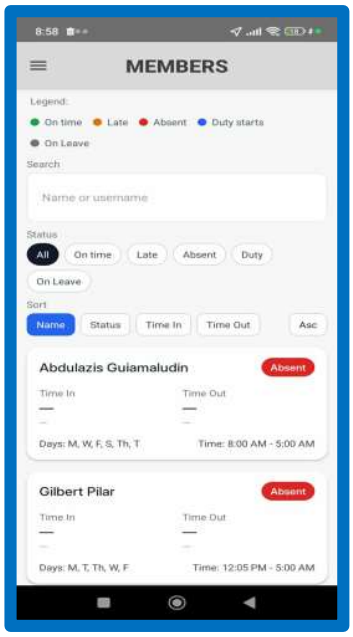
- ➢ **Tap Payslip**
- ➢ Select:
  - Year
  - Month
  - Cutoff
- ➢ Payslip automatically loads showing:
  - Gross pay
  - Deductions and Allowances
  - Net pay

- **Tap Settings**
  - Allows account customization.
- Options
  - Edit basic information
  - Change password
  - Notification settings
  - About the app

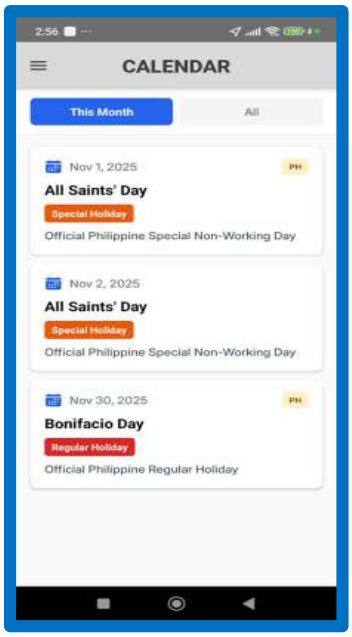
- **Tap Member**

Allows Head Employees to monitor their members

- Tap a security guard to view:
  - Attendance records
  - Recent time-ins/outs
  - Total hours worked



- **Tap Calendar**
- View:
  - Holidays
  - Company Events



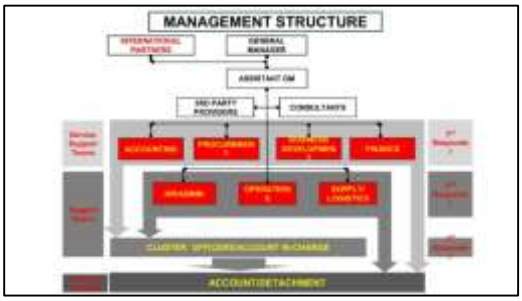
APPENDIX H. EXPENSES

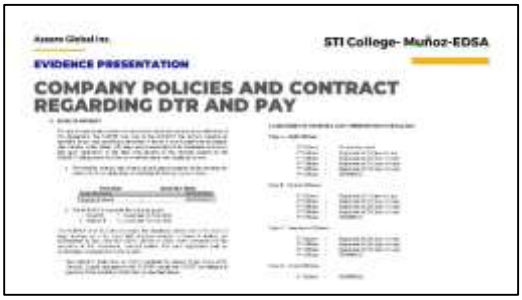
EXPENSES

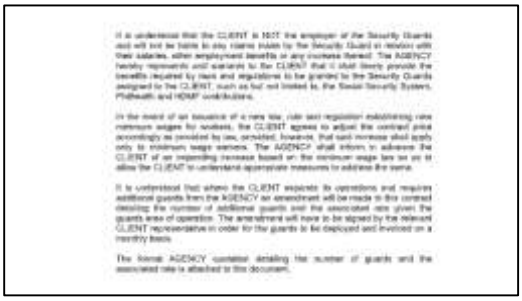
Quantity	Specifics	Actual Cost
(4) Reams	Bond paper	908.00 PHP
	Printing	800.00 PHP
4 PCS.	Blue Clear book	200.00 PHP
	Food	400.00 PHP
	Domain	671.63 PHP
TOTAL		2,979.63

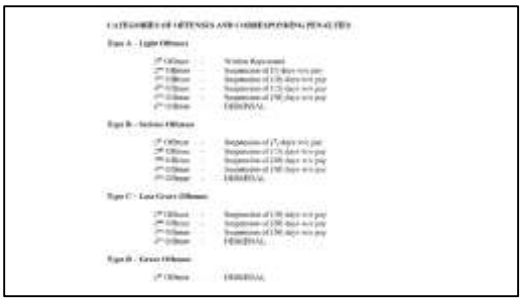
APPENDIX I. POWERPOINT PRESENTATION

POWERPOINT PRESENTATION

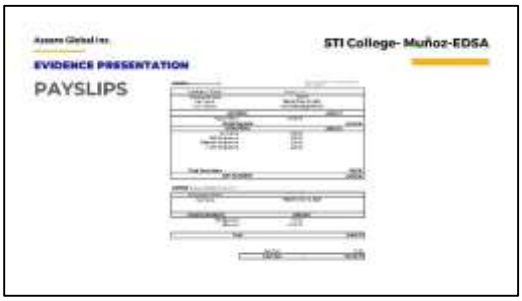


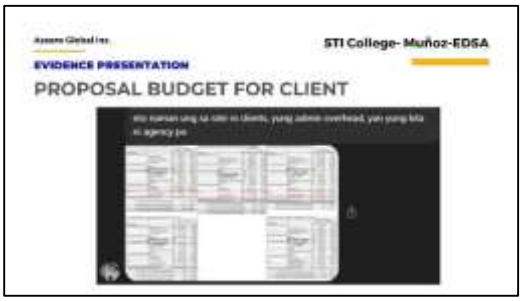


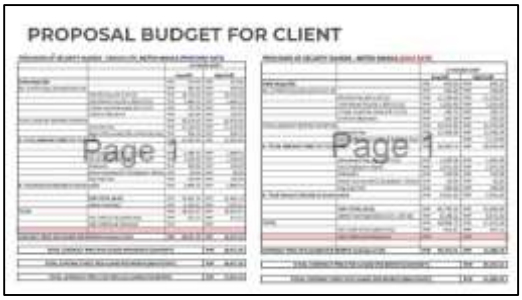


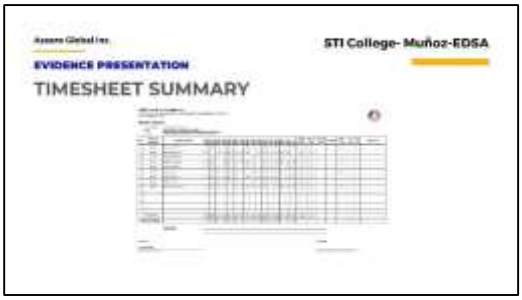


Assero Global Inc.
STI College- Muñoz-EDSA
EVIDENCE PRESENTATION
DESKTOP SPECIFICATIONS AT ASSERO

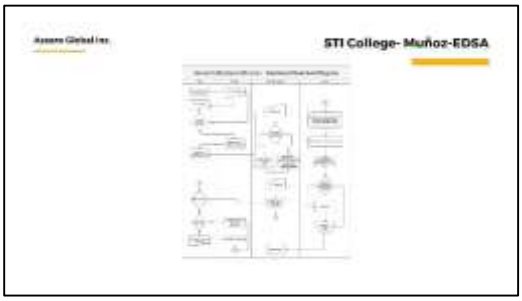

Assero Global Inc.
STI College- Muñoz-EDSA
EVIDENCE PRESENTATION
PAYSLIPS

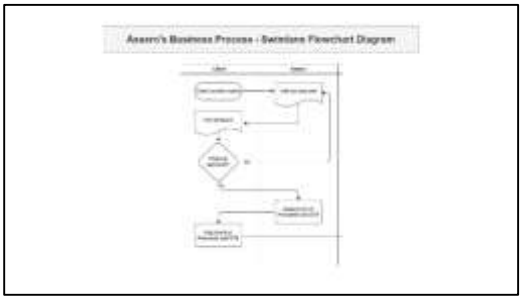

Assero Global Inc.
STI College- Muñoz-EDSA
EVIDENCE PRESENTATION
PROPOSAL BUDGET FOR CLIENT

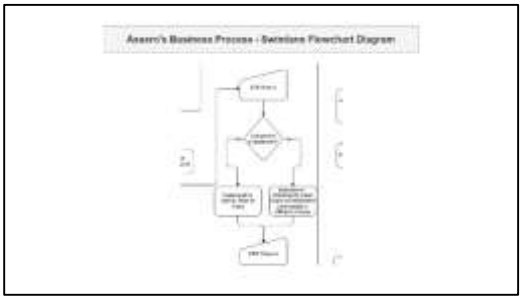

PROPOSAL BUDGET FOR CLIENT
Page 1
Page 1


Assero Global Inc.
STI College- Muñoz-EDSA
EVIDENCE PRESENTATION
TIMESHEET SUMMARY

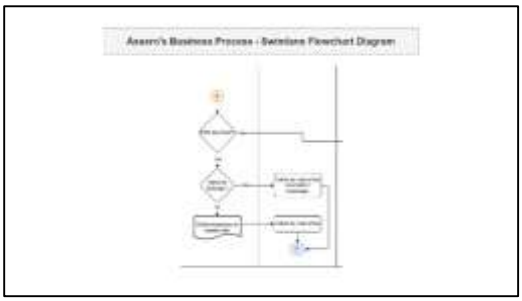
Assero Global Inc.
STI College- Muñoz-EDSA
THE PROCESS

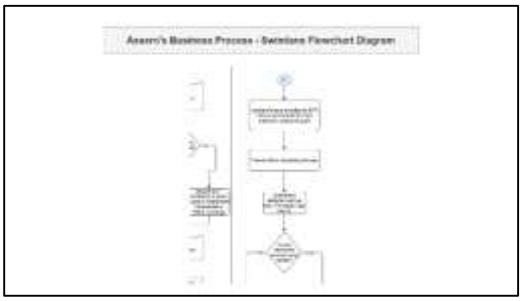

Assero Global Inc.
STI College- Muñoz-EDSA

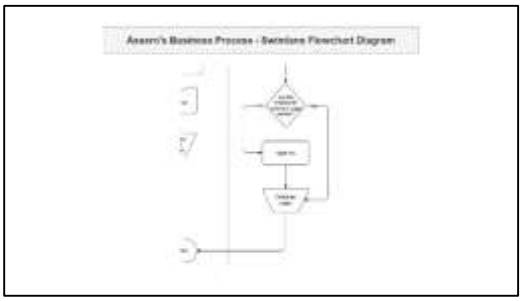

Assero's Business Process - Swimlane Flowchart Diagram


Assero's Business Process - Swimlane Flowchart Diagram

Assero's Business Process - Swimlane Flowchart Diagram


Assero's Business Process - Swimlane Flowchart Diagram

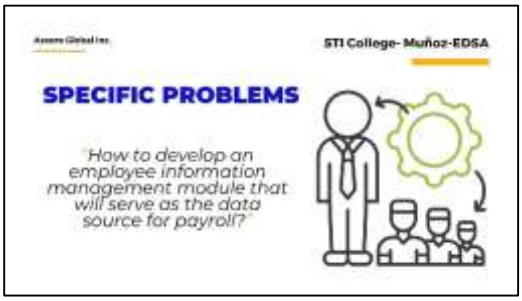

Assero's Business Process - Swimlane Flowchart Diagram

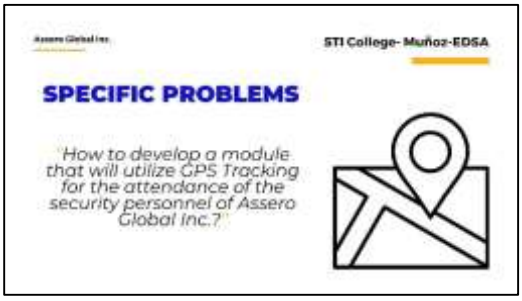

Assero's Business Process - Swimlane Flowchart Diagram

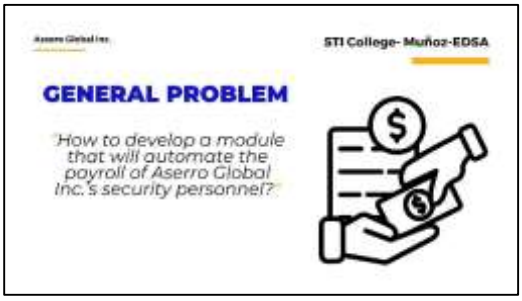
Assero's Business Process - Swimlane Flowchart Diagram

Assero Global Inc.
STI College- Muñoz-EDSA
PROBLEM PRESENTATION

Assero Global Inc.
STI College- Muñoz-EDSA
GENERAL PROBLEM
"How to develop a web-based timekeeping and payroll system with mobile solutions utilizing facial recognition and GPS tracking for the security guards of Assero Global Inc.?"


Assero Global Inc.
STI College- Muñoz-EDSA
SPECIFIC PROBLEMS
"How to develop an employee information management module that will serve as the data source for payroll?"


Assero Global Inc.
STI College- Muñoz-EDSA
SPECIFIC PROBLEMS
"How to develop a module that will utilize GPS Tracking for the attendance of the security personnel of Assero Global Inc.?"


Assero Global Inc.
STI College- Muñoz-EDSA
GENERAL PROBLEM
"How to develop a module that will automate the payroll of Aserro Global Inc.'s security personnel?"

STI College- Muñoz-EDSA

SPECIFIC PROBLEMS

How to develop a module that will provide the employee's payslip?

STI College- Muñoz-EDSA

SPECIFIC PROBLEMS

How to develop a module that will generate reports for Assero Global Inc.?

STI College- Muñoz-EDSA

OBJECTIVE PRESENTATION

STI College- Muñoz-EDSA

GENERAL OBJECTIVE

To develop a web-based timekeeping and payroll system with mobile solutions utilizing facial recognition and GPS tracking for the security guards of Assero Global Inc.

STI College- Muñoz-EDSA

SPECIFIC OBJECTIVES

To develop an employee information management module that will serve as the data source for payroll.

STI College- Muñoz-EDSA

SPECIFIC OBJECTIVES

To develop a module that will utilize GPS Tracking for the attendance of the security personnel of Assero Global Inc.

STI College- Muñoz-EDSA

SPECIFIC OBJECTIVES

To develop a module that will automate the payroll of Aserro Global Inc.'s security personnel.

STI College- Muñoz-EDSA

SPECIFIC OBJECTIVES

To develop a module that will provide the employee's payslip.

STI College- Muñoz-EDSA

SPECIFIC OBJECTIVES

To develop a module that will generate reports for Assero Global Inc.

STI College- Muñoz-EDSA

SCOPE AND LIMITATION

STI College- Muñoz-EDSA

SCOPE

- *Mobile Attendance Module*
- *Web-Based Monitoring Module*
- *Employee Payroll Module*
- *Payslip Preview Module*
- *Employee Records and Attendance*
- *Compliance Module*
- *Report Generation Module*
- *Cloud Storage and Backup Module*

STI College- Muñoz-EDSA

LIMITATION

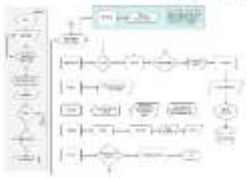
- *User Device Availability*
- *GPS Accuracy*
- *Regulatory Compliance*
- *Employee Device Compatibility*
- *Data Security*

STI College- Muñoz-EDSA

PROPOSED SOLUTION - FLOWCHART

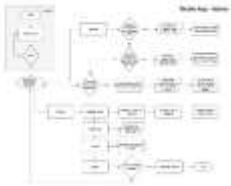
STI College- Muñoz-EDSA

PROPOSED SOLUTION - FLOWCHART



STI College- Muñoz-EDSA

PROPOSED SOLUTION - FLOWCHART



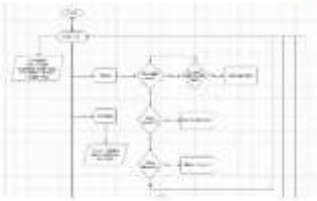
STI College- Muñoz-EDSA

PROPOSED SOLUTION - PROTOTYPE

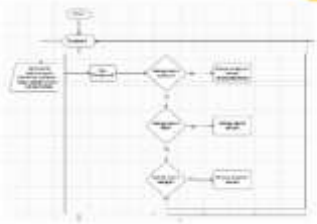
STI College- Muñoz-EDSA

PROPOSED SOLUTION - FLOWCHART (WEB)

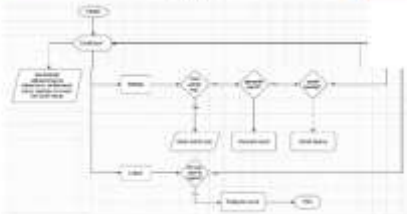
STI College- Muñoz-EDSA



STI College- Muñoz-EDSA



STI College- Muñoz-EDSA



APPENDIX J. PERSONAL CURRICULIM VITAE

Curriculum Vitae of

Daniel I. Tarnate

#76 Blk. 10 Pook Palaris, UP Diliman, Quezon City
deejaytarnate21@gmail.com
+639508300893

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2022 - Present	STI College Muñoz-EDSA
	2012 - 2013	Alternative Learning System
High School		Balara High School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization

SKILLS

Skills	Level of Competency	Date Acquired
Python	Beginner	2025
HTML	Beginner	2024
Microsoft Office	Intermediate	2023
CSS	Beginner	2023
Java	Beginner	2022

TRAINING, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar. or Workshop
2025	Beyond Firewalls: Strengthening People and Systems for a Secure Digital Future
2022	Salesforce Webinar

Curriculum Vitae of

Jason D. Selerio

Marilao, Bulacan
jselerio15@gmail.com
+639278586049

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2017-Present	STI College Muñoz-EDSA
	2015 - 2017	Access Computer College
	2013 - 2014	Lyceum of the Philippines University
High School	2012	Manuel Luis Quezon High School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization

SKILLS

Skills	Level of Competency	Date Acquired
Computer Programming	Intermediate	2017
Program Testing	Intermediate	2017

TRAINING, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar. or Workshop
2025	Beyond Firewalls: Strengthening People and Systems for a Secure Digital Future
2022	Salesforce Webinar

Curriculum Vitae of

Jimwell Andrey T. Concepcion

Baseco Port Area, Manila City
jimwell.concepcion@gmail.com
+639473694339

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2022 - Present	STI College Muñoz-EDSA
	2022	Asia Institute of Maritime Studies
High School	-	Technological Institute of the Philippines

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2024 - Present	Frontend Developer	Freelance/Self-Employed
2023 - 2024	Data Entry	Freelance/Self-Employed

SKILLS

Skills	Level of Competency	Date Acquired
Java	Intermediate	2017
C#	Intermediate	2017
Python	Intermediate	2017
Microsoft Office	Intermediate	2017

TRAINING, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar. or Workshop
2025	Beyond Firewalls: Strengthening People and Systems for a Secure Digital Future
2022	Salesforce Webinar

Curriculum Vitae of

Lhorea Anne R. Oliveros

Apolonio Samson, Quezon City
oliveros.lhoreaanne@gmail.com
+639668231386

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2022 - Present	STI College Muñoz-EDSA
High School	2016 - 2022	San Fracisco High School
Elementary	2010 - 2016	Apolonio Samson Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2023	Social Media Manager	Alberto Sports
2022	Sales Attendant	Dabe's Printing Services

SKILLS

Skills	Level of Competency	Date Acquired
CSS	Beginner	2022
Python	Beginner	2022
HTML	Advanced	2022
Open Broadcaster Software	Intermediate	2020
Canva	Intermediate	2018
Microsoft Office	Intermediate	2016

TRAINING, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar. or Workshop
2025	Beyond Firewalls: Strengthening People and Systems for a Secure Digital Future
2022	Salesforce Webinar

Curriculum Vitae of

Maria Christina S. Resurreccion

#15 Cotabato St. Bago Bantay,Quezon City
mariachristinaresurreccion29@gmail.com
+639620675101

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2022 - Present	STI College Muñoz-EDSA
High School	2016 - 2022	San Fracisco High School
Elementary	2015	Diosdado P. Macapagal Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2020	Academic Tutor	27 Cagayan Cotabato, Ramon Magsaysay, Quezon City

SKILLS

Skills	Level of Competency	Date Acquired
Visual Studio	Intermediate	2023
Ibis Paint	Intermediate	2020
Microsoft Office	Intermediate	2017

TRAINING, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar. or Workshop
2025	Ctrl+Alt+Defend: Online Cyber Hygiene Training
2022	Contact Center Services NCII